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Abstract

This project, titled "**HireNy**" operates within the domain of recruitment technology and addresses the challenges faced by job seekers and organizations in navigating the complexities of the modern job market. The primary objective of the project is to develop a streamlined recruitment platform that connects students, fresh graduates, and companies, enhancing the hiring process through innovative, user-centered features.

The project identifies several critical challenges that motivate its development. Job seekers frequently encounter difficulties in finding relevant opportunities, creating compelling profiles, and often difficulties in application processes. Fresh graduates, in particular, face significant barriers to entering the job market due to limited experience and inadequate guidance. Meanwhile, organizations face challenges in managing high volumes of applications, efficiently evaluating candidates, maintaining clear communication, and handling recruitment-related tasks that often require external support.

To address these issues, "**HireNy**" introduces a dual-sided platform equipped with specialized software features tailored to the distinct needs of job seekers and organizations.

For Job Seekers: The platform provides a seamless job application experience, personalized job notifications, career-enhancing resources, and courses. It includes AI-powered tools for CV creation, direct recruiter communication, organization reviews for transparency, and salary insights for market understanding.

For Organizations: The platform streamlines hiring with features like job posting, candidate filtering, interview scheduling, and technical assessment creation.

B2B Services: Organizations can connect with external service providers for a variety of specialized tasks, such as recruitment automation, event organization, custom tool integration, or other business needs, enhancing efficiency and enabling strategic focus.

The significance of this project lies in its potential to enhance recruitment efficiency and foster user engagement for both job seekers and organizations. By bridging critical gaps in the hiring process, "**HireNy**" offers a comprehensive solution that streamlines workflows, empowers professional growth, and creates a supportive environment for all users. Its innovative features and B2B integrations aim to redefine the service experience making it more accessible, transparent, and efficient.

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List of Abbreviation

Abbreviation	Definition
AI	Artificial Intelligence
CV	Curriculum Vitae
B2B	Business-to-Business
HR	Human Resources
MFA	Multi-Factor Authentication
TLS	Transport Layer Security
JWT	JSON Web Token
OAuth	Open Authorization
API	Application Programming Interface

Chapter 1: Introduction

1.1. Background and Motivation

The job market is a dynamic yet challenging environment, with both job seekers and organizations facing persistent obstacles that hinder effective employment connections. Job seekers often struggle to find positions that align with their skills, experience, and career aspirations. This challenge is further compounded by lengthy, impersonal application processes that can be discouraging. Fresh graduates, in particular, encounter significant barriers when entering the job market, as limited experience and insufficient guidance often hinder their ability to secure initial roles.

Organizations, on the other hand, grapple with the daunting task of managing an overwhelming volume of applications, making it difficult to efficiently identify suitable candidates. The recruitment process involves several time-intensive activities, such as scheduling interviews, designing technical assessments, and maintaining consistent communication with candidates. These challenges not only delay the hiring process but also result in suboptimal candidate experiences, affecting the overall recruitment outcome.

These challenges highlight the pressing need for a system that effectively bridges the gap between job seekers and organizations by facilitating a streamlined, engaging, and efficient hiring process. The development of the **"HireNy"** platform is driven by these challenges, aiming to provide a dual-sided solution tailored to the unique needs of both job seekers and organizations. By incorporating user-focused features, the platform seeks to empower job seekers with better access to relevant opportunities while equipping organizations with advanced tools to optimize their hiring processes.

This study is significant as it proposes a comprehensive system that not only simplifies recruitment but also supports the professional growth of fresh graduates and assists startups in building structured and efficient hiring environments. By addressing critical pain points in the job market, **"HireNy"** offers a transformative approach to recruitment, fostering a more inclusive, effective, and engaging ecosystem for all stakeholders involved.

1.2. Problem Statement

The job market is a complex ecosystem filled with challenges for both job seekers and organizations. Job seekers often face difficulties in identifying suitable roles that align with their skills and experience, selecting the most suitable course to enhance their skills, navigating cumbersome application processes, receiving timely feedback, and building profiles that effectively showcase their potential. At the same time, organizations grapple with sourcing, evaluating, and selecting candidates efficiently while managing various recruitment tasks, including technical test creation, interview scheduling, and service coordination. These challenges highlight the pressing need for a dual-sided platform that bridges the gaps and caters to the needs of both groups.

Challenges Faced by Job Seekers

Job seekers encounter several significant challenges that hinder their ability to secure meaningful employment, including:

1. Difficulty finding relevant opportunities:

- Job seekers often spend excessive time searching for job listings that match their skills, location preferences, and salary expectations, leading to inefficiencies and frustration.

2. Limited access to skill-enhancement resources:

- Affordable and effective training resources and courses are frequently unavailable, leaving many job seekers unprepared for competitive roles in the job market.

3. Inefficient application processes:

- Lengthy and impersonal application systems often leave candidates uncertain about their application status or unclear on specific requirements.

4. Lack of profile-building support:

- Job seekers need assistance in creating standout CVs and professional portfolios that effectively highlight their qualifications and experiences.

5. Barriers for fresh graduates:

- Fresh graduates, in particular, face significant hurdles navigating the job market due to limited experience and lack of proper mentorship, resulting in difficulties securing their first job or even landing an interview.

6. Challenges in assessing career readiness:

- Many job seekers lack the tools and feedback to assess their preparedness for desired roles, leaving gaps in their understanding of skill deficiencies or areas for improvement.

Challenges Faced by Organizations

Organizations also face a unique set of challenges that hinder their ability to efficiently source and select the right candidates. These include:

1. Overwhelming application volumes:

- Organizations often receive an excessive number of applications for job openings, making it difficult for recruiters to identify qualified candidates promptly.

2. Time-consuming recruitment tasks:

- Tasks such as scheduling interviews, creating technical assessments, and maintaining consistent communication with candidates require significant time and resources.

3. Difficulty in filtering and shortlisting candidates:

- The lack of effective tools for filtering candidates based on specific criteria adds complexity to the recruitment process.

4. Difficulty in External Support Services:

- Organizations often need to collaborate with external companies to complete critical tasks across various domains, such as recruitment, marketing, design, and more.
- Despite the critical nature of these services, no centralized platform exists that enables businesses to seamlessly access and manage external service providers. This lack of integration adds logistical complexity, inefficiency, and unnecessary overhead.

5. Lack of data-driven insights:

- Many organizations lack access to analytics and insights that could help optimize hiring decisions and improve recruitment strategies.

1.3. Objectives

The aim of this project is to develop "**HireNy**" a comprehensive job-seeking and recruitment platform designed to bridge the gap between job seekers and recruiters. By streamlining the hiring process and enhancing user engagement, the platform offers a seamless connection between talent and opportunity. It incorporates user-centric features that address the unique needs of both groups while creating a dynamic, efficient, and user-friendly experience.

Key Features for Job Seekers

The platform simplifies the job search process and empowers job seekers to achieve their career goals through the following features:

1. Customized Job Search:

- Filter opportunities based on personal preferences, including skills, location, and salary expectations.

2. Streamlined Application Process:

- Apply for jobs easily with an intuitive and efficient system.

3. AI-Powered CV Creation:

- Build professional CVs using intelligent tools that highlight key skills and experiences.

4. Portfolio Management:

- Create and manage portfolios to effectively showcase talents and accomplishments.

5. Direct Communication:

- Engage with recruiters through integrated communication tools for better clarity and updates.

6. AI Assistant Support:

- Access a virtual assistant for guidance, answering FAQs, and personalized job search support.

7. Organization Insights:

- Leave and view reviews and comments about companies to gain insights into work culture and make informed career decisions.

8. Salary Insights:

- Access salary range data for various roles in their region to set realistic expectations and negotiate effectively.

Key Features for Organizations

Organizations benefit from features that simplify hiring workflows, save time, and optimize the candidate selection process:

1. Job Management:

- Post job openings and manage listings effortlessly.

2. Candidate Filtering:

- Filter and rank candidates based on qualifications and relevance to the role.

3. Interview Scheduling:

- Utilize our built-in features to simplify interview scheduling and coordination, ensuring a seamless and efficient process.

4. Technical Assessments:

- Create and manage technical tests to evaluate candidate competencies.

5. Services Integration:

- Provide and request various professional services, such as marketing, design, advertising, and other industry-specific solutions.

1.4. Project Scope and Limitations

Scope of "HireNy"

The scope of "HireNy" is defined by its mission to create a comprehensive dual-sided recruitment platform that addresses challenges faced by job seekers and organizations in the modern job market. The platform facilitates efficient connections between talent and opportunity, ensuring a user-friendly and transparent experience.

The platform serves:

1. Job Seekers:

- Enabling access to personalized job recommendations based on skills, location, and salary preferences.
- Providing AI-powered tools for creating professional CVs and managing portfolios.
- Offering insights into salary ranges for various roles, assisting users in setting realistic expectations and negotiating effectively.
- Allowing job seekers to leave and review organization feedback to foster informed decision-making.
- Streamlining communication with recruiters for real-time updates on hiring progress.
- Access skill-building resources and courses to develop their skills and prepare for career opportunities.

2. Organizations:

- Simplifying job posting and candidate management workflows.
- Filtering and ranking candidates based on their qualifications and job relevance.
- Facilitating interview scheduling and creating technical assessments.
- Supporting external service integration (e.g., marketing, design, advertising, training services, advertising).

The chosen technology stack includes:

- **Backend:** Django, providing a secure, scalable, and robust foundation.
- **Frontend:** React, ensuring a dynamic and responsive user interface.
- **Mobile Development:** Flutter, enables consistent and efficient applications for both Android and iOS platforms.

Data collection methods include user interactions, analytics, and feedback. Testing will leverage iterative development cycles, incorporating user feedback sessions to refine features and optimize usability.

Limitations

While "HireNy" aims to provide a comprehensive solution for job seekers and organizations, the project acknowledges several limitations:

1. Technical Limitations:

- Advanced AI features, such as tailored job recommendations and automated candidate shortlisting, may require iterative improvements as more data is collected and processed.

2. Time Constraints:

- Non-critical features such as advanced analytics, additional integrations, or expanded reporting tools may be deferred to future updates to meet project deadlines.

3. Geographical Scope:

- The platform's initial launch will focus on specific regions, with gradual expansion planned based on market demand and user adoption rates.

Project Plan

To view Clearer picture of the use case diagram, visit this [Google Drive link](#).

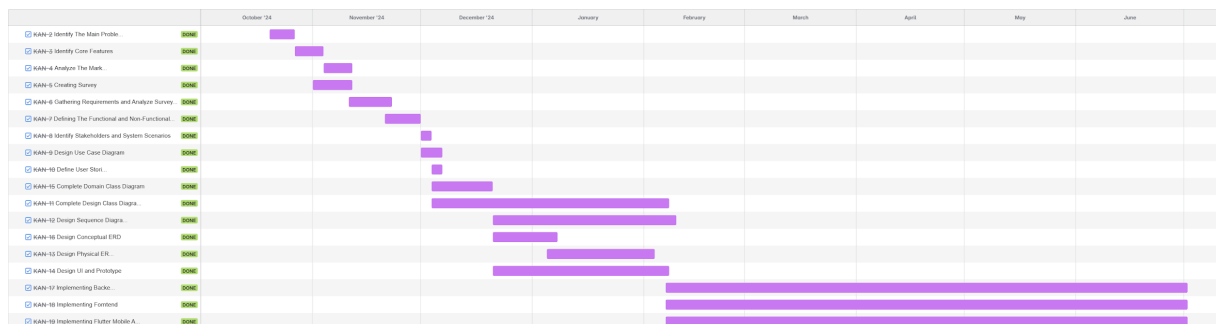


Figure 1: Project Plan

1.5. Project Methodology

In the implementation of "HireNy" we have chosen the Hybrid methodology, combining elements of the **Waterfall** and **Agile** approaches. This methodology is particularly suited for the project as structured step-by-step progression is essential, yet flexibility is required to adapt and refine deliverables throughout the development lifecycle. By leveraging the strengths of both methodologies, we can deliver a robust and evolving recruitment platform.

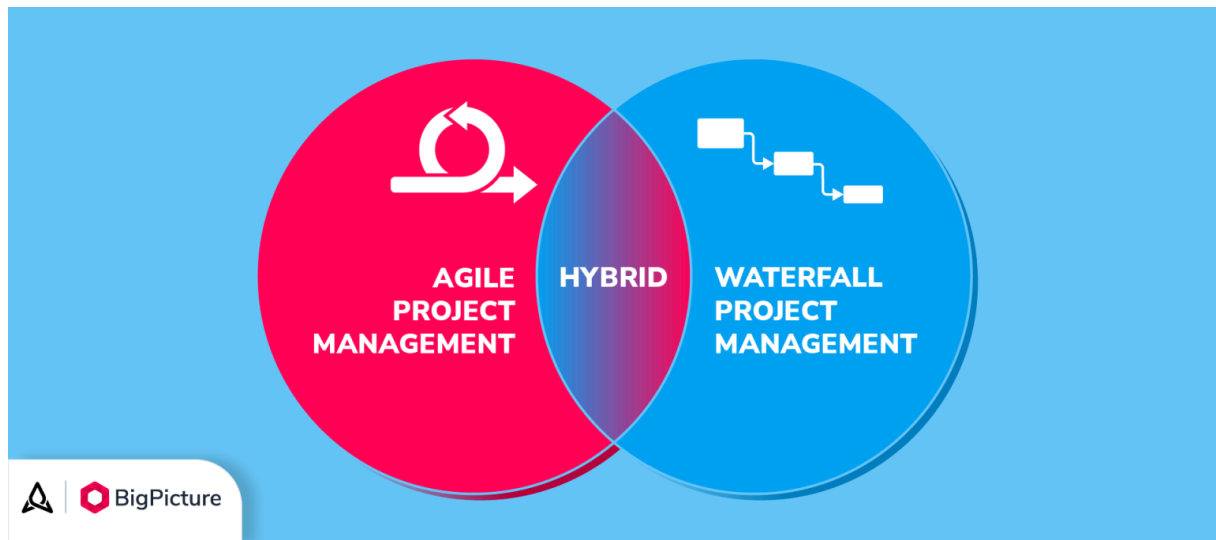


Figure 2: Hybrid Methodology Agile & Waterfall ([Image Source](#))

Why Choose Hybrid (Waterfall and Agile)?

Key Strengths of Waterfall:

- **Sequential Phases:** Waterfall ensures a structured, step-by-step progression, where each phase (e.g., Requirements, Design, Development, Testing, Deployment) must be completed before moving to the next.
- **Clear Milestones:** It establishes well-defined objectives and deliverables for each phase, ensuring systematic progress.
- **Comprehensive Documentation:** Detailed planning and documentation upfront provide a strong foundation and clarity for the entire project lifecycle.

Key Strengths of Agile:

- **Flexibility and Adaptability:** Agile allows for iterative adjustments and refinements within each phase, enabling us to respond to evolving requirements and feedback.

- **Collaborative Approach:** Agile emphasizes team collaboration and continuous stakeholder involvement, fostering better communication and alignment.
- **Incremental Progress:** Features are developed and delivered in smaller iterations (sprints), allowing early validation and reducing risks.

Enhanced Risk Management: By combining Waterfall's structured planning with Agile's iterative adaptability, potential risks or issues can be addressed early in each phase. This reduces the likelihood of delays or rework later in the project lifecycle.

Improved Quality and Collaboration: The hybrid model promotes collaboration between cross-functional teams at each phase while enabling iterative refinements. This fosters clear communication, early validation, and continuous improvement of deliverables.

Hybrid Methodology in Action

Our project will adhere to the principles of the Hybrid (Waterfall and Agile) methodology, ensuring structured progress while allowing for iterative refinements within each phase.

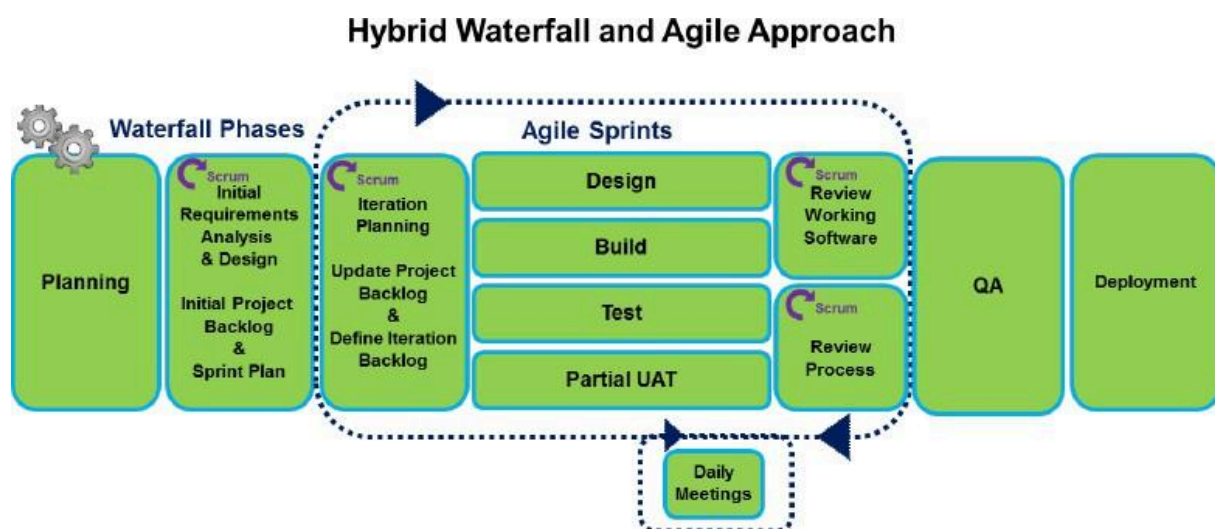


Figure 3: Hybrid Methodology in Action ([Image Source](#))

Breaking Down the Project: The project will be divided into distinct phases, following the step-by-step approach of the Waterfall model (e.g., Requirements Analysis, Design, Development, Testing, Deployment). Within each phase, Agile practices will allow for ongoing modifications, refinements, and improvements based on stakeholder feedback.

1. Requirements and Planning (Waterfall):

- The project begins with a detailed analysis of requirements and planning.

- Clear objectives and milestones will be established to guide subsequent phases.

2. Design and Development (Agile):

- Within the design and development phases, Agile iterations (sprints) will allow for the creation and refinement of functional components.
- Each sprint focuses on specific features, enabling early testing and validation while incorporating feedback.

3. Testing and Validation (Agile):

- Testing will be performed iteratively within each phase to identify and resolve issues early.
- Feedback from stakeholders and users will inform further refinements before moving to the next phase.

4. Deployment and Feedback (Waterfall with Agile Adjustments):

- The platform will be deployed in stages, with opportunities for Agile enhancements based on user feedback.
- This ensures that the final product is polished, functional, and meets stakeholder expectations.

Regular Reviews and Feedback: At the end of each phase, regular reviews will be conducted with team members and stakeholders to evaluate outcomes and gather feedback. This feedback loop ensures that modifications and improvements are seamlessly integrated within each phase.

Adapting to Change: The hybrid nature of this methodology provides a balance between upfront planning and flexibility. While major milestones are addressed sequentially, Agile iterations within each phase allow for adaptability to evolving requirements, market conditions, and technological advancements.

Incremental Delivery: Functional components of the platform will be delivered incrementally throughout the design, development, and testing phases. This approach ensures early validation of features, reduces risks, and enables the platform to evolve in alignment with project goals and user needs.

By adopting the Hybrid (Waterfall and Agile) methodology, we ensure a structured yet flexible approach to the development of "**HireNy**" balancing systematic progress with iterative refinement to deliver a high-quality recruitment platform.

1.6. Project Report Outline

This project report aims to provide a comprehensive overview of "**HireNy**". By offering a detailed examination of our approach and findings, this report serves as a valuable resource for stakeholders, project team members, and anyone interested in understanding the complexities of "**HireNy**", from planning to execution.

It includes:

Chapter 2: Market and Literature Survey

A comprehensive review of the literature and market analysis to identify gaps and opportunities relevant to the project.

Chapter 3: HireNy Analysis

Focuses on the analysis of requirements, problem definition, and system architecture, incorporating diagrams and models to structure the proposed solution.

Chapter 4: HireNy Design and Implementation

Covers the system's design and implementation, including UI development, prototyping, tools, technologies, and methodologies used.

Chapter 5: Testing, Evaluation, and Sustainability

Focuses on verifying system functionality through testing and performance evaluation. And AI effectiveness. Also discusses cost sustainability, infrastructure scalability, and outlines the financial model to maintain long-term operation through monetized features.

Chapter 6: Conclusion and Future Work

Summarizes the project's achievements and outlines the roadmap for future enhancements. Highlights completed features like AI-driven job matching and resume parsing, then proposes technical improvements, ecosystem expansion, accessibility goals, and a monetization strategy through paid features for job seekers and organizations.

Chapter 2: Market and Literature Survey

2.1. Literature Review

Overview of Digital Recruitment Platforms

Digital recruitment platforms such as **LinkedIn**, **Wuzzuf**, and **Bayt** have become essential tools for connecting job seekers with organizations. They provide services like:

- Job postings and applications.
- Candidate management tools for organizations.
- Features like organization reviews, salary insights, and networking opportunities.

Despite these advancements, several limitations persist:

1. Limited Tailored Support for Job Seekers:

Many platforms fall short in offering tailored career guidance, upskilling resources, and job application support, leaving fresh graduates and inexperienced professionals without the tools they need to succeed.

2. Inefficient Recruitment Workflows:

Organizations struggle with time-consuming candidate filtering, limited assessment tools, and fragmented hiring processes, leading to delays and inefficiencies.

3. Lack of Focused Job & Recruitment Solutions

Some platforms operate more as social networking sites rather than dedicated job and recruitment solutions. Their primary focus on connecting people often results in less targeted job matching.

4. Gaps in Upskilling Resources:

Few platforms provide integrated training, professional development resources, or learning opportunities aligned with industry demands, making it harder for job seekers to meet employer expectations.

5. Limited Access to B2B Services:

Businesses often lack a centralized platform to efficiently request and manage essential services like advertising, design, advertising, and other industry-specific solutions, adding complexity to their operations.

Key Gaps in Existing Literature

Academic and market studies highlight the need for a centralized platform that:

- Offer real-time, personalized job alerts and recommendations.

- Enhances recruitment workflows with advanced tools for CV shortlisting, technical test creation, and interview scheduling.
- Enables seamless collaboration between businesses to exchange and access professional services efficiently.

2.2. Market Analysis

2.2.1. Comparing Current Platforms

A detailed comparison of **LinkedIn**, **Wuzzuf**, **Bayt**, and **HireNy** highlights the strengths and gaps in the current market offerings.

Table 1: Comparison Between Current Platforms

Feature	HireNy	LinkedIn	Wuzzuf	Bayt	Limitations of Competitors
Skill Development via Courses	✓ Providing courses to help in skill development.	⚠ Premium access required.	✗ No upskilling tools.	✗ No upskilling tools.	Limited access to skill development courses.
Direct Communication Between Users	✓ Secure chat for direct communication between users.	✓ Messaging feature available.	✗ No direct communication feature.	✗ No direct communication feature.	Lack of direct interaction between users.
Salary Insights	✓ Detailed salary data.	✗ No salary insights.	✗ No salary insights.	⚠ Limited salary insights and not accurate	Limited or no access to salary data.
Organization Reviews	✓ Create, edit, and delete reviews for organizations.	✗ No company review feature.	✗ No company review feature.	✗ No company review feature.	Limited or no ability to review companies.
Automated Candidate Ranking	✓ Automated ranking based on qualifications and preferences.	✗ No automated ranking feature.	✗ No automated ranking feature.	✗ No automated ranking feature.	Competitors rely on manual sorting.
B2B Services	✓ Seamless collaboration and service exchange.	✗ No dedicated B2B features.	✗ No dedicated B2B features.	✗ No dedicated B2B features.	No platform facilitates direct business collaborations.
Interactive Chatbot	✓ Chatbot support for career guidance.	✗ No chatbot assistance.	✗ No chatbot assistance.	✗ No chatbot assistance.	Lack of automated career support for users.

Resume/CV Builder/CoverLetter	✓ Resume/CV Builder/CoverLetter	⚠ Basic CV builder available Other features are not available .	✗ No CV, resume or cover letter tools.	⚠ Premium access required for CV and Other features are not available .	Limited resume/CV builder and cover letter tools.
Custom Assessments	✓ Create and manage technical assessments with customizable questions.	✗ No customizable assessment tools.	✗ No customizable assessment tools.	✗ No customizable assessment tools.	No ability to create tailored assessments for candidates.
Calendar Integration	✓ View and manage upcoming interviews, assessments, and deadlines.	✗ No dedicated calendar for job-related events.	✗ No dedicated calendar for job-related events.	✗ No dedicated calendar for job-related events.	Lack of an integrated calendar for tracking important dates.

2.2.2. Market Gaps

1. Tailored Job Matching:

Platforms struggle to provide accurate, relevant job suggestions based on specific skills, salary expectations, and career goals.

2. Integrated Training:

Few platforms connect users with courses or certifications to improve employability directly within the job search ecosystem.

3. Recruitment Efficiency:

Tools like interview scheduling, candidate ranking, and technical test creation are limited.

4. Comprehensive Reviews and Salary Insights:

Users lack access to transparent organization reviews and localized salary benchmarks.

2.3. Survey Insights

To gain a deeper understanding of the challenges faced by job seekers and organizations, we conducted a comprehensive survey targeting various participants, including students, fresh graduates, mid-career professionals, and recruiters. Below are the highlights and findings of the survey.



Figure 4: Survey Analysis 1

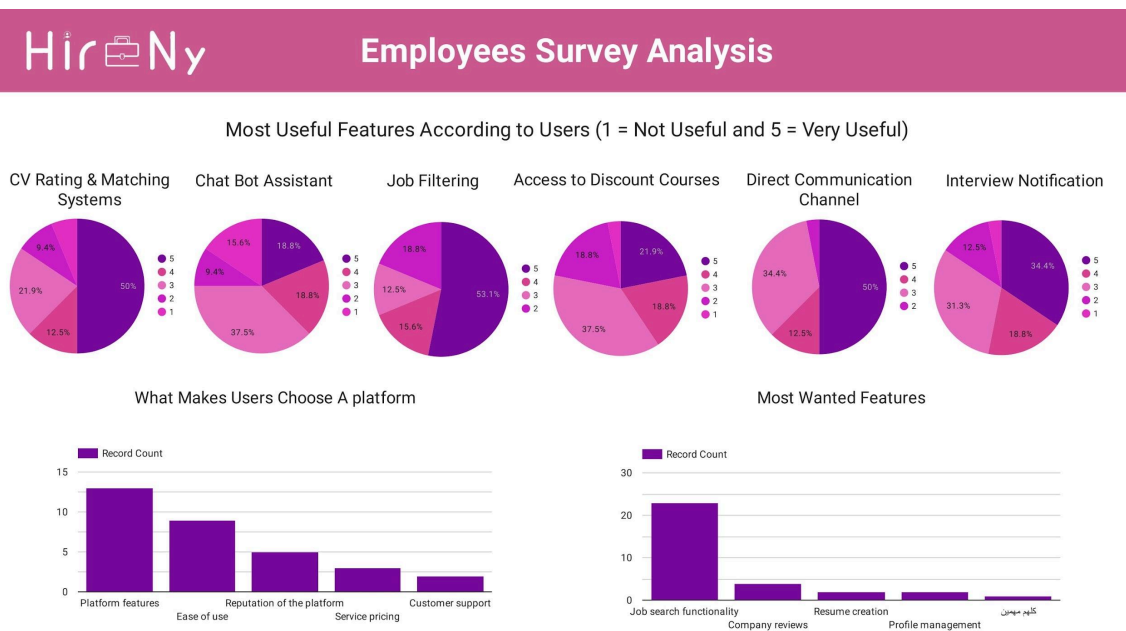


Figure 5: Survey Analysis 2

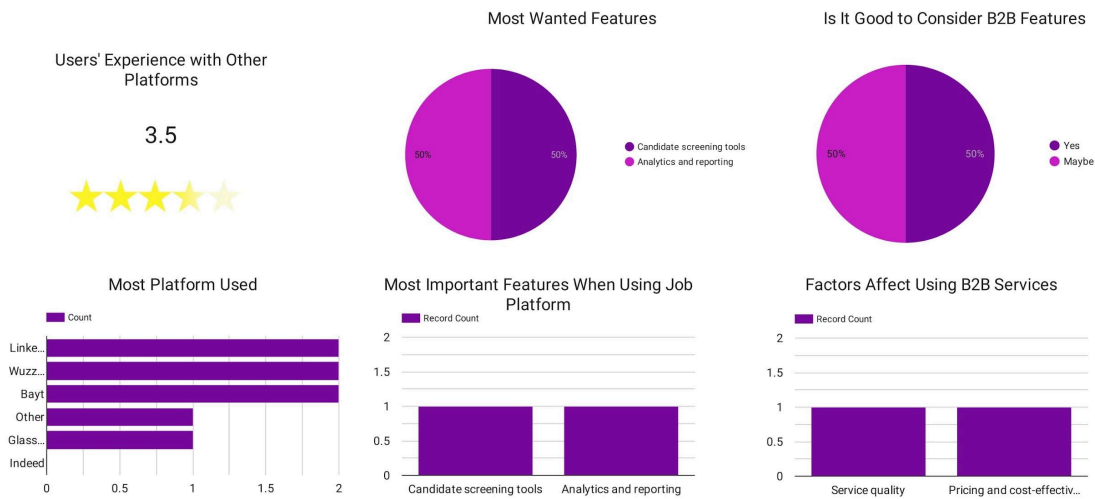


Figure 6: Survey Analysis 3

For detailed insights and analysis, you can access the complete interactive survey report [here](#).

2.4. Opportunities for "HireNy"

Based on the literature review and survey analysis, several opportunities for "HireNy" emerge:

1. Enhanced Job Seeker Experience:

- Advanced filters for skills, location, and salary preferences to improve job matching.
- Real-time notifications for job opportunities and employer updates.
- AI-powered tools for CV creation and portfolio management tailored to job requirements.
- Chatbot assistance for career guidance and job applications.

2. Upskilling Feature:

- Providing industry-aligned courses to help job seekers enhance their skills and stay competitive.
- Integrated learning modules with certifications linked to job applications.

3. Empowering Recruitment Process:

- AI-powered automated candidate ranking.
- Smart shortlisting, filtering, and matching tools.
- Customizable technical assessments for evaluating candidates.
- Integrated interview scheduling and feedback collection system.

4. Transparent Employer Insights:

- Include detailed organization reviews and salary benchmarks for better decision-making.

5. B2B Feature:

- Enable businesses to collaborate on various professional tasks, such as:
- Outsourcing services.
- Employee training.
- Technical assessments.
- Advertising and branding.
- Other industry-specific solutions.

Chapter 3: Analysis

3.1. Main Problem

The job market operates as a dynamic and multifaceted ecosystem, presenting significant challenges for both job seekers and recruiters. Job seekers often struggle to find roles that truly match their skills and experience, contend with lengthy and inefficient application processes, and receive inadequate feedback. They also face difficulties in crafting profiles that effectively highlight their qualifications and potential. Meanwhile, recruiters face their own set of hurdles, such as identifying qualified candidates, evaluating them accurately, and managing the recruitment process. This includes tasks like designing technical assessments, scheduling interviews, and coordinating related services.

3.2. Stakeholders

1. Anonymous User

An anonymous user can explore basic features of the platform, such as exploring the system and viewing public information, but must create an account to access personalized features like applying for jobs or managing profiles.

2. Job Seekers (Students, Fresh Graduates, Employees)

Job seekers benefit from a variety of tools to enhance their job search and career development. They can create and manage profiles, communicate directly with HR, and use chatbots for support. Features include searching and applying for jobs, tracking applications, viewing organization reviews, view calendar, and exploring organization pages. Job seekers can also take technical tests, explore courses, and manage their enrolled courses. Additionally, They can explore salary insights to help them make informed career decisions.

3. Organization's Representatives (Recruiters, Coordinators, etc.)

Organization representatives handle recruitment activities efficiently through features like job posting, managing posts, view calendar, and tracking applications. They can view and filter candidate profiles, set interview times, and communicate directly with candidates via a secure chat. They can also create and manage assessments, offer and request service integrations, and access organizations' pages.

4. Organization's Admin

Same as Organization's Representative but with additional features like managing organization representatives and viewing detailed statistics on recruitment activities.

5. Admin

Admins oversee platform integrity by verifying organizations, and handling reported issues.

6. Super Admin

The super admin holds ultimate authority over platform operations. In addition to performing all admin tasks, they manage other admins, handle detailed reports, and ensure smooth operation of all platform functionalities.

3.3. System Architecture

To view Clearer picture of the system architecture diagram, visit this [Google Drive link](#).

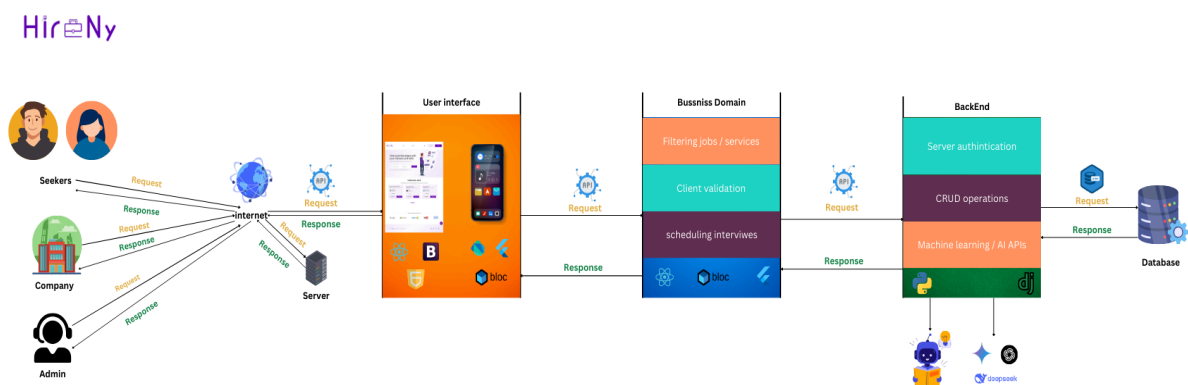


Figure 7: System Architecture Diagram

3.4. Functional Requirement

Functional requirements define the specific behaviors, features, and capabilities a system must have to meet the needs of its users and achieve its objectives. These requirements outline *what the system should do*, describing the essential tasks, operations, and interactions that must be supported.

Functional requirements are typically derived from the system's purpose and the needs of stakeholders, ensuring the system delivers value by fulfilling its intended use cases. By capturing functional requirements effectively, the development process becomes structured and aligned, minimizing misunderstandings and ensuring the final product meets its goals.

Table 2: Functional Requirements

Function Requirement ID	Function Requirement Name	Function Description
Functions of Anonymous User		
FR01	Create a profile	Anonymous users can create a basic profile (either for job seekers or companies) to explore job opportunities, view company profiles, and interact with features like browsing jobs and accessing salary insights. This will allow them to register for future actions like applying for jobs or creating detailed company profiles once they become a registered user.
FR02	Login	All users (Anonymous Users, Registered Users, Organization Representatives, Admins, Super Admins) can log into the platform using their registered credentials (email and password) to access their respective features and dashboards.
FR03	Forget password	All users (Anonymous Users, Registered Users, Organization Representatives, Admins, and Super Admins) can request a password reset if they forget their password. They will receive a reset link via email to securely create a new password.
Common Functions Between Registered Users (Admin / Super Admin / Job Seeker / Organization's Representative , Organization's Admin)		
FR04	Receive notifications	All registered users can receive notifications with any updates in the platform including (new job post, new interview, new technical assessment, etc.)

Function Requirement ID	Function Requirement Name	Function Description
FR05	View profile (Job seeker profile / organization page)	Allows users to view detailed information about job seeker profiles or organization pages, including skills, experience, or company details.
FR06	Manage profile	Enables users to update and manage their personal information.
FR07	Direct communication between users	Enables users to communicate with each other directly through the platform via secure chat.
Common Functions Between Job Seeker and Organization's Representative		
FR08	Use Chatbot	Enables users to interact with a chatbot for seamless engagement and support.
FR09	Search for (job/service) post	For job seekers, this feature enables them to search for job posts based on their preferences. For organization representatives, this feature enables them to search for services offered by other organizations.
FR10	Apply for (job/service) application	For job seekers, this feature enables them to apply for job posts. For organization representatives, this feature enables them to apply for service posts.
FR11	View my (job/service) applications	Enables users to view their applications for postings they have applied for and track their status.

Function Requirement ID	Function Requirement Name	Function Description
FR12	Manage (job/service) application	Enables users to manage their applications for job postings, such as withdrawing applications and updating applications.
FR13	View Calendar	For seekers, they can view the calendar to see the upcoming interviews, deadlines for assessments, and other important dates. For companies, they can view the calendar to see the upcoming interviews and other important dates.
Functions of Job Seeker		
FR14	Take assessments	Allows users to complete assessments created by companies to evaluate their skills for a job post.
FR15	Explore Courses	Job seekers can browse and discover a variety of courses offered by partnered educational platforms.
FR16	Enroll In Courses	Users can enroll for courses of interest to enhance their skills and advance their career goals.
FR17	View My Courses	Users should be able to view a list of all the courses they have enrolled in, along with their current progress.
FR18	Unenroll From Course	Job seekers can remove courses they no longer wish to follow.

Function Requirement ID	Function Requirement Name	Function Description
FR19	View Salary Insights	Users can view updated salary insights about each position based on experience level and location.
FR20	Manage reviews	Enables job seekers to Create, edit, and delete reviews for organization or courses.
FR21	Create CV, resume and cover letter	Enables job seekers to create CV, resume, and cover letter on the platform.
Common Functions Between Organization's Admin and Organization's Representative		
FR22	Create Post	Enables organization Admin or representatives to create job or service posts on the platform.
FR23	Set interview time	Enables users to schedule and manage interview times for job postings.
FR24	View My Posts	View a list of the organization's active posts.
FR25	Manage Posts	Update: Edit and update the details of an existing post (e.g., role responsibilities, job requirements, etc.). Delete: Remove a post from the platform when it's no longer required or has been filled.
FR26	Track applications for each post (job/service)	Enables organization Admin or representatives to track the applications for each job or service post they have created on the platform.

Function Requirement ID	Function Requirement Name	Function Description
FR27	Filter Candidates	The organization Admin or representatives filter list of candidates based on skills, experience, and qualifications
FR28	Automated Candidate Ranking	Enables organization Admin or representatives to view the ranking of job seekers based on their qualifications and preferences.
FR29	Create Assessment	Create a new technical assessment for candidates, including questions, tests, or challenges.
FR30	View My Assessments	View a list of existing technical assessments created by the organization.
FR31	Manage Assessments	Update: Edit and update the content or structure of an existing assessment (e.g., add or remove questions). Delete: Remove an assessment from the platform when it's no longer needed or outdated.
FR32	Search for a Job Seekers	Allows organization Admin or representatives to search for job seekers based on criteria such as skills, experience, education, and location to find suitable candidates for job opportunities.
FR33	Search for companies	Enables organization Admin or representatives to search for other companies on the platform for collaboration or other purposes.
Functions of the Organization's Admin		

Function Requirement ID	Function Requirement Name	Function Description
FR34	Manage organization representative	Allows organization Admin to view, add, or remove other representatives within their organization.
FR35	View Statistics	An organization's admin can access, and review statistics related to postings, including the number of applicants and their application statuses.
Common Functions Between Admin and Super Admin		
FR36	Verify Organization.	Admins review the organization's details and approve or reject requests to create new organization profiles.
FR37	Manage Courses	Admins can manage courses, including adding, editing, and deleting courses.
FR38	Resolve Issues Reported By Candidates/Companies	Admins can address and resolve complaints or issues reported by candidates or companies.
Super Admin's function		
FR39	Manage Admins	Super admin can view existing admins, add new admin, or delete an admin.
FR40	Manage Reports	Admins can monitor data , and platform performance. They can also generate comprehensive reports on system usage, user trends, and overall platform analytics.

3.5. Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational criteria that a system must satisfy to ensure a high-quality user experience. While functional requirements describe what the system does, non-functional requirements focus on how the system performs and operates under various conditions.

These requirements address aspects such as reliability, scalability, usability, security, and maintainability, ensuring the system not only works but works efficiently, effectively, and robustly. They provide guidelines for the system's architecture, design, and performance benchmarks.

Table 3: Non-Functional Requirements

Non-Functional Requirement Name	Non-Functional Requirement Description
Performance Requirements	
Response Time	Django backend APIs should respond to requests within 2 seconds under normal conditions, ensuring minimal wait times for users. React frontend pages should load key components, like the homepage, within 1.5 seconds.
Concurrent Users	The backend should handle 10,000 simultaneous API calls without degradation, ensuring stability during heavy traffic. React and Flutter should maintain smooth performance for all users.
Availability	
Uptime	The system must meet “three nines” availability, 99.9% of the time, minimizing downtime to less than 53 minutes annually. This guarantees uninterrupted user access.
Error Tolerance	The system must be reliable and stable, with minimal errors or bugs, displaying retry options and avoiding crashes, to ensure a good user experience.
Security	

Non-Functional Requirement Name	Non-Functional Requirement Description
Data Encryption	All communication between the backend and frontend must use HTTPS (TLS 1.2+), protecting sensitive data during transmission.
User Authentication	Secure authentication through Django with OAuth 2.0, JWT tokens, and Django Authentication to validate user sessions across web and mobile apps.
Usability	
The Interactive Assistance	platform includes prompts and a chatbot to guide users and answer questions.
User-Friendly interface	The platform should allow users to complete essential tasks like applying for jobs or creating a CV in just three simple steps, ensuring quick and intuitive navigation.
Maintainability	
Scalable Architecture	allows individual components to be updated without impacting the entire system
Comprehensive Testing	Implement testing frameworks to ensure updates don't affect existing functionality.
Consistent Coding Standards	Adhere to standardized coding practices to improve readability and make the system easier to maintain.
Scalability	
System Expansion	The architecture should allow for easy addition of servers or resources as the user base grows, ensuring the system can scale smoothly.
Database Scalability	Ensure the database can scale to handle a growing volume of data without compromising speed or reliability.

Non-Functional Requirement Name	Non-Functional Requirement Description
Portability	
Cross-Browser Compatibility	React frontend must perform well on Chrome, Firefox, Edge, and Safari.
Multi-Platform Mobile Support	Flutter apps should perform well on Android (API 21+) and iOS (version 12+).

3.6. Functional Requirements Overview

Functions of Anonymous User

- Create a profile
- Login
- Forgot password

Common Functions Between Registered Users (Admin / Super Admin / Job Seeker / Organization's Representative / Organization's Admin)

- Receive notifications
- View profile (job seeker profile / org page)
- Manage profile
- Direct communication between user

Common Functions Between Job Seeker and Organization's Representative

- Use chatbot
- Search for (job/service) post
- Apply for (job/service) application
- View my (job/service) applications
- Manage (job/service) application
- View Calendar

Functions of Job Seeker

- Take assessments
- Explore courses
- Enroll in courses
- View my courses

- Unenroll From Course
- View salary insights
- Manage reviews
- Create CV, resume and cover letter

Common Functions Between Organization's Admin and Organization's Representative

- Create post (job/service)
- Set Interview Time
- View my posts
- Manage posts
- Track applications for each post (job/service)
- Filter candidates
- Automated Candidate Ranking
- Create assessment
- View my assessments
- Manage assessments
- Search for job seekers
- Search for companies

Functions of the Organization's Admin

- Manage organization representative
- View statistics

Common Functions Between Admin and Super Admin

- Verify organization
- Manage Courses
- Resolve issues reported by candidates/companies

Functions of Super Admin

- Manage admins
- Manage reports

3.7. User Stories

1. Anonymous User

User Story 1: Create a Profile

As an anonymous user, I want to create a profile (either as a job seeker or a company) so that I can explore job opportunities, view company profiles, and interact with other platform features.

User Story 2: Login

As an anonymous user, I want to log in using my credentials (email and password) so that I can access personalized features like job applications or managing my profile.

User Story 3: Forgot Password

As an anonymous user, I want to reset my password if I forget it so that I can get access to my account by receiving a password reset link.

2. Common Functions Between Registered Users (Admin / Super Admin / Job Seeker / Organization's Representative / Organization's Admin)

User Story 4: Receive Notifications

As a registered user, I want to receive notifications about postings, application status, messages, and other relevant updates so that I can stay informed about platform activities.

User Story 5: View Profile

As a user, I want to view any profile or page on the platform, including my own.

User Story 6: Manage Profile

As a registered user, I want to edit my personal or organizational profile, update contact details, and upload new cv or documents so that my profile is always current and accurate.

User Story 7: Direct Communication Between Users

As a registered user, I want to communicate directly with other users through the platform so that I can interact with job seekers, organization representatives.

3. Common Functions Between Job Seeker and Organization Representative

User Story 8: Use Chatbot

As a job seeker or organization representative, I want to use the chatbot for quick assistance and FAQs, so I can resolve issues or get information without waiting for human support.

User Story 9: Search for (Job/Service) Post

As a job seeker, I want to search for job posts based on my preferences so that I can find relevant job opportunities.

As an organization representative, I want to search for service posts based on my preferences so that I can find relevant service providers.

User Story 10: Apply for (Job/Service) Application

As a job seeker, I want to apply for job postings that match my qualifications and preferences so that I can be considered for relevant job opportunities.

As an organization representative, I want to apply for service postings that meet my requirements so that I can engage with service providers.

User Story 11: View My (Job/Service) Applications

As a job seeker or organization representative, I want to view my applications for postings I have applied for so that I can track the status of my applications.

User Story 12: Manage (Job/Service) Application

As a job seeker or organization representative, I want to manage my applications for job postings by withdrawing applications, updating applications, and tracking the status of my applications so that I can stay organized and informed.

User Story 13: View Calendar

As a seeker, I want to view the calendar so that I can see the upcoming interviews, deadlines for assessments, and other important dates.

As a company, I want to view the calendar so that I can see the upcoming interviews and other important dates.

4. Job Seeker

User Story 14: Take assessments

As a job seeker, I want to take technical tests to assess my skills and qualifications so that I can demonstrate my expertise to potential employers.

User Story 15: Explore Courses

As a job seeker, I want to explore courses related to my field of interest so that I can enhance my skills and knowledge.

User Story 16: Enroll in Courses

As a job seeker, I want to enroll in courses that align with my career goals and interests so that I can acquire new skills and qualifications.

User Story 17: View My Courses

As a job seeker, I want to view the courses I have enrolled in so that I can track my progress and manage my learning journey.

User Story 18: Unenroll From Course

As a job seeker, I want to unenroll from a course that I no longer need or it doesn't match my preference.

User Story 19: View Salary Insights

As a job seeker, I want to view salary insights for different job roles and industries so that I can make informed decisions about my career path.

User Story 20: Manage Reviews

As a job seeker, I want to create, edit, and delete reviews for organizations or courses so that I can share my feedback and experiences with other users.

User Story 21: Create CV, Resume, and Cover Letter

As a job seeker, I want to create a CV, resume, and cover letter on the platform so that I can showcase my qualifications and experiences to potential employers.

5. Common Functions Between Organization's Admin and Organization's Representative**User Story 22: Create Post (Job/Service)**

As an organization Admin or representatives, I want to create job or service posts on the platform so that I can attract job seekers or service providers.

User Story 23: Set Interview Time

As an organization Admin or representatives, I want to schedule and manage interview times for job postings so that I can coordinate and plan interviews effectively.

User Story 24: View My Posts

As an organization Admin or representatives, I want to view the job or service posts I have created so that I can track and manage my postings.

User Story 25: Manage Posts

As an organization Admin or representatives, I want to manage my job or service posts by updating, deleting, or reposting them so that I can keep my postings relevant and up-to-date.

User Story 26: Track Applications for Each Post (Job/Service)

As an organization representative, I want to track the applications for each job or service post so that I can review and manage candidate applications effectively.

User Story 27: Filter Candidates

As an organization Admin or representatives, I want to filter job seekers based on their qualifications, skills, and preferences so that I can identify suitable candidates for job postings.

User Story 28: Automated Candidate Ranking

As an organization Admin or representatives, I want to view the ranking of job seekers based on their qualifications and preferences so that I can identify suitable candidates for job postings.

User Story 29: Create Assessment

As an organization Admin or representatives, I want to create assessments for job seekers to evaluate their skills and qualifications so that I can assess candidate suitability for job postings.

User Story 30: View My Assessments

As an organization Admin or representatives, I want to view the assessments I have created for job seekers so that I can track and manage candidate evaluations.

User Story 31: Manage Assessments

As an organization Admin or representatives, I want to manage the assessments I have created by updating and deleting them so that I can evaluate candidates effectively.

User Story 32: Search for Job Seekers

As an organization Admin or representatives, I want to search for job seekers based on their qualifications, skills, and preferences so that I can identify potential candidates for job postings.

User Story 33: Search for Companies

As an organization Admin or representatives, I want to search for companies based on their profiles, services, and preferences so that I can identify potential service providers.

6. Functions of the Organization Admin**User Story 34: Manage Organization Representative**

As an organization admin, I want to add , update, or delete organization representatives so that I can manage the users who represent the organization on the platform.

User Story 35: View Statistics

As an organization admin, I want to view statistics related to postings, applications, and candidate evaluations so that I can analyze and track platform activities.

7. Common Functions Between Admin and Super Admin**User Story 36: Verify Organization**

As an admin, I want to verify new organization registrations to ensure the platform maintains high-quality, legitimate organizations.

User Story 37: Manage Courses

As an admin or super admin, I want to manage courses so that I can add, edit, or delete courses.

User Story 38: Resolve Issues Reported by Candidates/Companies

As an admin, I want to resolve complaints or issues reported by candidates or companies, ensuring a smooth and respectful experience for all users.

8. Functions of Super Admin**User Story 39: Manage Admins**

As a super admin, I want to view existing admins, add new admins, or delete admins as needed to ensure proper platform management.

User Story 40: Manage Reports

As a super admin, I want to generate and view reports on user activity, platform performance, and other metrics to ensure smooth system operations.

3.8. Use Case Diagram

To view Clearer picture of the use case diagram, visit this [Google Drive link](#), [LucidChart link](#).

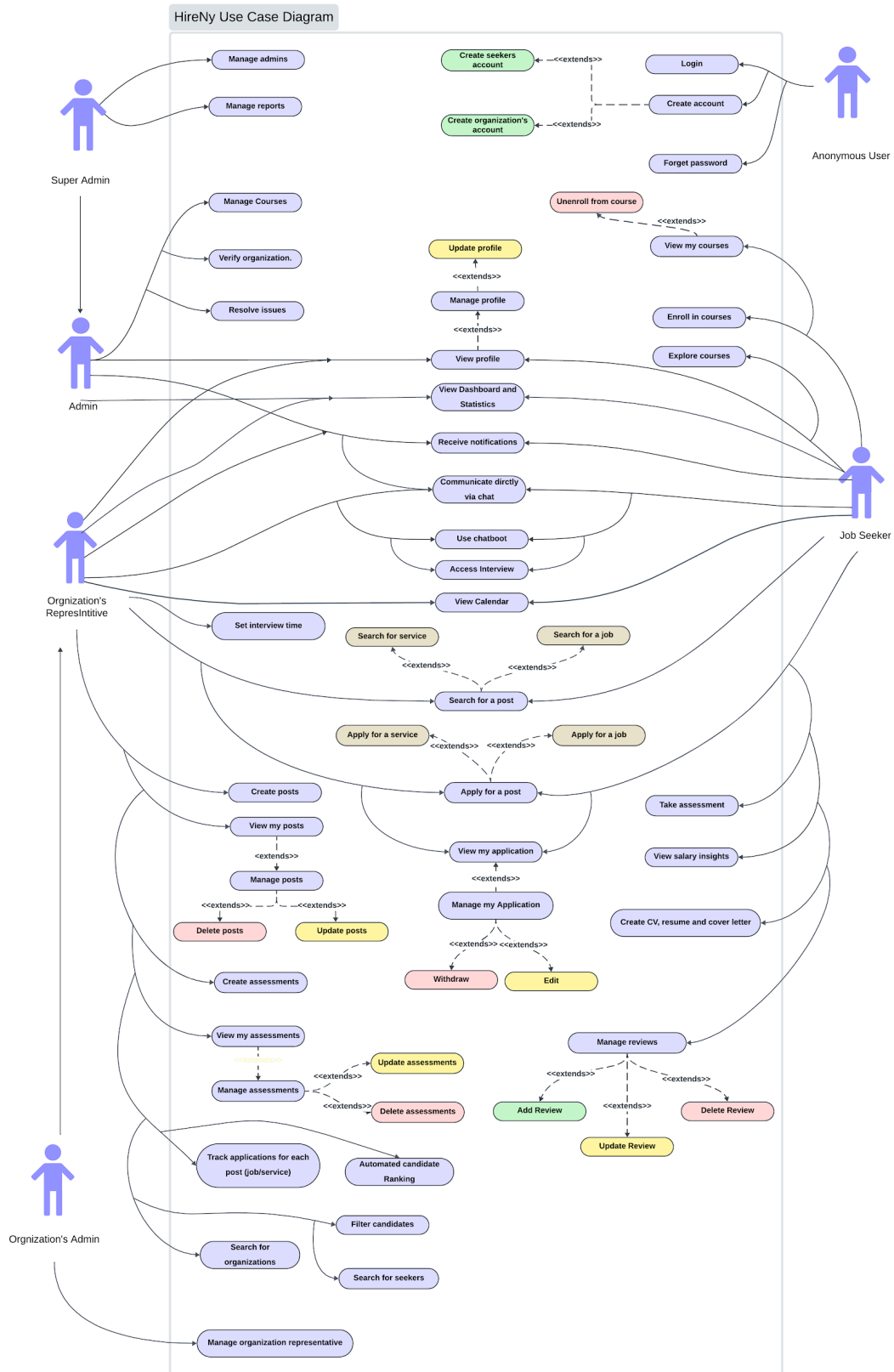


Figure 8: Use Case Diagram

3.9. Use Case Tables / Description

Table 4: Use Case 1 (Create a Profile)

Use Case ID:	001	
Use Case Name:	Create a Profile	
Description	Allows anonymous users (Job Seekers or Companies) to create a basic profile. The profile will be verified for companies before being activated, ensuring that only valid companies can post jobs or interact on the platform.	
Actors:	Anonymous User	
Pre-conditions:	The user is anonymous, and not logged in.	
Post-conditions:	The profile is created. The system or admin will verify the profile before it is fully activated for companies.	
The flow of events:	User Action	System Action
	1. The user accesses the registration form.	
		2. The system displays the registration form.
	3. User fills in the necessary information (email, username, password).	
		4. System validates the entered data (e.g., checks for valid email, strong password)
	5. The user submits the form.	
		6. The system processes the form and creates the profile: • If the user is a company, the system triggers a verification request (e.g., requesting business documents) • System notifies the user (company) that verification is in process (via email or on-platform notification) • Once verified, the company profile is activated and the user is notified (via email or on-platform notification)

Exceptions:	User Action	System Action
	1. User enters incomplete or invalid data	
		2. The system checks the form and notifies users of errors (e.g., missing required fields, invalid email format, weak password).
	3. User attempts to use an existing username or email	
		4. The system detects existing credentials and notifies the user (e.g., "The email address is already registered" or "Username already taken").
	5. The system encounters an internal error	
		6. The system notifies the user and logs the error for investigation (e.g., "An unexpected error occurred. Please try again later").
Includes:	None	
Notes and Issues:	1. Password Requirements: ○ Passwords must meet specific criteria (e.g., a minimum of 8 characters, must include at least one number, one uppercase letter, and one lowercase letter). 2. Username Uniqueness: ○ The system should enforce username uniqueness and notify users if the chosen username is already taken. 3. Email Validation: ○ The system must validate the email format during registration to avoid incorrect entries (e.g., missing "@", or incorrect domain). 4. Account Verification: ○ For company accounts, email verification or additional manual verification steps should be implemented to ensure the authenticity of company profiles. 5. Security: ○ All user data (including email and password) should be encrypted during transmission (e.g., using HTTPS) and securely stored (e.g., using hash functions for passwords). 6. Error Handling: ○ Clear and user-friendly error messages should be shown in case of invalid form submissions (e.g., incomplete fields, invalid credentials).	

	7. User Guidance: ○ Clear instructions should be provided to users regarding the requirements for registration (e.g., password complexity, valid email format, etc.).
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Table 5: Use Case 2 (Login)

Use Case ID:	002	
Use Case Name:	Login	
Description	Allows anonymous users who are registered (Job Seekers, Organization Representatives, Admins, and Super Admins) to log into the platform using their registered credentials (eg: email and password) to access their respective features and profiles.	
Actors:	Anonymous User	
Pre-conditions:	• The user is registered and has valid credentials (eg: email and password). • The user is not logged in yet.	
Post-conditions:	The user successfully logs into the platform and gains access to their dashboard and other platform features.	
The flow of events:	User Action	System Action
	1. The user accesses the login page.	
		2. The system displays the login form (fields eg: email and password and other authentication ways).
	3. The user enters their email and password.	
		4. The system validates the entered credentials (eg, checks if the email exists and the password matches).
	5. The user submits the form.	
		6. The system verifies the credentials and logs the user in.
	7. The system redirects the user to their home page.	
Exceptions:	User Action	System Action
	1. The user enters an incorrect email or password.	

		2. The system checks the credentials and notifies the user of the error (e.g., "Invalid email or password").
	3. The user tries to log in without having an account.	
		4. The system displays an error message, prompting the user to register.
	5. The system encounters an internal error.	
		6. The system notifies the user and logs the error for investigation.
Includes:	None	
Notes and Issues:	1. Email & Password Validation: ○ The system should securely validate email format and password strength before allowing login. 2. Security: ○ Passwords should be encrypted during transmission (HTTPS) and securely stored in the system. ○ Consider implementing multi-factor authentication (MFA) for enhanced security. 3. Error Handling: ○ Clear error messages should be displayed for incorrect login attempts to guide the user.	

Table 6: Use Case 3 (Forget Password)

Use Case ID:	003	
Use Case Name:	Forget Password	
Description	Allows anonymous users to reset their password in case they forget it. The system will send a password reset link via email to the user, allowing them to securely create a new password.	
Actors:	Anonymous User	
Pre-conditions:	The user has an account and has forgotten their password.	
Post-conditions:	The user receives a password reset link via email and can create a new password.	
The flow of events:	User Action	System Action
	1. The user clicks on the "Forgot Password" Button.	

		2. The system displays the password reset form (email input).
	3. The user enters their registered email address	
		4. The system validates the email and checks if it matches an existing account.
	5. The user submits the form.	
		6. The system sends a password reset link to the user's email address.
	7. The user clicks the reset link in their email	
		8. The system displays the page where the user can create a new password.
	9. The user enters and confirms the new password.	
		10. The system validates the new password and updates it in the system.
	11. The system notifies the user that the password has been successfully reset.	
Exceptions:	User Action	System Action
	1. The user enters an email that is not registered.	
		2. The system notifies the user that the email is not recognized and should register as a new user.
	3. User clicks on the register.	
		4. The system navigates to the registration page.
Includes:	None	
Notes and Issues:	● Email Verification: ○ The system should check if the email exists in the system and ensure that the password reset link is only sent to registered email addresses. ● Password Validation:	

	○ The new password should meet the system's security requirements (e.g., a minimum of 8 characters, at least one number, one uppercase letter, etc.). ● Security: ○ The password reset link should have an expiration time (e.g., 1 hour) to prevent misuse. ○ The system should ensure the password is encrypted during transmission (HTTPS) and securely stored in the system. ● User Experience: ○ Clear instructions should be provided to users on how to reset their password and what to do if they don't receive the reset email.
--	--

Table 7: Use Case 4 (Receive Notifications)

Use Case ID:	004	
Use Case Name:	Receive Notifications	
Description	Enables registered users to receive notifications for updates, including new job posts, application statuses, interview schedules, and organizational posts to stay informed.	
Actors:	Registered User (Job Seeker, Organization Representative, Admin, Super Admin)	
Pre-conditions:	● The user is registered and logged into the platform. ● Notifications are enabled in the user's account settings.	
Post-conditions:	The user receives notifications.	
The flow of events:	User Action	System Action
	1. A relevant event occurs on the platform (e.g., new job post, interview update).	
		2. The system generates a notification for the event based on the user's preferences.
Includes:	None	
Notes and Issues:	● Notifications should support in-app. ● Ensure notifications with sensitive data are encrypted and follow privacy standards. ● Log and retry failed notifications.	

Table 8: Use Case 5 (View Profile)

Use Case ID:	005	
Use Case Name:	View Profile	
Description	The platform enables all users to view any profile or page, including their own. Admins and Super Admins also have management privileges to oversee users and content.	
Actors:	Admin / Super Admin / Job Seeker / Org. Representative	
Pre-conditions:	• The user must be logged into the platform. • The user must have an active profile or portfolio available on the platform.	
Post-conditions:	The system displays the user's profile or portfolio as it would appear to others.	
The flow of events:	User Action	System Action
	1. The user (Job Seeker, Organization Representative, Admin, or Super Admin) selects the "View Profile" option for any user on the platform.	
		2. The system displays the selected user's profile (job seeker profile or organization portfolio).
	3. The user reviews the information displayed.	
		4. The system allows the user to navigate back to the dashboard or continue browsing other profiles.
Exceptions:	User Action	System Action
	1. User is not logged in.	
		2. The system prompts the user to log in before viewing their profile or portfolio.
Includes:	None	
Notes and Issues:	• Ensure profiles are displayed clearly with sensitive information protected based on privacy settings. • Implement appropriate pagination for profiles with extensive content.	

Table 9: Use Case 6 (Manage Profile)

Use Case ID:	006	
Use Case Name:	Manage Profile	
Description	The platform allows users to manage their profiles, including adding, editing, and deleting information.	
Actors:	Job Seeker, Organization Representative (HR, Recruiter)	
Pre-conditions:	- The user (job seeker or organization representative) must be logged into the platform. - The user must have access to profile management features.	
Post-conditions:	- The system securely saves the updated, added, or deleted profile/portfolio information. - Notifications (if applicable) are sent to the user to confirm the successful update or deletion of information. - The user's profile or portfolio reflects the changes immediately.	
The flow of events:	User Action	System Action
	1. The user (Job Seeker, Organization Representative, Admin, or Super Admin) selects the "Manage Profile" or "Manage Portfolio" option	
		2. The system displays editable fields for personal information, uploaded files, contact details, and portfolio sections (for organization reps).
	3. The user deletes outdated or irrelevant information if necessary.	
		4. The system confirms deletion and updates the profile or portfolio view accordingly.
Exceptions:	User Action	System Action
	1. User enters invalid input (e.g., incorrect email format).	
		2. The system notifies the user and prompts for valid input.
	3. User leaves mandatory fields empty.	

		4. The system highlights missing fields and requests completion before saving.
	5. User tries to delete critical profile elements.	
		6. The system warns the user about the action and asks for confirmation.
Includes:	None	
Notes and Issues:	• Validate changes to profile information and require user confirmation for updates. • Ensure all profile changes are transmitted securely and maintain an audit trail for tracking.	

Table 10: Use Case 7 (Direct Communication Between User)

Use Case ID:	007	
Use Case Name:	Direct Communication Between Candidates and Org. Representative	
Description	Allows job seekers and Org. Representatives to communicate directly with each other using a secure, encrypted messaging system. This feature helps job seekers ask questions, get updates, or clarify doubts about their job application or interview process. The communication ensures privacy and data security.	
Actors:	Job Seeker, Org. Representative	
Pre-conditions:	The user must be logged into the platform	
Post-conditions:	• Successful conversation between job seeker and HR representative. • Chat history is stored securely.	
The flow of events:	User Action	System Action
	For Job seeker	
	1. The job seeker clicks the "Chat with HR" button.	
		4. System opens the secure chat interface with the relevant HR representative.
	5. Job seeker types and send a message.	
		6. The system sends the message to the HR representative in real-time.
	7. HR representative replies.	

		8. The system displays HR's reply in the chat interface for the job seeker to view.
	9. The job seeker and HR continue exchanging messages.	
		10. The system maintains a real-time secure chat log between both parties.
	For HR	
	1. HR accesses a job seeker profile.	
		2. System displays this profile with a "Chat" button.
	5. HR clicks the "Chat" button.	
		6. The system opens the secure chat interface with the relevant applicant.
	7. Job seekers type and send a message.	
		8. The system sends the message to the HR representative in real-time.
	9. HR representative replies.	
		10. The system displays HR's reply in the chat interface for the job seeker to view.
Exceptions:	11. The job seeker and HR continue exchanging messages.	
		12. The system maintains a real-time secure chat log between both parties.
	User Action	System Action
	1. The user tries to send an empty message.	
		2. The system notifies the user that the message cannot be empty and prompts them to type a message.
	5. Technical issue during chat.	

		6. The system notifies both parties about the issue and prompts them to try again later.
Includes:	None	
Notes and Issues:	• The system should ensure end-to-end encryption of all messages exchanged. • The system must store chat history for future reference and troubleshooting, while also complying with privacy policies.	

Table 11: Use Case 8 (Use Chatbot)

Use Case ID:	008	
Use Case Name:	Use Chatbot	
Description	Allows both job seekers and organization representatives (HR, recruiters) to interact with an AI-powered chatbot for assistance. Job seekers can use the chatbot for asking questions, and getting guidance on various tasks, while organization representatives can manage job posts, schedule interviews, and get guidance on recruitment processes.	
Actors:	Job Seeker, Organization Representative	
Pre-conditions:	The user (job seeker or organization representative) must be logged into the platform	
Post-conditions:	The user successfully interacts with the chatbot to get assistance with their respective tasks.	
The flow of events:	User Action	System Action
	1. The User accesses the chatbot interface.	
		2. The system opens the chatbot interface and welcomes the user.
	3. The User asks a question or requests assistance.	
		4. The system interprets the query and provides a relevant response or guidance.
	5. The User interacts with the chatbot for clarification.	
		6. The system continues assisting the user by providing detailed information or following the user's requests.

Exceptions:	User Action	System Action
	1. The chatbot cannot understand the user's query.	
		2. The system asks the user to rephrase the query or offers suggestions based on common tasks.
	3. The chatbot doesn't have access to the requested data.	
		4. The system notifies the user and suggests alternative methods to retrieve the information.
Includes:	None	
Notes and Issues:		

Table 12: Use Case 9 (Search for a Post)

Use Case ID:	009	
Use Case Name:	Search for a post	
Description	Allows job seekers and org. representative to search for available job/service opportunities based on various filters like location.	
Actors:	Job Seeker, Organization Representative	
Pre-conditions:	The user is logged into the platform	
Post-conditions:	The user is presented with a list of job opportunities that match the search criteria.	
The flow of events:	User Action	System Action
	1. The user accesses the post search page.	
		2. The system displays the search form with available filters.
	3. The user selects the filters.	
		4. The system processes the selected filters and prepares a list of matching post opportunities.
	5. The user selects a post listing to view more details.	

		6. The system displays detailed information about the post.
Exceptions:	User Action	System Action
	1. The user selects filters that yield no results	
		2. The system notifies the user that no post match the criteria and suggests modifying the search.
	3. The user submits an invalid filter (e.g., incorrect location).	
		4. The system validates the entered filter and prompts the user to correct the input if necessary.
	5. The system encounters an internal error	
		6. The system notifies the user of the error and logs it for investigation.
Includes:	None	
Notes and Issues:	1. Search Functionality: ○ The search should be fast and optimized to display results without long loading times. ○ The system should allow users to combine multiple filters for a more refined search experience (e.g., location + category). 2. Filter Validation: ○ The system should ensure that filters are valid. 3. Post Listings Updates: ○ Ensure post listings are up to date and removed once they're no longer available.	

Table 13: Use Case 10 (Apply For a Post Application)

Use Case ID:	010
Use Case Name:	Apply for a post application
Description	Allows users to apply for available post openings by submitting their application. The user can track the application status and receive updates based on the company's review process.
Actors:	Job Seeker and Organization's Representative
Pre-conditions:	The user is logged in and has browsed through post listings.
Post-conditions:	The user successfully submits their application for a post, and the application status is recorded in the system.

The flow of events:	User Action	System Action
	1. The user selects a post listing they are interested in applying for.	
		2. The system displays the post details and application form.
	3. The user fills in the required details	
		4. The system validates the application form.
	5. The user submits the application.	
		6. The system processes the application and sends a confirmation to the user.
	7. The user receives a notification confirming the application has been submitted.	
		8. The system stores the application and updates the post status to reflect that the user has applied.
Exceptions:	User Action	System Action
	1. The user enters incomplete or invalid details in the form.	
		2. The system prompts the user to correct the invalid fields.
	3. The system encounters an internal error while processing the application.	
		4. The system notifies the user of the error and suggests they try again later.
Includes:	None	
Notes and Issues:	1. Application Confirmation: The system should send a confirmation email upon submission of the application, including the next steps or contact details for follow-up.	

	2. Validation: ○ The system should validate the user's information to ensure the correct format.
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Table 14: Use Case 11 (View My Application)

Use Case ID:	011	
Use Case Name:	View My Application	
Description	Allows users to view and track the status of their applications and any additional feedback provided by the company.	
Actors:	Job Seeker and Organization's Representative	
Pre-conditions:	The user is logged in and has submitted one or more applications.	
Post-conditions:	The user can view the current status of their applications and any relevant updates or feedback.	
The flow of events:	User Action	System Action
	1. The user navigates to the "My Applications" section	
		2. The system retrieves and displays a list of all applications the user has submitted, with status indicators.
Exceptions:	User Action	System Action
	1. The user tries to view an application that no longer exists or was deleted.	
		2. The system notifies the user that the application is no longer available or has been removed.
	3. The user encounters a technical issue while viewing applications.	
		4. The system notifies the user of an error and suggests they try again later.
Includes:	None	
Notes and Issues:	Ensure the system provides timely updates on the status of applications. And the status field should be visible.	

Table 15: Use Case 12 (Manage Application)

Use Case ID:	012	
Use Case Name:	Manage Application	
Description	Allows users to manage their applications. This includes editing application details or withdrawing an application.	
Actors:	Job Seeker and Organization's Representative	
Pre-conditions:	The user is logged in and has submitted at least one application.	
Post-conditions:	The user's application is either updated or withdrawn, and the system reflects the new status.	
The flow of events:	User Action	System Action
	1. The user navigates to the "My Applications" section.	
		2. The system displays a list of the user's applications with current statuses.
	3. The user selects a specific application to manage.	
		4. The system displays the application details.
	5. The user clicks on "Edit" to update application details or "Withdraw" to cancel the application.	
		6. If the user edits, the system validates the new information. If the user withdraws, the system removes the application from the active list and updates its status.
	7. The user receives a confirmation of the action (either successful update or withdrawal).	
		8. The system updates the application in the database and notifies the company (if the application was withdrawn).
Exceptions:	User Action	System Action

	1. The user tries to edit an application after the deadline or after it has been reviewed.	
		2. The system disables the "Edit" option and informs the user that editing is no longer possible.
	3. The user tries to withdraw an application after the company has contacted him.	
		4. The system disables the "Withdraw" option and notifies the user that the application is under review.
	5. The system encounters an error while processing the update/withdrawal.	
		6. The system notifies the user of the error and suggests they try again later.
Includes:	None	
Notes and Issues:	Ensure that the system checks if the post is still open or if the application is still under review before allowing edits or withdrawals.	

Table 16: Use Case 13 (View my calendar)

Use Case ID:	013	
Use Case Name:	View My Calendar	
Description	Allows job seekers and org. representative to take overview about their schedule organized in a calendar.	
Actors:	Job Seeker and Organization's Representative	
Pre-conditions:	The user is logged in the system.	
Post-conditions:	The user can view their schedule and any other details about specific tasks in the calendar.	
The flow of events:	User Action	System Action
	1. The user navigates to the "My Calendar" section.	
		2. The system displays a calendar showing the tasks of the user highlighted.
	3. The user selects a specific task from the calendar.	

		4. The system displays information about this task.
Exceptions:	User Action	System Action
	1. The user tries to select a task that is no longer available.	
		2. The system displays a message to inform the user that the task was deleted.
Includes:	None	
Notes and Issues:		

Table 17: Use Case 14 (Take Assessment)

Use Case ID:	014	
Use Case Name:	Take Assessment	
Description	Job Seeker	
Actors:	Allows job seekers to take a required technical test for a job application. The results are sent to the employer for evaluation.	
Pre-conditions:	• The job seeker has applied for a job that requires a technical test. • The job seeker is invited by the employer to take the technical test.	
Post-conditions:	• The job seeker successfully completes the technical test. • The results of the test are saved and made available for review by the employer.	
The flow of events:	User Action	System Action
	1. The user applies for a job that needs a technical test.	
		2. The system notifies the user about the upcoming technical test.
	3. The user clicks the link to start the test.	
		4. The system verifies the user's eligibility and provides access to the test.
	5. The user completes the test.	
		6. The system records the user's responses or submissions and tracks

		the progress of the test.
	7. The user submits the completed test.	
		8. The system processes the test results, saves them, and shows the result.
Exceptions:	User Action	System Action
	1. The user attempts to start the test after the deadline.	
		2. The system notifies the user that the test is no longer available and provides a reason (e.g., the test deadline has passed).
	3. The user experiences technical issues during the test (e.g., connection loss).	
		4. The system prompts the user to resume the test if the issue is resolved or notifies them of any problems if they cannot complete the test.
	5. The user submits the test without answering all questions.	
		6. The system prompts the user to complete any unanswered questions before submitting.
Includes:	None	
Notes and Issues:	• The system should provide a clear and easy-to-understand interface for taking the test. • Test completion time should be monitored and clearly communicated to users.	

Table 18: Use Case 15 (Explore Courses)

Use Case ID:	015
Use Case Name:	Explore Courses
Description	Allows job seekers to browse available courses offered on the platform. They can filter courses based on categories, industry, or duration to find relevant learning opportunities.
Actors:	Job seeker

Pre-conditions:	Job seekers must be logged into the platform.	
Post-conditions:	The job seeker has explored courses and selected ones they are interested in.	
The flow of events:	User Action	System Action
	1. Job seeker clicks on the "Explore Courses" section.	
		2. The system displays a list of available courses with filters like category, skill level, etc.
	3. Job seeker applies filters (e.g., "Marketing," "Graphic design")	
		4. System updates the list of courses based on the applied filters and shows relevant courses.
	5. Job seeker browses course details.	
		6. System displays detailed information about each course, including the syllabus, duration, instructor, and pricing
Exceptions:	User Action	System Action
	1. The user tries to access explore courses without being logged in.	
		2. The system prompts the user to log in before proceeding to explore courses.
	3. No courses available based on selected filters	
		4. The system notifies the user that no courses match the current filters and suggests trying different criteria.
Includes:	None	
Notes and Issues:	• Ensure that course listings are updated regularly to reflect the availability and status of courses. • Provide adequate descriptions and reviews for each course to help job seekers make informed decisions.	

Table 19: Use Case 16 (Enroll in Courses)

Use Case ID:	016	
Use Case Name:	Enroll in Courses	
Description	Allows job seekers to enroll in courses to enhance their skills. After selecting a course, job seekers can proceed with the enrollment process.	
Actors:	Job Seeker	
Pre-conditions:	• The job seeker must be logged in to their account. • The job seeker must have explored courses and selected one they are interested in.	
Post-conditions:	The job seeker is successfully enrolled in the course and can access course materials.	
The flow of events:	User Action	System Action
	1. Job seeker selects a course to enroll in.	
		2. The system displays the course enrollment page with details, and the option to start enrollment.
	3. The job seeker clicks the "Enroll" button.	
		4. The system navigates the user to the content page..
Exceptions:	User Action	System Action
	1.If the course is no longer available	
		2. the system displays an error message: "The selected course is no longer available. Please choose another course."
Includes:	None	
Notes and Issues:	Ensure course materials are accessible only to enrolled users and provide progress tracking features.	

Table 20: Use Case 17 (View My Courses)

Use Case ID:	017
Use Case Name:	View My Courses

Description	Allows job seekers to view a list of all the courses they have enrolled in.	
Actors:	Job Seeker	
Pre-conditions:	• The job seeker must be logged into the platform. • The job seeker must have enrolled in one or more courses.	
Post-conditions:	The job seeker can see a list of their enrolled courses, along with their current progress and other course details.	
The flow of events:	User Action	System Action
	1. Job seeker navigates to the "My Courses" section.	
		2. The system displays a list of courses that the user has enrolled in, showing relevant information such as course name.
	3. Job seeker selects a course from the list.	
Exceptions:	User Action	System Action
	1. The user has not enrolled in any courses.	
		2. The system notifies the user that they have not enrolled in any courses and suggests exploring available courses.
Includes:	None	
Notes and Issues:		

Table 21: Use Case 18 (Unenroll From Course)

Use Case ID:	018
Use Case Name:	Unenroll From Course
Description	Allows job seekers to remove courses from their account. This feature enables job seekers to decide whether to continue with or withdraw from a course.
Actors:	Job Seeker
Pre-conditions:	• Job seekers must be logged into the platform.

	The job seeker must be enrolled in at least one course.	
Post-conditions:	The job seeker has successfully removed a course.	
The flow of events:	User Action	System Action
	1. Job seeker accesses the "My Courses" section	
		2. The system displays a list of courses with options to remove.
	5. Job seeker clicks on remove course.	
		6. The system successfully removes the selected course from the account.
Exceptions:	User Action	System Action
	1. The user is not enrolled in the selected course.	
		2. The system notifies the user that they are not enrolled in the course and cannot remove it.
Includes:	None	
Notes and Issues:		

Table 22: Use Case 19 (View Salary Insights)

Use Case ID:	019	
Use Case Name:	View Salary Insights	
Description	Allows job seekers to view salary insights for different positions based on factors like experience and location. This feature helps job seekers make informed decisions about their careers and job applications.	
Actors:	Job Seeker	
Pre-conditions:	Job seekers must be logged into the platform	
Post-conditions:	The job seeker can view updated salary data for relevant job positions.	
The flow of events:	User Action	System Action
	1. Job seeker navigates to the "Salary Insights" section.	
		2. The system displays a form to enter search criteria.

	3. The job seeker fills the form (e.g., location, experience level, industry).	
		4. The system updates the salary data to reflect the selected filters.
Exceptions:	User Action	System Action
	1. No salary data is available for the selected criteria.	
		2. The system notifies the user that no data is available for the selected criteria and suggests trying different options.
Includes:	None	
Notes and Issues:	Salary data should be regularly updated to reflect current market trends.	

Table 23: Use Case 20 (Manage Reviews)

Use Case ID:	020	
Use Case Name:	Manage Reviews	
Description	Enables users to view, edit, or delete reviews about organizations or courses. Reviews provide insights into company culture, work environment, employee experiences, and course quality.	
Actors:	Job Seeker	
Pre-conditions:	The user is logged in and browsing.	
Post-conditions:	Reviews are viewed, edited, or deleted, and changes are reflected in the system.	
The flow of events:	User Action	System Action
	1. The user clicks on the "Reviews" section.	
		2. The system displays a list of reviews.
	3. The user submits a new review.	
		4. The system validates and adds the review.
	5. The user edits their previously submitted review.	

		6. The system updates the review after validation.
	7. The user deletes their review.	
		8. The system removes the review and updates the review count.
Exceptions:	User Action	System Action
	1. The user attempts to view reviews for an organization or course with no reviews.	
		2. The system displays a message indicating no reviews are available.
	3. The user submits a review with invalid or inappropriate content.	
		4. The system rejects the review and notifies the user with the reason.
	5. The user attempts to edit or delete a review they didn't submit.	
		6. The system denies access and displays an error message.
	7. The system encounters an error while processing reviews.	
		8. The system notifies the user and logs the issue for further investigation.
Includes:	None	
Notes and Issues:	Ensure that reviews go through a moderation process to avoid inappropriate content.	

Table 24: Use Case 21 (Create CV, resume and cover letter)

Use Case ID:	021
Use Case Name:	Create CV, resume and cover letter
Description	Enable the user to Create CV, resume and cover letter using enhanced AI tools based on user information recorded on the system.

Actors:	Job Seeker	
Pre-conditions:	The user is logged in and browsing.	
Post-conditions:	The user will be able to download the generated document.	
The flow of events:	User Action	<i>System Action</i>
	1. The user clicks on the AI tools section .	
		2. The system navigates the user to the documents creation page.
	3. The user chooses the type of document he wishes to generate.	
		4. The system generates the specified documents based on account information, and provides a download link.
	5. the user clicks on the download link.	
		6. the system starts to download the document on the user's device.
Exceptions:	User Action	<i>System Action</i>
	1. The system encounters an error while generating the document.	
		8. The system notifies the user and logs the issue for further investigation.
Includes:	None	
Notes and Issues:		

Table 25: Use Case 22 (Create Post)

Use Case ID:	022
Use Case Name:	Create Post
Description	Enables organization Admin or representatives to create job or service posts on the platform.
Actors:	Organization Representative , Organization Admin

Pre-conditions:	The organization Admin or representatives must be logged into the platform.	
Post-conditions:	The post is created successfully and submitted .	
The flow of events:	User Action	System Action
	1. The organization Admin or representatives access the "Create Job Post" or "Create Service Post (Offer Service)" section.	
		2. The system displays the post creation form.
	3. The organization Admin or representatives fills details like role, requirements, and salary.	
		4. The system validates the form and shows a preview of the job post.
	5. The organization Admin or representatives submits the job post.	
Exceptions:		6. The system saves the post and submits it.
	User Action	System Action
	1. The user enters incomplete or invalid data.	
		2. The system notifies the user of the errors and prompts them to correct the missing or incorrect fields.
Includes:	None	
Notes and Issues:	Include clear validation for post fields (e.g., salary format, requirements).	

Table 26: Use Case 23 (Set interview time)

Use Case ID:	023
Use Case Name:	Set interview time
Description	Enables users to schedule and manage interview times for job postings.
Actors:	Organization Representative , Organization Admin

Pre-conditions:	The organization Admin or representatives must be logged into the platform.	
Post-conditions:	• The interview time was scheduled successfully. • The system notified the job seeker.	
The flow of events:	User Action	System Action
	1. the user clicks on people who applied for a job.	
		2. The system displays all job seekers who applied for this job.
	3. The user selects the candidates and clicks on schedule interview.	
		4. The system displays a form for interview time.
	5. the user enters interview time and clicks submit.	
Exceptions:	User Action	System Action
	1. The user tries to schedule an interview time that is already taken for another job.	
		2. the system notify the user to try scheduling another time.
Includes:	None	
Notes and Issues:		

Table 27: Use Case 24 (View My Posts)

Use Case ID:	024
Use Case Name:	View My Posts
Description	Allows organization Admin or representatives to view a list of all job posts they have created. They can view the number of applicants.
Actors:	Organization Representative , Organization Admin
Pre-conditions:	The organization Admin or representatives must be logged into the platform.

Post-conditions:	The organization Admin or representatives views a list of their posts with applicant details.	
The flow of events:	User Action	System Action
	1. The organization representative navigates to the "My Job Posts" or "My Service Posts" section.	
		2. The system displays a list of posts.
	3. The organization representative clicks on a specific post to view details.	
Exceptions:		4. The system displays the details of the selected post, including the number of applicants and other relevant information.
	User Action	System Action
	1. No job posts are available for the representative.	
		2. The system notifies the user that no posts are currently available .
Includes:	None	
Notes and Issues:	- Ensure that job post status is updated in real-time to avoid discrepancies. - Provide easy access to applicant details for each post.	

Table 28: Use Case 25 (Manage Posts)

Use Case ID:	025	
Use Case Name:	Manage Posts	
Description	Allows users to manage their job posts. This includes editing existing posts, updating job requirements, or deleting posts that are no longer needed.	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	- The organization representative must be logged into the platform. - The job post must exist and be under the representative's organization.	
Post-conditions:	The job post is updated or deleted successfully.	
The flow of events:	User Action	System Action

	1. The user accesses the "My Job Posts" or "My Service Posts" section.	
		2. The system displays a list of all the organization's posts.
	3. The user selects a post to edit or delete.	
		4. The system saves the changes and updates the job post accordingly.
Exceptions:	User Action	System Action
	1. User attempts to delete a post that has applicants.	
		2. The system warns the user that deleting the post will affect applicants and asks for confirmation.
	3. The user edits a post with invalid data.	
		4. System checks for errors and notifies the user of invalid data (e.g., missing job requirements, salary field not filled).
Includes:	None	
Notes and Issues:		

Table 29: Use Case 26 (Track Applicant Numbers for Each Job)

Use Case ID:	026	
Use Case Name:	Track Applicant Numbers for Each Job	
Description	Allows users to view the number of applicants for each job post they have created. This helps them assess the level of interest in specific job postings.	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	- The organization representative must be logged into the platform. - The organization should have one or more active job posts with open applications.	
Post-conditions:	The representative successfully views the number of applicants for each job post.	
The flow of events:	User Action	System Action

	1. The user navigates to the "Job Posts" section.	
		2. The system displays a list of job posts with the number of applicants for each post.
	3. The user selects a job post to view more details.	
		4. The system shows the details of the job post along with the applicant list and their statuses ("Applied," "Interviewing").
	5. The organization representative uses filtering or sorting options such as by applicant status.	
		6. The system filters and sorts applicant data, showing counts based on selected criteria like applied or shortlisted.
Exceptions:	User Action	System Action
	1. No applicants for a job post	
		2. The system displays a message saying "No applicants yet" for the selected job post.
	3. The system encounters a data retrieval issue.	
		4. The system notifies the user that there is an issue retrieving the data and prompts them to try again.
	5. Incorrect filters applied such as filtering by status with no matching applicants.	
		6. The system notifies the user: "No applicants match the selected criteria. Try adjusting your filters."
Includes:	None	
Notes and Issues:	Include filtering or sorting options for applicants based on status (e.g., applied, interview, rejected).	

Table 30: Use Case 27 (Filter Candidates)

Use Case ID:	027	
Use Case Name:	Filter Candidates	
Description	Allows users to filter applicants based on specific criteria (e.g., skills, experience, location). This helps representatives quickly find suitable candidates.	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	- The user must be logged into the platform. - Applicants must have applied for a job post.	
Post-conditions:	The user can view filtered lists of candidates based on chosen criteria.	
The flow of events:	User Action	System Action
	1. The user accesses the "Filter" options.	
		2. The system displays a list of filter criteria, including skills, location, years of experience, etc.
	3. The user selects desired filter criteria.	
		4. The system filters the list of applicants based on the selected criteria and displays the filtered results.
Exceptions:	User Action	System Action
	1. No candidates match the selected filters.	
		2. The system displays a message saying "No candidates match the selected criteria."
Includes:	None	
Notes and Issues:	- Provide multiple filtering options for comprehensive candidate search. - Ensure that filtering results are updated in real time.	

Table 31: Use Case 28 (Automated Candidate Ranking)

Use Case ID:	028
Use Case Name:	Automated Candidate Ranking

Description	Allows users to view candidates by their matching score (How much their profile match the job they applied for)	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	• The user must be logged into the platform. • there should be at least one job seeker applied for this job.	
Post-conditions:	The user would be able to identify the most relevant candidates for this job.	
The flow of events:	User Action	System Action
	1. The user accesses the candidates page who applied for this job.	
		2. The system displays all candidates with their matching score.
	3. The user can sort these candidates by matching scores.	
		4. The system applies the sorting immediately.
Exceptions:	User Action	System Action
	1. No candidates applied for the job.	
		2. The system displays a message saying "No candidates applied for the job".
Includes:	None	
Notes and Issues:	Allow representatives to preview the assessment before finalizing it.	

Table 32: Use Case 29 (Create Assessment)

Use Case ID:	029
Use Case Name:	Create Assessment
Description	Allows users to create a technical assessment for candidates. This includes adding questions, challenges, or tests to assess the candidate's skills and qualifications for the job position.
Actors:	Organization Representative, Organization Admin

Pre-conditions:	• The organization representative or organization admin must be logged into the platform. • The job post for which the assessment is being created must exist.	
Post-conditions:	The assessment is created and linked to the corresponding job post for candidate participation.	
The flow of events:	User Action	System Action
	1. The organization representative accesses the "Create Assessment" section.	
		2. The system displays a form to input assessment details (e.g., test type, time limit).
	3. The organization representative creates the assessment by adding questions and challenges.	
		4. The system saves the assessment and associates it with the selected job post.
Exceptions:	User Action	System Action
	1. Assessment details are incomplete.	
		2. The system notifies the representative that required fields are missing (e.g., no questions added).
	3. The system encounters an error while saving the assessment.	
		4. The system notifies the representative of the error and asks to try again later.
Includes:	None	
Notes and Issues:	Allow representatives to preview the assessment before finalizing it.	

Table 33: Use Case 30 (View My Assessments)

Use Case ID:	030
Use Case Name:	View My Assessments
Description	Allows organization representatives or organization admin to view the

	assessments they have created. This includes viewing the list of candidates who have taken the assessment.	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	• The organization representatives or organization admin must be logged into the platform. • The organization representatives or organization admin must have created at least one assessment.	
Post-conditions:	The organization representatives or organization admin can view the details of assessments, including questions and candidate progress.	
The flow of events:	User Action	System Action
	1. The organization representative accesses the "My Assessments" section.	
		2. The system displays a list of all assessments created by the representative.
	3. The organization representative selects an assessment to view.	
		4. The system displays the assessment details, including questions and the status of candidates who have completed it.
	5. The organization representative views the list of candidates who have completed the assessment.	
Exceptions:	User Action	System Action
	1. Assessment data is missing or incomplete.	
		2. The system notifies the representative that the assessment is incomplete or has missing data.
	3. The system encounters an error while generating a report.	

		4. The system notifies the representative of the error and provides troubleshooting steps or a retry option.
Includes:	None	
Notes and Issues:	• Allow representatives to track the completion status of candidates who have taken the assessments. • Include options for reviewing the results of assessments and generating reports.	

Table 34: Use Case 31 (Manage Assessments)

Use Case ID:	031	
Use Case Name:	Manage Assessments	
Description	Allows organization representatives or organization admin to manage their technical assessments by editing, deleting, or updating them based on feedback and evolving requirements.	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	• The organization representatives or organization admin must be logged into the platform. • The organization representatives or organization admin must have created at least one assessment.	
Post-conditions:	The representative can modify or delete assessments, and any changes are reflected in the system immediately.	
The flow of events:	User Action	System Action
	1. The organization representative navigates to the "My Assessments" section.	
		2. The system displays a list of all assessments created by the representative with options to edit or delete each assessment.
	3. The organization representative selects an assessment to edit or delete.	
		4. The system displays the assessment details and provides options for modification or deletion.

	5. The organization representative makes the required changes.	
		6. The system saves the updates to the assessment or deletes it if selected.
Exceptions:	User Action	System Action
	1. The system encounters an error while managing assessments.	
		2. The system notifies the representative of the error and asks them to try again later.
Includes:	None	
Notes and Issues:	Allow assessment changes to be reflected in candidate experiences immediately.	

Table 35: Use Case 32 (Search for Job Seekers)

Use Case ID:	032	
Use Case Name:	Search for Job Seekers	
Description	The platform allows organization representatives or organization admin to search for job seekers using various filters. The representative can search for job seekers based on specific criteria, such as skills, experience, location, and job preferences, to find the most suitable candidates for open positions.	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	- The organization representatives or organization admin must be logged into the platform. - The organization representatives or organization admin must have access to the job seeker database.	
Post-conditions:	- The system displays a list of job seekers that match the search criteria. - The organization representatives or organization admin can view detailed profiles of the job seekers who match the search filters.	
The flow of events:	User Action	System Action
	1. The organization representatives or	

	organization admin selects the "Search for Seekers" option from the dashboard	
		2. The system presents various search filters (e.g., skills, experience, location, salary expectation, etc.).
	3. The organization representatives or organization admin enters or selects the desired search criteria.	
		4. The system searches the job seeker database and displays a list of seekers that match the search criteria.
Exceptions	User Action	System Action
	1. The organization representatives or organization admin enters invalid criteria (e.g., nonsensical location or skill).	
		2. The system notifies the organization representatives or organization admin of invalid search criteria and prompts for valid input.
	3. No job seekers match the search criteria.	
		4. The system notifies the organization representatives or organization admin that no matching results were found and suggests refining the search.
Includes	None	
Notes and Issues:	• The system should provide options to save frequent search filters for future use. • Consider implementing fuzzy matching to handle variations in user input (e.g., skill names or job titles). • Results should be sorted based on relevance to the selected filters.	

Table 36: Use Case 33 (Search for Companies)

Use Case ID:	033	
Use Case Name:	Search for Companies	
Description	The platform allows organization representatives or organization admin to search for other companies using various filters. The representative can search for companies based on specific criteria, such as location, and category.	
Actors:	Organization Representative, Organization Admin	
Pre-conditions:	- The organization representatives or organization admin must be logged into the platform. - The organization representatives or organization admin must have access to the other organizations database.	
Post-conditions:	- The system displays a list of companies that match the search criteria. - The organization representatives or organization admin can view detailed profiles of the companies who match the search filters.	
The flow of events:	User Action	System Action
	1. The organization representatives or organization admin selects the "Search for organizations" option from the dashboard	
		2. The system presents various search filters (e.g., category, location, size, etc.).
	3. The organization representatives or organization admin enters or selects the desired search criteria.	
		4. The system searches the organization's database and displays a list of organizations that match the search criteria.
Exceptions:	User Action	System Action

	1. The organization representatives or organization admin enters invalid criteria (e.g., nonsensical location).	
		2. The system notifies the organization representatives or organization admin of invalid search criteria and prompts for valid input.
	3. No organization matches the search criteria.	
		4. The system notifies the organization representatives or organization admin that no matching results were found and suggests refining the search.
Includes:	None	
Notes and Issues:	• The system should provide options to save frequent search filters for future use. • Consider implementing fuzzy matching to handle variations in user input (e.g., skill names or job titles). • Results should be sorted based on relevance to the selected filters.	

Table 37: Use Case 34 (Manage Organization Representative)

Use Case ID:	034
Use Case Name:	Manage Organization Representative
Description	Allows organization Admin to view, add, or remove other representatives within their organization.
Actors:	Organization Admin
Pre-conditions:	• The user must be logged into the platform. • The user must have the appropriate permissions to manage organization representatives.
Post-conditions:	• The system successfully saves any updates to the representative's information. • If the representative manages team members, changes are reflected in the organization's representative list and permissions.

The flow of events:	User Action	System Action
	1. The user selects the "Organization Representatives" option.	
		2. The system displays a list of current representatives.
	3. The user can add, edit, or remove team members.	
		4. The system confirms the action and refreshes the team member list accordingly.
Exceptions	User Action	System Action
	1. The user enters invalid data (e.g., incorrect email format).	
		2. The system notifies the Admin and prompts for valid input.
Includes	None	
Notes and Issues:	Ensure all team changes are logged and tracked to avoid confusion or issues with team management.	

Table 38: Use Case 35 (View Statistics)

Use Case ID:	035	
Use Case Name:	View Statistics	
Description	Allows organization representatives to view statistics and analytics related to their job posts, applicants, and overall company activity on the platform.	
Actors:	Organization Admin	
Pre-conditions:	- The organization Admin must be logged into the platform. - The organization should have active job posts and applicants.	
Post-conditions:	The organization Admin can view detailed statistics on applicant trends, post-performance, and overall activity.	
The flow of events:	User Action	System Action
	1. The organization representative navigates to the "Dashboard" section.	

		2. The system displays detailed charts, graphs, and other data related to job posts, applicants, and company performance.
	3. The organization representative filters or adjusts time ranges.	
		4. The system updates the statistics accordingly to display data for the selected period.
Exceptions:	User Action	System Action
	1. No data is available for the selected period.	
		2. The system notifies the representative that no data exists for the specified time range.
Includes:	None	
Notes and Issues:	• Provide visualizations such as pie charts, bar graphs, or line charts for better data interpretation. • Include options for downloading reports.	

Table 39: Use Case 36 (Verify Organization)

Use Case ID:	036	
Use Case Name:	Verify Organization	
Description	Admins review the details of organizations that wish to create profiles and approve or reject their requests. This ensures that only legitimate organizations can post job listings and interact with candidates on the platform.	
Actors:	Admin, Super Admin	
Pre-conditions:	• The organization has submitted the required details for profile creation. • Admin or Super Admin is logged into the platform.	
Post-conditions:	• The organization profile is either approved or rejected. • The organization is notified of the decision.	
The flow of events:	User Action	System Action
	1. Admin navigates to the "verify organization" section.	

		2. The system displays the list of pending organization requests for approval.
	3. Admin reviews the organization's details.	
		4. The system displays the full details of the organization's submission.
	5. Admin approves or rejects the organization.	
		6. The system updates the organization's status and notifies the organization of the decision.
Exceptions:	User Action	System Action
	1. Admin rejects the organization.	
		2. The system prompts for a reason and notifies the organization.
Includes:	None	
Notes and Issues:	Admins should validate organizations thoroughly before approving.	

Table 40: Use Case 37 (Manage Courses)

Use Case ID:	037	
Use Case Name:	Manage Courses	
Description	The Admin and Super Admin can manage courses in the system by adding, updating or deleting courses or content.	
Actors:	Admin, Super Admin	
Pre-conditions:	Admin or Super Admin is logged into the platform.	
Post-conditions:	The course updating should be reflected in the job seeker view.	
The flow of events:	User Action	System Action
	1. The Admin and Super Admin access the manage courses section.	
		2. The system displays a list of all courses and its lessons.

	3. The Admin and Super Admin select to add, edit, or delete a course or lesson.	
		4. The system allows the Admin and Super Admin to manage courses and lessons.
	5. The Admin and Super Admin confirms the action.	
		6. The system updates the course or lesson accordingly.
Exceptions:	User Action	System Action
	1. The Admin and Super Admin add a course or lesson that already exists.	
		2. The system displays an error message that the course or lesson already exists.
Includes:	None	
Notes and Issues:	The Admin and Super Admin can also view the course details and lessons.	

Table 41: Use Case 38 (Resolve Issues Reported by Candidates/Companies)

Use Case ID:	038	
Use Case Name:	Resolve Issues Reported by Candidates/Companies	
Description	Admin and Super Admin addresses issues or complaints reported by candidates or companies, ensuring that any problems or disputes are resolved effectively.	
Actors:	Admin, Super Admin	
Pre-conditions:	- A complaint or issue has been reported by a candidate or company. - Admin, Super Admin is logged into the platform.	
Post-conditions:	The reported issue is resolved, and both parties are notified of the resolution.	
The flow of events:	User Action	System Action
	1. Admin, Super Admin accesses the "Chat icon" section.	

		2. The system displays a list of issues reported by candidates/companies.
	5. Admin, Super Admin takes action to resolve the issue.	
		6. The system sends a notification to the user who reported the issue if the issue is resolved.
Exceptions:	User Action	System Action
	1. Admin, Super Admin cannot resolve the issue immediately.	
		2. The system sends a notification to the user who reported the issue that the issue is being reviewed.
Includes:	None	
Notes and Issues:	- Keep detailed logs of resolved issues for future reference. - Ensure communication with users is prompt.	

Table 42: Use Case 39 (Manage Admins)

Use Case ID:	039	
Use Case Name:	Manage Admins	
Description	Super Admin can manage other admin users, including adding, editing, or removing admins .	
Actors:	Super Admin	
Pre-conditions:	Super Admin is logged into the platform	
Post-conditions:	Admin accounts are created, modified, or deleted as needed.	
The flow of events:	User Action	System Action
	1. Super Admin navigates to the "Admins" section.	
		2. The system displays a list of all admin accounts.
	3. Super Admin selects to add, edit, or delete an admin.	
		4. The system allows the Super Admin to manage other admin accounts.
	5. Super Admin confirms the action	

		6. The system updates the admin account accordingly.
Exceptions:	User Action	System Action
	1. Super Admin tries to delete the only admin account	
		2. System prompts for confirmation and ensures at least one active admin before deletion.
Includes:	None	
Notes and Issues:	Track changes to admin accounts for audit purposes.	

Table 43: Use Case 40 (Manage Reports)

Use Case ID:	040	
Use Case Name:	Manage Reports	
Description	Super Admin can manage reports related to platform activity, including user behavior, and postings.	
Actors:	Super Admin	
Pre-conditions:	Super Admin is logged into the platform.	
Post-conditions:	Reports are generated, and reviewed, and any necessary actions are taken based on the data.	
The flow of events:	User Action	System Action
	1. Super Admin navigates to the "Dashboard" section.	
		2. The system displays available reports such as the number of posts, etc.
Exceptions:	User Action	System Action
	1. No data is available for the selected report.	
		2. The system notifies the Super Admin that there is no data to be presented.
Includes:	None	
Notes and Issues:	• Reports should be exportable (e.g., PDF formats).	

3.10. Domain Class Diagram

To view Clearer picture of the domain class diagram, visit this [Google Drive link](#), [LucidChart link](#).

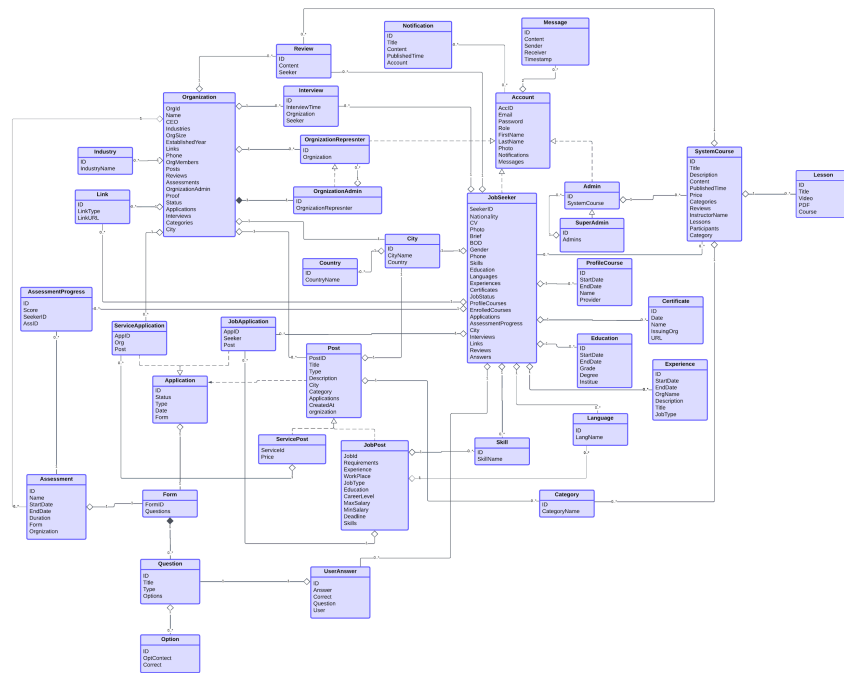


Figure 9: Domain Class Diagram

3.11. Conceptual ERD

To view Clearer picture of the conceptual ER diagram, visit this [Google Drive link](#), [LucidChart link](#).

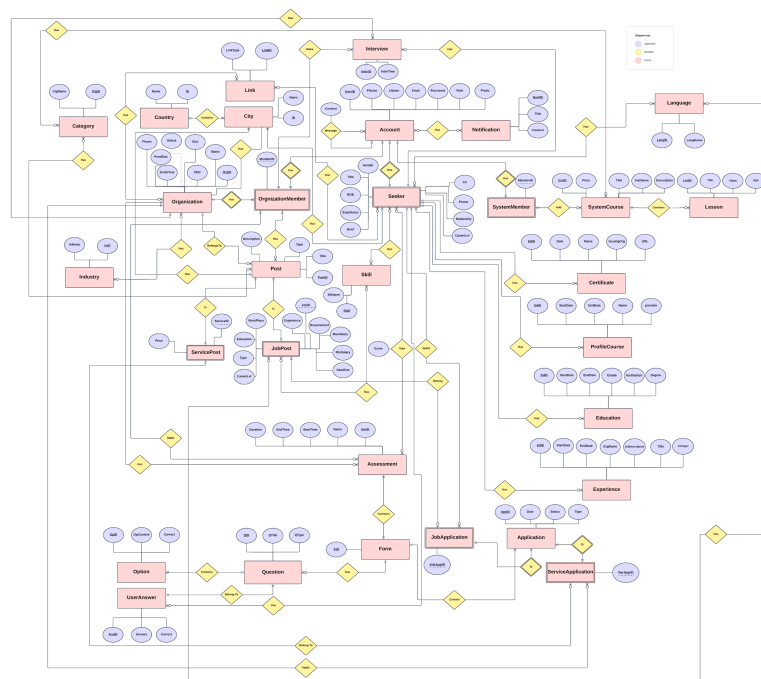


Figure 10: Conceptual ER Diagram

3.12. Physical ERD

To view Clearer picture of the ER diagram, visit this [Google Drive link](#), [LucidChart link](#).

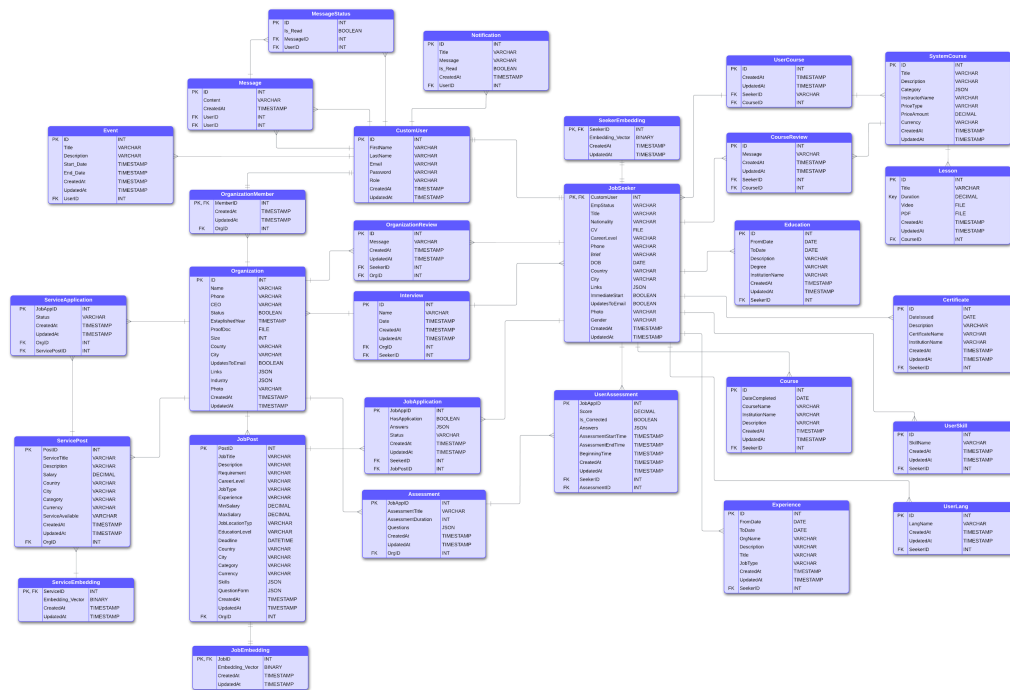


Figure 11: Physical ER Diagram

3.13. System Sequence Diagram

To view Clearer picture of the system sequence diagrams, visit this [Google Drive link](#), [LucidChart link](#).

1. Add admin

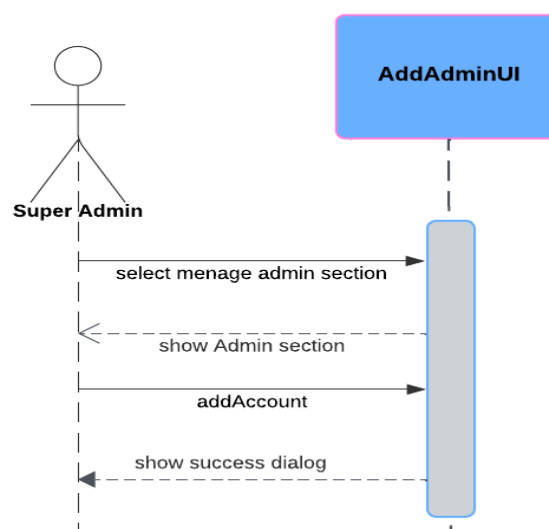


Figure 12: Add Admin System Sequence Diagram

2. Add lesson

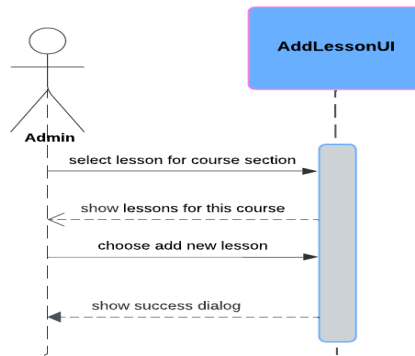


Figure 13: Add lesson System Sequence Diagram

3. Add review

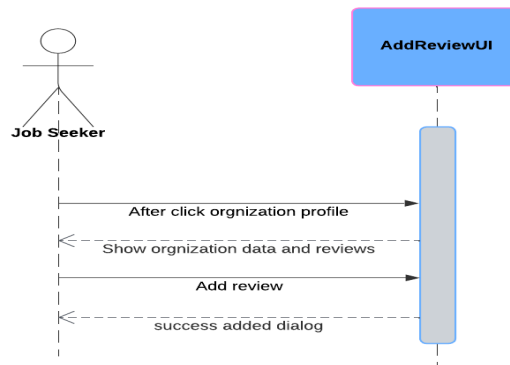


Figure 14: Add Review System Sequence Diagram

4. Apply for a job

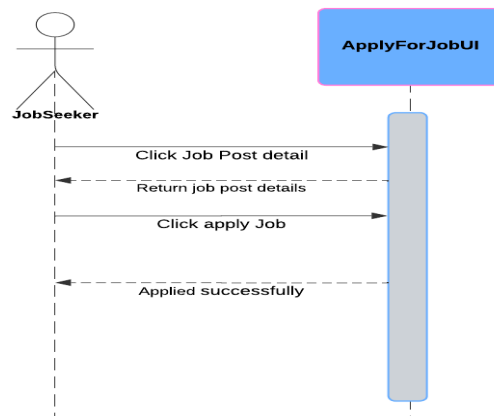


Figure 15: Apply For a Job System Sequence Diagram

5. Ask for a service

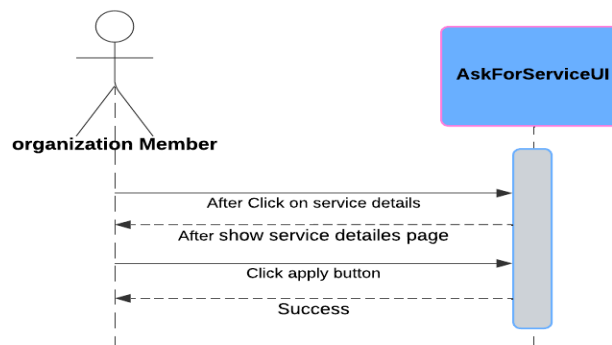


Figure 16: Ask For a Service System Sequence Diagram

6. Enroll in a course

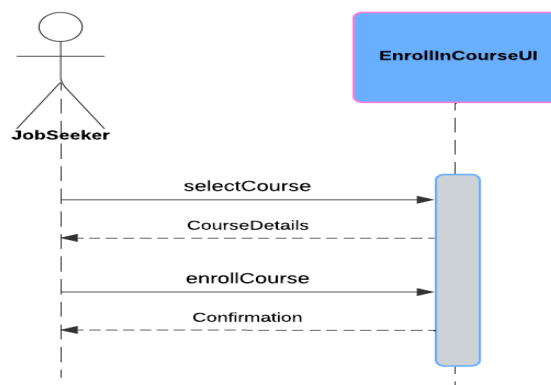


Figure 17: Enroll In a Course System Sequence Diagram

7. Explore organizations

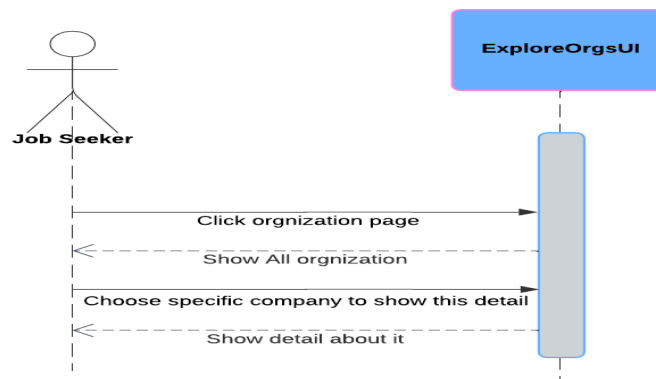


Figure 18: Explore Organizations System Sequence Diagram

8. Get applicants for a job

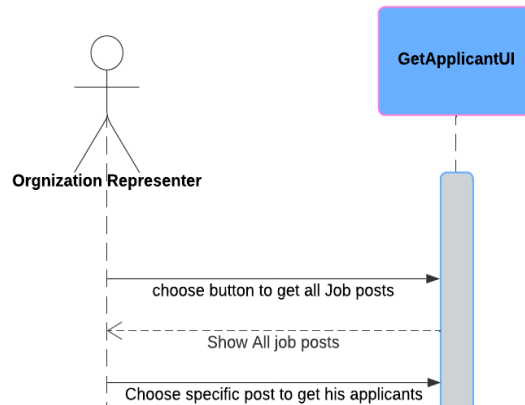


Figure 19: Get Applicants For a Job System Sequence Diagram

9. Offer a service

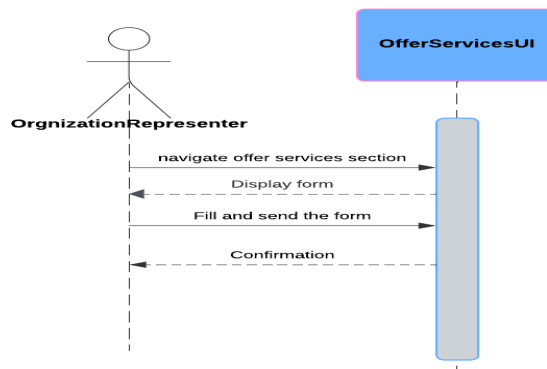


Figure 20: Offer a Service System Sequence Diagram

10. Search for a job

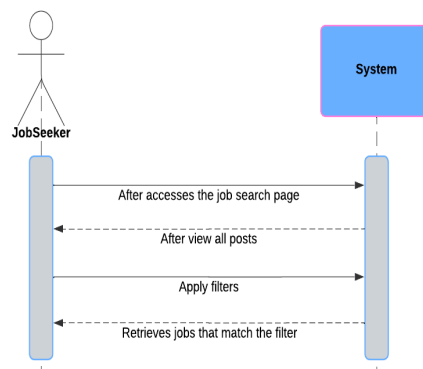


Figure 21: Search For a Job System Sequence Diagram

11. Set interview time

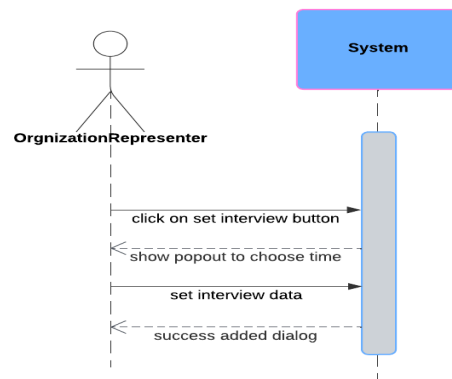


Figure 22: Set Interview Time System Sequence Diagram

12. Verify organization

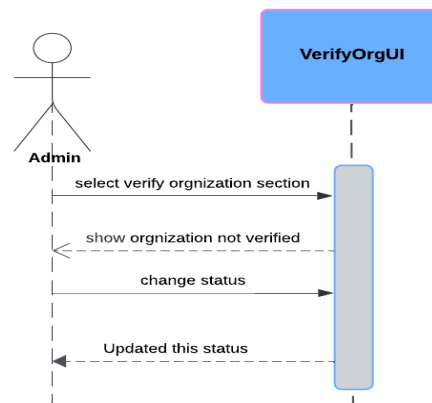


Figure 23: Verify Organization System Sequence Diagram

Chapter 4: HireNy Design and Implementation

4.1. Tools and programming languages

1. Canva

Canva is a versatile and user-friendly tool for designing creative graphics and logos.

2. Figma

A tool for creating user interface design for web and mobile applications.

3. LucidChart

An online diagramming tool used for creating flowcharts, organizational charts, wireframes, and other visual representations to simplify complex ideas and processes.

4. GitHub

We selected GitHub as the primary tool for collaboration in our project. GitHub is a widely used web-based platform that provides Git repository hosting, allowing us to securely store and track changes easily, manage branches and manage our project's source code.

5. Jira

A project management tool widely used for tracking tasks, issues, and project workflows. It supports Agile methodologies such as Scrum and Kanban, helping teams organize and prioritize work efficiently.

6. Visual Studio Code

Selected for its efficiency in writing and managing source code, it provides a lightweight yet

powerful code editing environment with a wide range of features and extensibility options.

7. Android Studio

An integrated development environment (IDE) for Android application development. It includes tools for building, testing, and emulating Android apps on virtual devices.

Selected to make an emulator.

8. Dart

Dart is an open-source, object-oriented programming language developed by Google. It was designed with a focus on creating high-performance, scalable web and mobile

applications. Dart is known for its simplicity, efficiency, and versatility, making it suitable for a wide range of development tasks.

- **Flutter**

An open-source UI toolkit that uses Dart to build natively compiled applications for mobile, web, and desktop platforms from a single codebase.

9. JavaScript

JavaScript is a versatile, high-level programming language widely used for web development. It enables dynamic and interactive content on websites and is supported by all modern browsers. JavaScript can be used for both client-side and server-side development, making it a fundamental technology for building web applications.

- **React**

React is a JavaScript library developed by Facebook for building fast and scalable user interfaces. It follows a component-based architecture, allowing developers to create reusable UI components. React efficiently updates and renders components using a virtual DOM.

- **Redux**

Redux is a state management library for JavaScript applications, often used with React. It provides a centralized store for managing application state, ensuring predictable behavior across different components. Redux follows a unidirectional data flow, making state management more predictable and easier to debug.

10. Python

Python is a high-level, interpreted programming language known for its simplicity and readability. Python supports multiple programming paradigms, including procedural, object-oriented, and functional programming styles. It has a large and comprehensive standard library that makes it suitable for various tasks, from web development to data analysis, scientific computing, artificial intelligence, and more.

- **Django**

A high-level Python web framework designed for building secure, scalable, and maintainable web applications quickly. It follows the "batteries-included" philosophy, offering built-in features for database management, templating, and authentication.

11. Postman

A widely used API testing tool that allows developers to design, test, and document APIs efficiently. It provides an intuitive interface for sending requests, analyzing responses, and automating API testing workflows.

4.2. Design Class Diagram

To view Clearer picture of the design class diagram, visit this [Google Drive link](#), [LucidChart link](#).

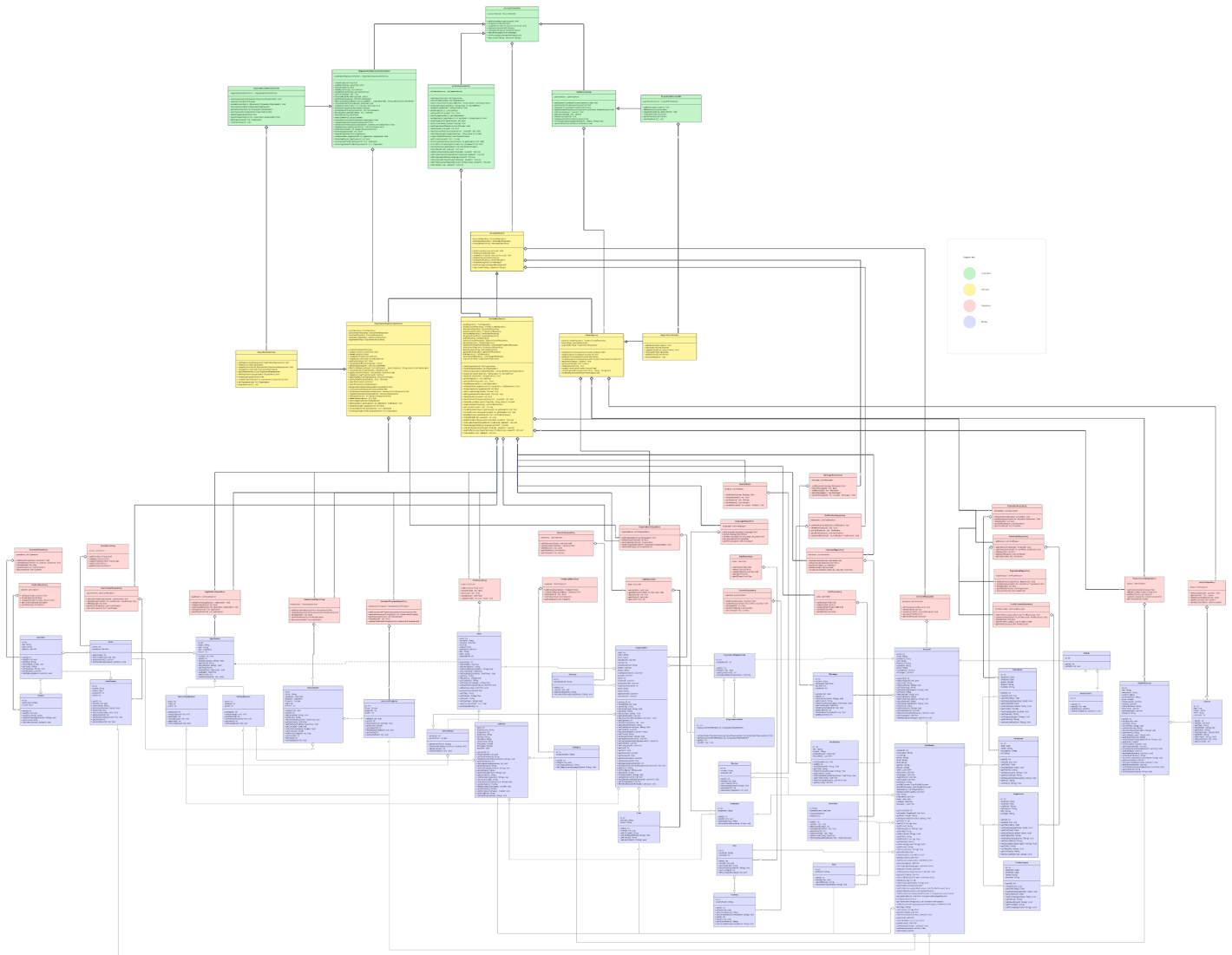


Figure 24: Design Class Diagram

4.3. Detailed Sequence Diagram

To view Clearer picture of the detailed sequence diagrams, visit this [Google Drive link](#), [LucidChart link](#).

1. Add admin

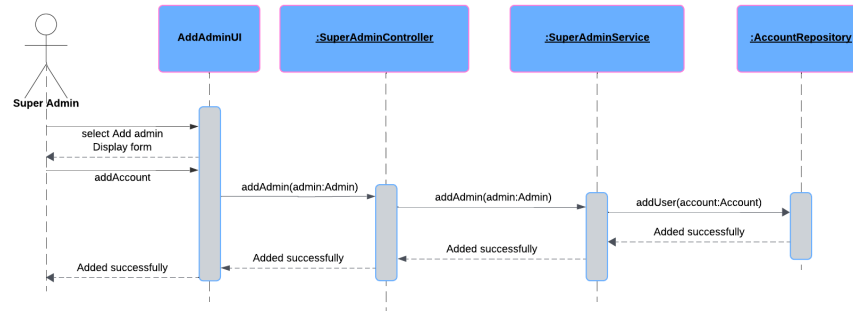


Figure 25: Add Admin Detailed System Sequence

2. Add lesson

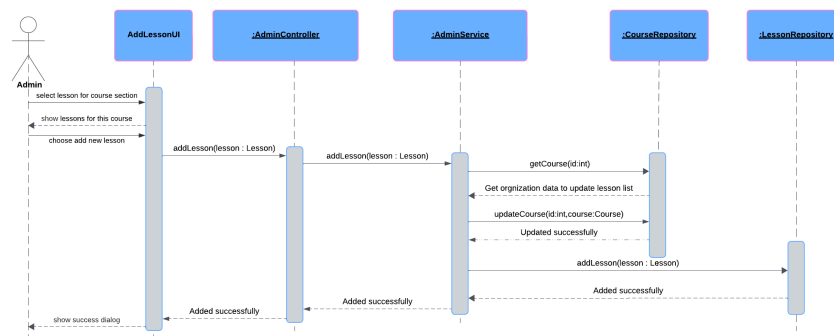


Figure 26: Add lesson Detailed Sequence Diagram

3. Add review

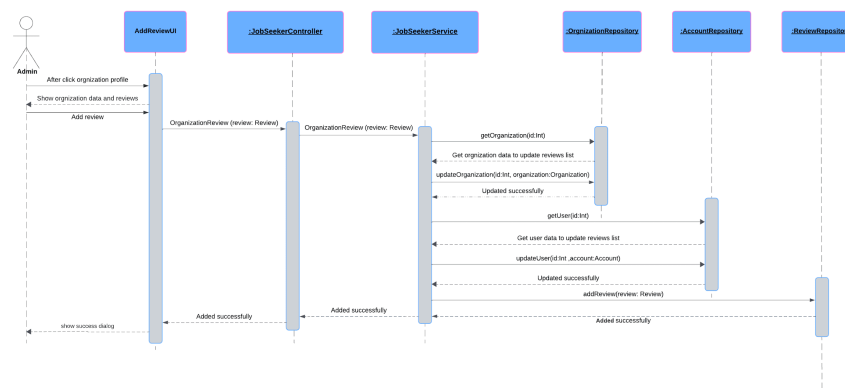


Figure 27: Add Review Detailed Sequence Diagram

4. Apply for a job

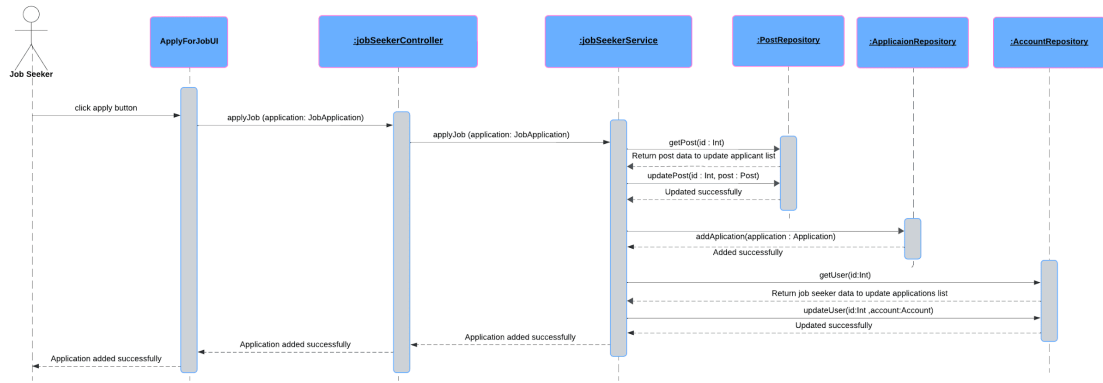


Figure 28: Apply For a Job Detailed Sequence Diagram

5. Ask for a service

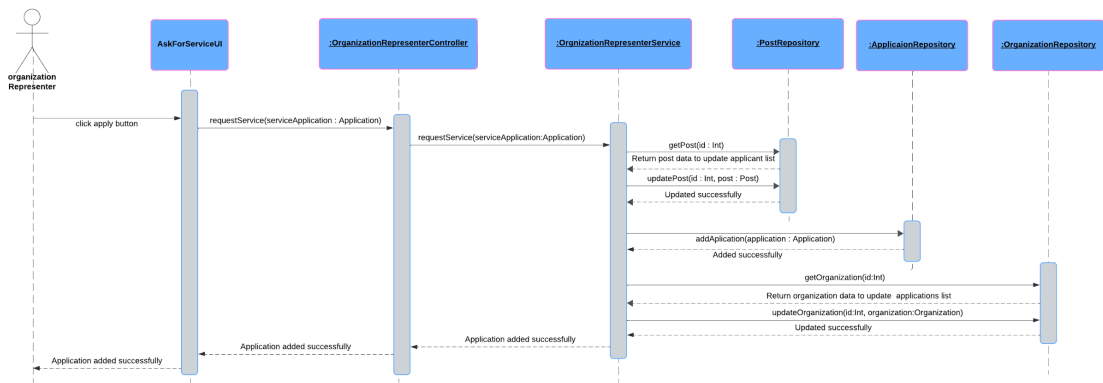


Figure 29: Ask For a Service Detailed Sequence Diagram

6. Enroll in a course

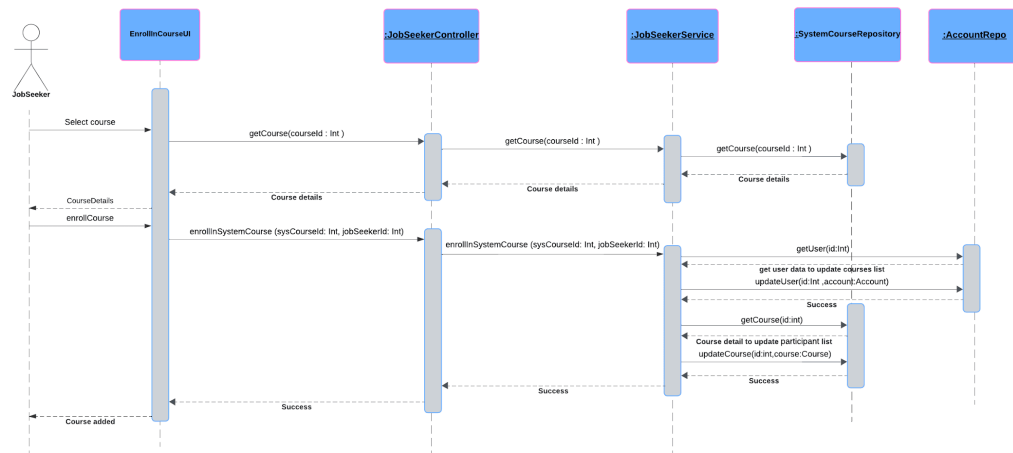


Figure 30: Enroll In a Course Detailed Sequence Diagram

7. Explore organizations

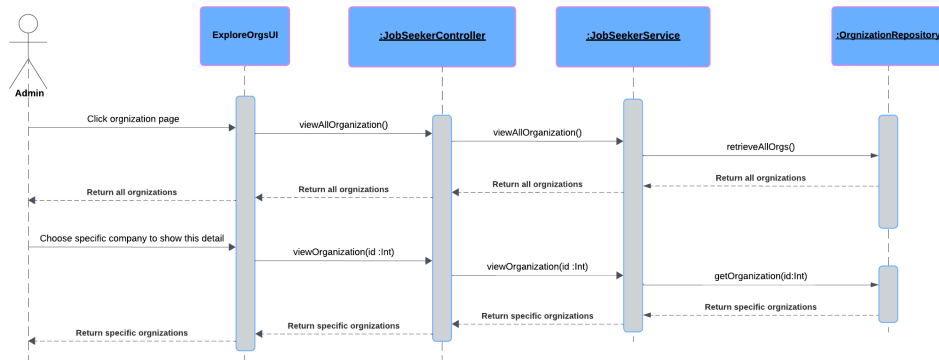


Figure 31: Explore Organizations Detailed Sequence Diagram

8. Get applicants for a job

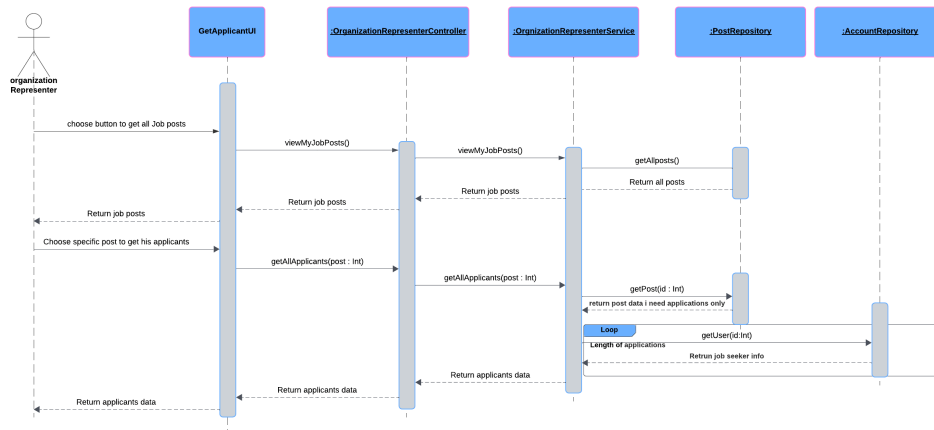


Figure 32: Get Applicants For a Job Detailed Sequence Diagram

9. Offer a service

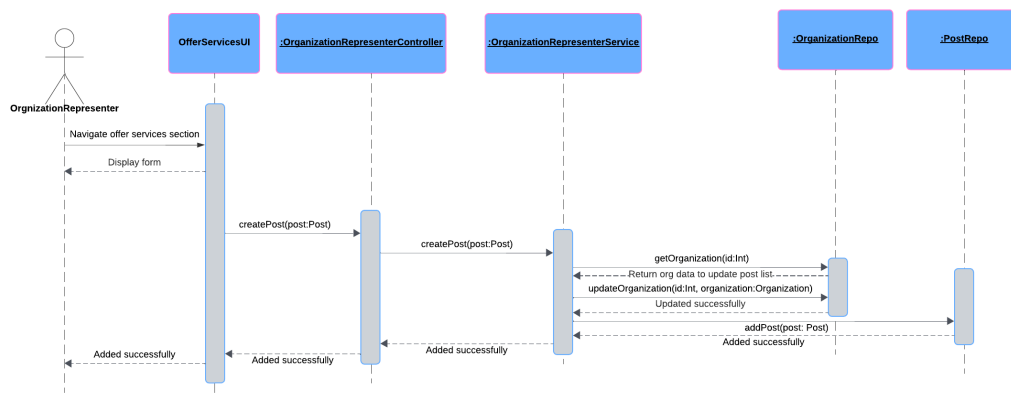


Figure 33: Offer a Service Detailed Sequence Diagram

10. Search for a job

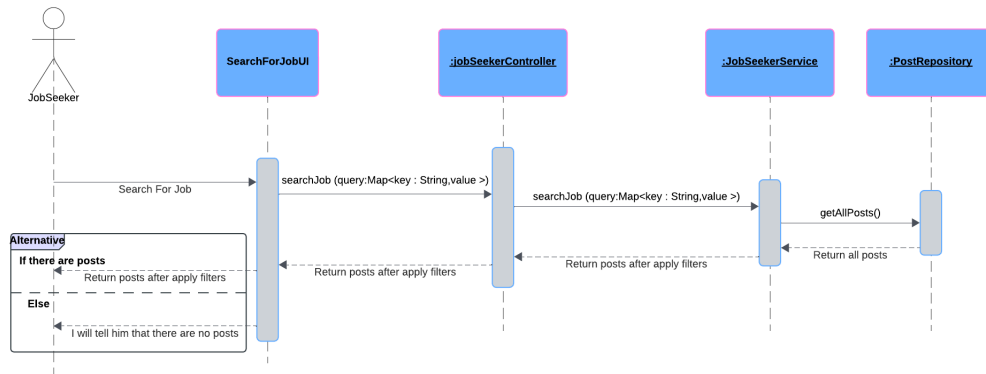


Figure 34: Search For a Job Detailed Sequence Diagram

11. Set interview time

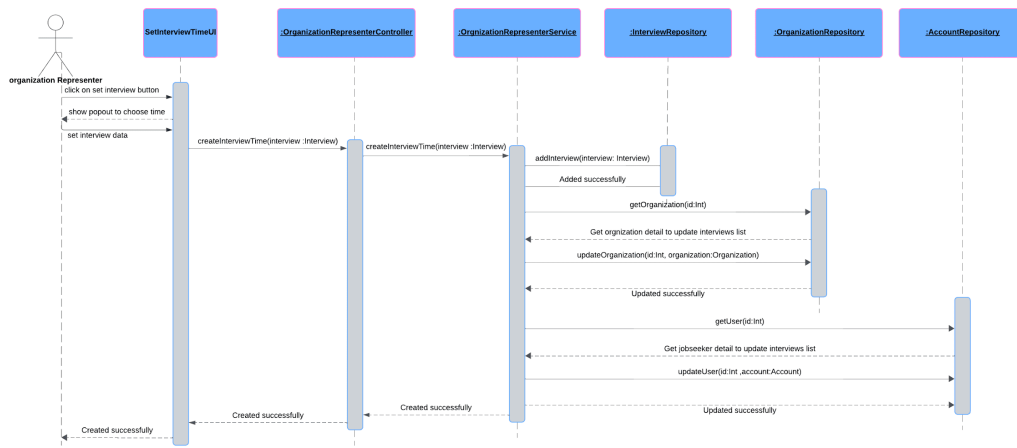


Figure 35: Set Interview Time Detailed Sequence Diagram

12. Verify organization

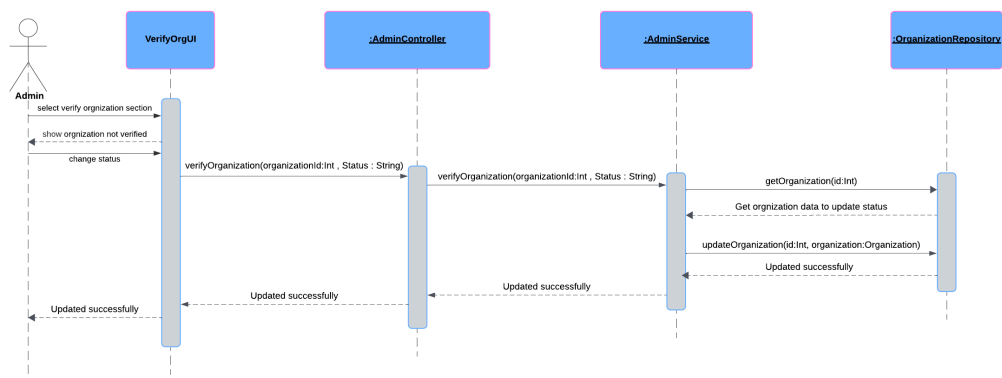


Figure 36: Verify Organization Detailed Sequence Diagram

4.4. System Prototype Web Screens

This Link to the Full System Prototype Screens [\[here\]](#)

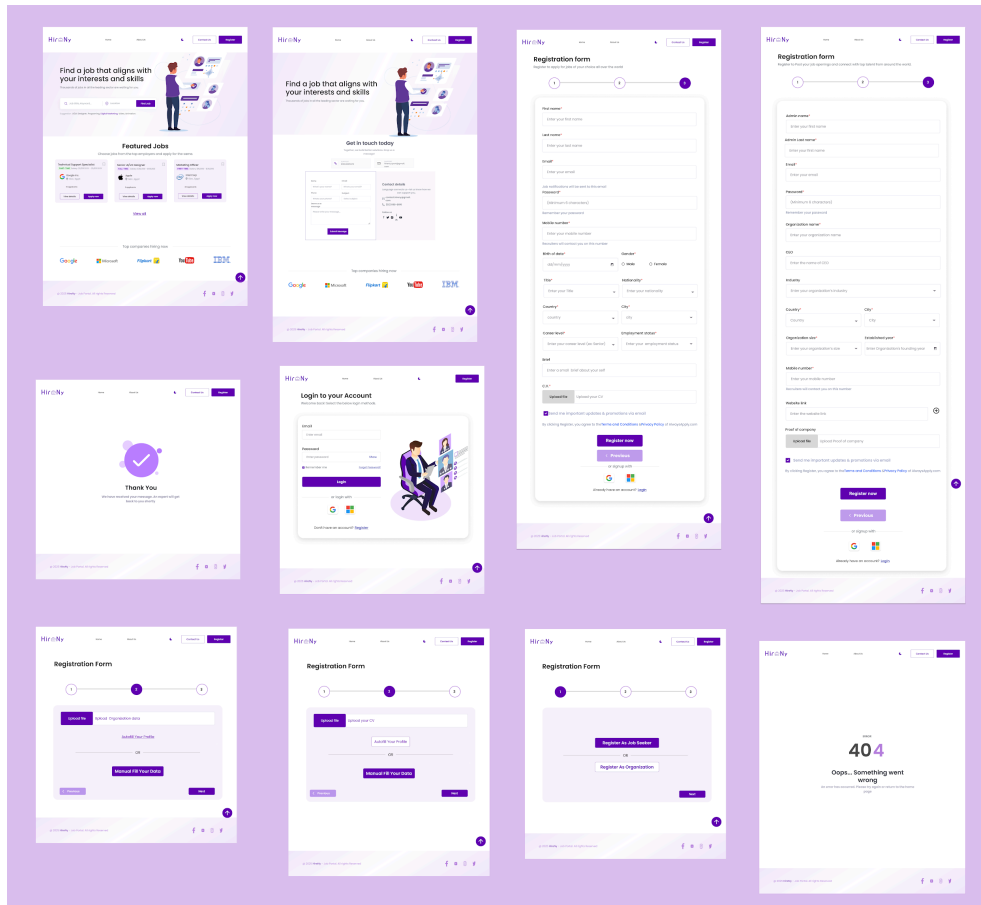


Figure 37: Anonymous UI Design

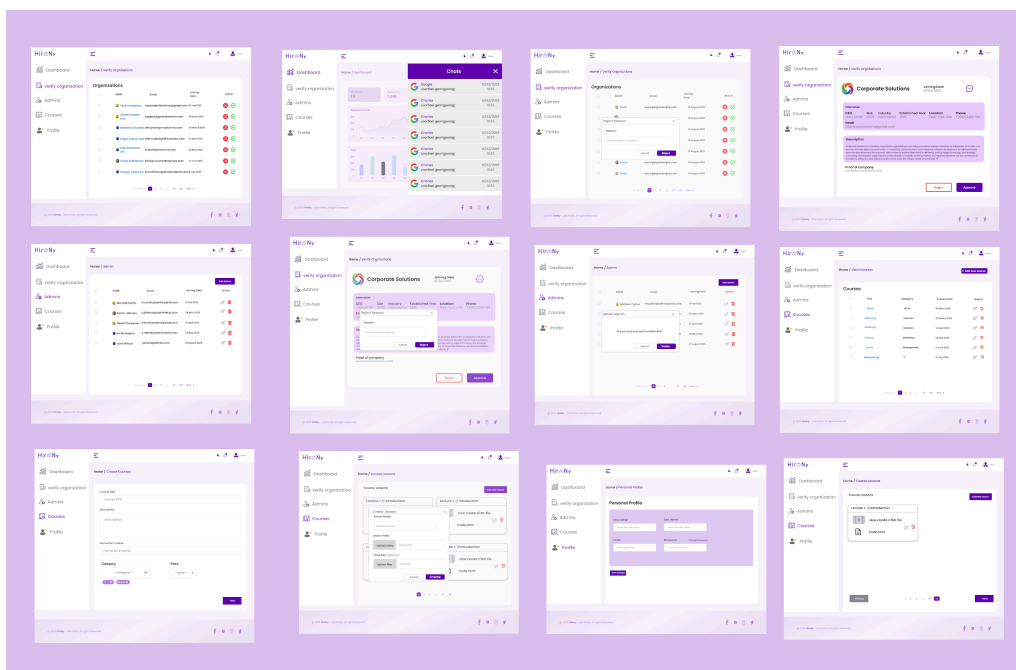


Figure 38: Admin UI Design

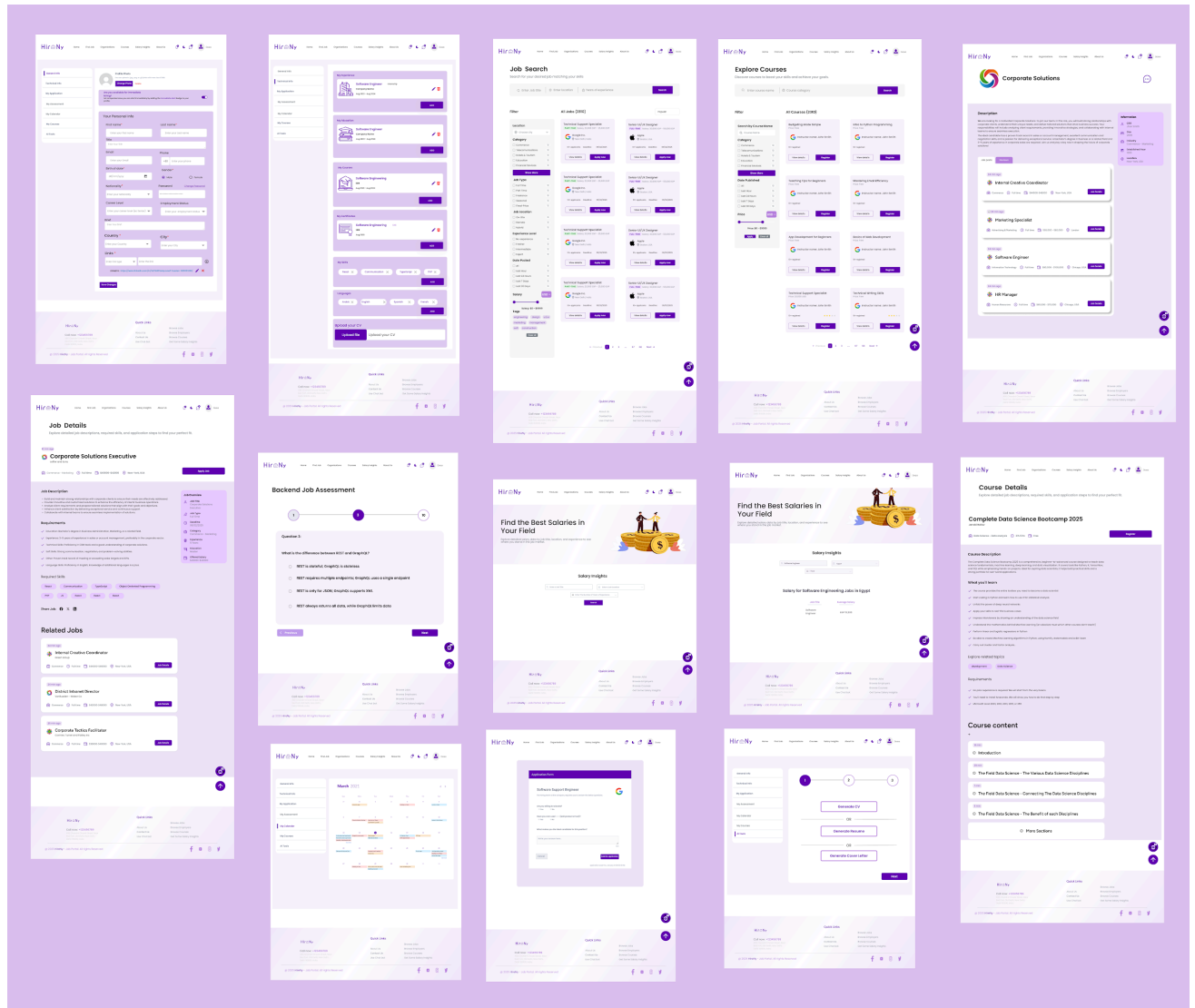


Figure 39: Job Seeker UI Design

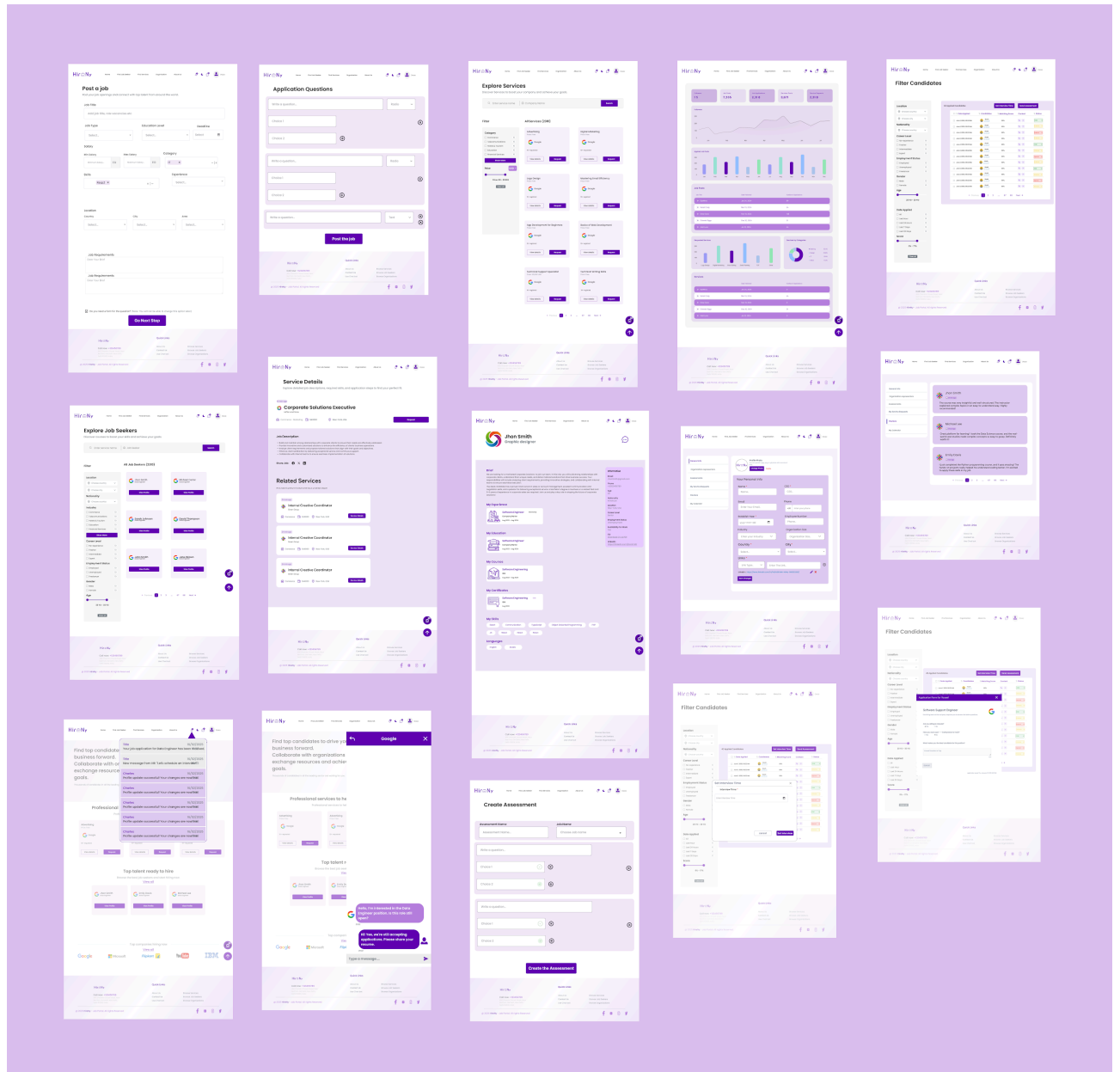


Figure 40: Organization UI Design

4.5. System Prototype Mobile Screens

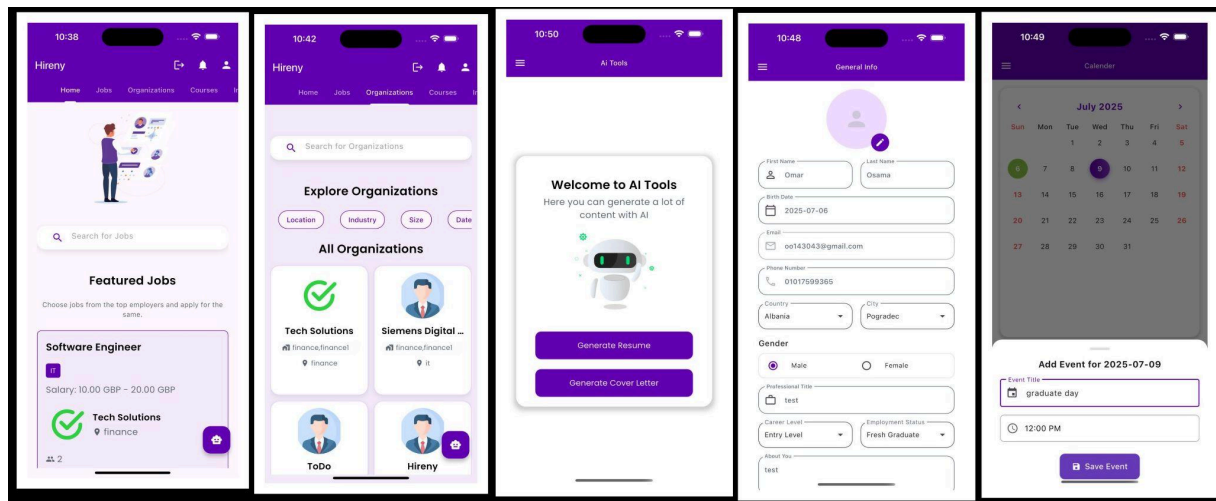


Figure 41: Mobile Design 1

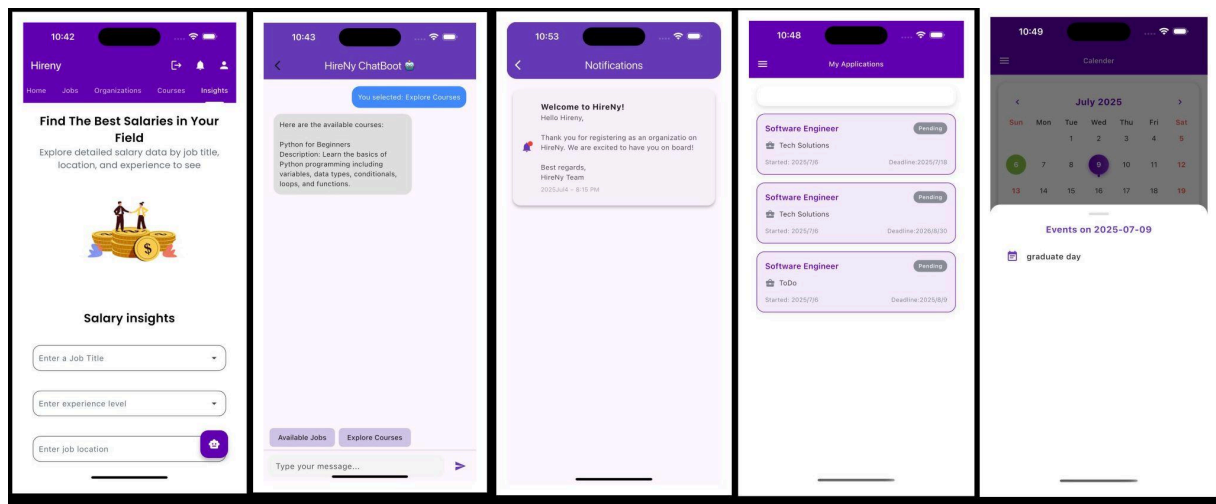


Figure 42: Mobile Design 2

4.6. AI Models Used

1. Salary Insight Model

- **Web Scraping from PayScale.com**

We used automated web scraping tool **BeautifulSoup** to extract salary-related data from [PayScale.com](https://www.payscale.com). This includes information like job titles, experience levels, countries, and corresponding salary ranges, which is used as training data for our predictive model.

- **Target Encoding & Standard Scaler**

- **Target Encoder** was used to convert categorical columns (like job title or country) into numerical format based on the target variable (salary).
- **Standard Scaler** was applied to normalize continuous features, ensuring consistent input ranges for better model performance

- **XGBoost (Extreme Gradient Boosting)**

A powerful gradient boosting decision tree algorithm optimized for speed and performance. We used it to train a salary prediction model with the scraped and processed features.

XGBoost helps capture complex nonlinear relationships and provides feature importance rankings for better interpretability.

- **Salary Insight Model – Outliers Effects**

We evaluated the effect of **outliers** on salary prediction by:

- Detecting outliers using **Isolation Forest** technique.
- Comparing the data with and without outlier removal.
This improved model robustness and reduced the impact of extreme/unrealistic values.

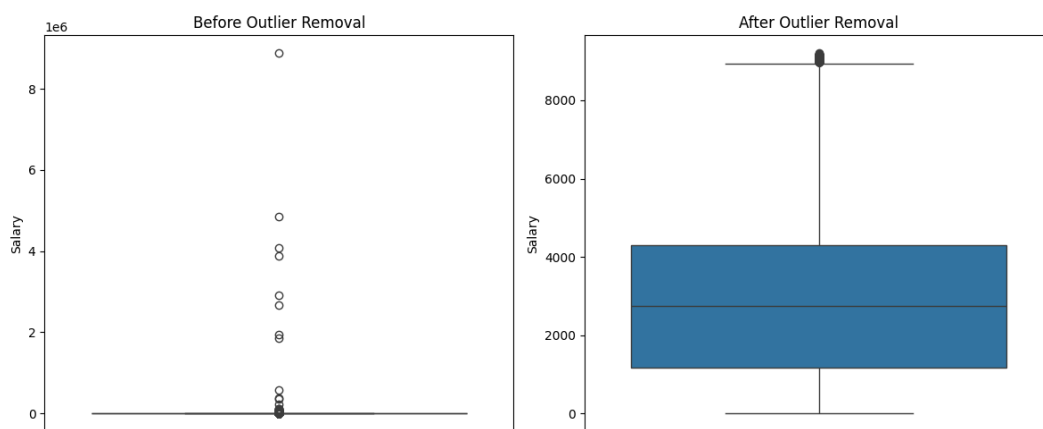


Figure 43: Difference Between Before & After Outlier Removal

2. Salary Insight Model – Other Models Evaluation

Alongside XGBoost, we evaluated and compared other regression models:

- **MLPRegressor (Neural Network)**
- **K-Nearest Neighbors**
- **Linear Regression**

NR	a1	a2	a3	a4	a5	a6
1	Model	Set	MAE	MSE	RMSE	R2
2	XGBoost	Train	676.967287195304	993514.3110955597	996.7518804073358	0.7733574054257375
3	XGBoost	Test	687.8462053241434	1034358.7400331112	1017.0342865572975	0.7638541566501411
4	MLP	Train	777.9904966579618	1236427.8897656864	1111.9477909352067	0.7179434439837453
5	MLP	Test	780.7893204133939	1256268.771339295	1120.8339624312314	0.7131917225618336
6	Linear	Train	822.3510670346575	1423311.5419536985	1193.026211763052	0.6753111483616533
7	Linear	Test	824.0551794107496	1437232.7582740958	1198.8464281441954	0.6718773394017865
8	KNN	Train	629.98538435379	849851.4070579117	921.8738563696835	0.8061300921918308
9	KNN	Test	768.9032538970482	1249279.2019235848	1117.711591567156	0.7147874530375797

Figure 44: Regression Models Evaluation

Each model was assessed using MAE, MSE, RMSE and R^2 metrics to select the best-performing option.

3. Recommendation System

- **Sentence Transformer Embeddings**

Used pre-trained **SentenceTransformer** models to generate vector embeddings of user profiles and job descriptions, capturing semantic meaning rather than just keyword matching.

- **Save & Load Binary Embeddings**

To improve performance, embeddings are saved as binary fields in the database files and loaded when needed instead of re-generating them each time.

- **Cosine Similarity**

Similarity between job postings and user profiles is measured using **cosine similarity** between their respective embedding vectors. Higher similarity scores indicate better matching opportunities.

4. Profanity Detection (better_profanity)

Integrated the **better_profanity** Python package to automatically detect and censor inappropriate or offensive words in user-generated review to flag and review potentially harmful or disrespectful language in platform feedback.

5. Gemini API

- **Cover Letter Generation**

Used the **Gemini LLM API** to generate personalized cover letters based on the user's profile, skills, and job description. This provides a smart, AI-assisted writing tool for job seekers.

- **Job Title Recommendation**

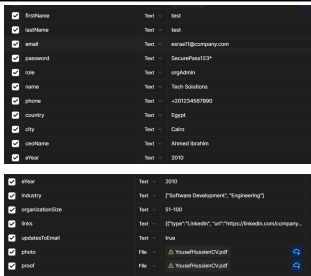
Used the **Gemini LLM API** to suggest relevant **job titles** based on the user's skills, education, and experience. This helps guide users who are unsure about suitable roles.

- **ChatBot Assistant**

A conversational **ChatBot** powered by a **Gemini LLM API** helps users with:

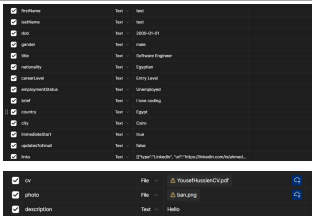
- Navigating the platform
- Understanding job recommendations
- Answering questions about applications, profiles, and system use

			"detail": "Authentication credentials were not provided." }
		{ "lastName": "User", "email": "admin@example.com", "password": "AdminPass123*" }	{ "detail": "Authentication credentials were not provided." }
		{ "firstName": "Admin", "email": "admin@example.com", "password": "AdminPass123*" }	{ "detail": "Authentication credentials were not provided." }
		{ "firstName": "Admin", "lastName": "User", "password": "AdminPass123*" }	{ "detail": "Authentication credentials were not provided." }
		{ "firstName": "Admin", "lastName": "User", "email": "admin@example.com", }	{ "detail": "Authentication credentials were not provided." }
		{ "firstName": "Admin", "lastName": "User", "email": "admin@example.com", "password": "AdminPass123*" }	{ "message": "Admin created successfully" }
5	sendOTP	Empty	{ "error": "Email is required."}
		{ "email": " esraamahmoud2334@gmail.com " }	{ "status": "Email sent successfully!" }
6	verifyOTP	{ "email": "esraamahmoud2334@gmail.com ", "otp":994022 }	{ "message": "OTP verified successfully" }

7	changePassword	<pre>{ "old_password": "Pass1234567*", "new_password": "SecurePass123**" }</pre>	<pre>{ "message": "Password changed successfully" }</pre>
8	ResetPassword	Empty	<pre>{ "error": "JSON parse error - Expecting value: line 2 column 1 (char 2)" }</pre>
		<pre>{ "new_password": "Pass5656*" }</pre>	<pre>{ "error": "Email or new password not provided" }</pre>
		<pre>{ "email": "esraa78@gmail.com", }</pre>	<pre>{ "error": "Email or new password not provided" }</pre>
		<pre>{ "email": "esraa78@gmail.com", "new_password": "Pass5656*" }</pre>	<pre>{ "message": "Password reset successfully" }</pre>
9	registerSeekerandOrgAdmin	Empty	<pre>{ "firstName": ["This field may not be null."], "lastName": ["This field may not be null."], "email": ["This field may not be null."], "password": ["This field may not be null."], "role": ["This field may not be null."] }</pre>
			<pre>{ "id": 19, "created_at": "2025-07-02T20:47:07.988912Z", "updated_at": "2025-07-02T20:47:07.988963Z", "user": { "id": 37, "firstName": "test", "lastName": "test", "email": "esraa1@company.com", "password": "pbkdf2_sha256\$878888\$0YKq1z7X1X2H1iB4i4zW\$09XDL1qM1U0eAC/z11bK7adAp7a1e33q9k45/oi=", "role": "orgAdmin", "organization": { "name": "Tech Solutions", "phone": "+201234567890", "country": "Egypt", "city": "Cairo", "ceoName": "Ahmed Ibrahim", "eYear": 2019, "industry": ["Software Development", "Engineering"], "organizationSize": "51-100", "links": {} } } }</pre>
10	addOrgRep	Empty	<pre>{</pre>

			<pre>"firstName": ["This field may not be null."], "lastName": ["This field may not be null."], "email": ["This field may not be null."], "password": ["This field may not be null."] }</pre>
		<pre>{ "lastName": "User", "email": "represnter@example.com", "password": "RepresnterPass123*" }</pre>	<pre>{ "firstName": ["This field may not be null."] }</pre>
		<pre>{ "firstName": "Admin", "email": "represnter@example.com", "password": "RepresnterPass123*" }</pre>	<pre>{ "lastName": ["This field may not be null."] }</pre>
		<pre>{ "firstName": "Admin", "lastName": "User", "password": "RepresnterPass123*" }</pre>	<pre>{ "email": ["This field may not be null."] }</pre>
		<pre>{ "firstName": "Admin", "lastName": "User", "email": "represnter@example.com", "password": "RepresnterPass123*" }</pre>	<pre>{ "id": 15, "created_at": "2025-07-02T20:59:23.942706Z", "updated_at": "2025-07-02T20:59:23.942617Z", "user": { "id": 39, "firstName": "Admin", "lastName": "User", "email": "represnter@example.com", "role": "orgRep", "password": "pbkdf2_sha256\$576898\$seAkoAmrJauJue1A8q4DB:u1\$2yKt5DX8IpeA197K+ELc/PdsvmcCLmpZ6HncmCL110=" }, "organization": 11 }</pre>
11	verificationOrg	<pre>{ "verificationStatus": false, "reason": "... " }</pre>	<pre>{ "message": "Organization deleted due to unverified status" }</pre>
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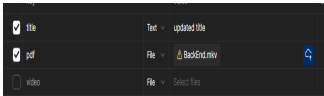
12	getAllNonVerifiedOrganizations		[Not-Verified-Organizations] if there is org <pre> { "id": 6, "name": "Siemens Digital Industries Software", "phone": "201234567890", "country": "it", "city": "finance", "ceoName": "Ahmed Ibrahim", "eYear": 2017, "industry": "finance,finance1", "organizationSize": "health", "links": [{ "type": "LinkedIn", "url": "https://mail.google.com/mail/u/0/#inbox" }], "updatesToEmail": false, "verificationStatus": false, "photo": "/media/photos/default.png", "created_at": "2025-06-24T17:26:25.719136Z", "role": "orgRep" } </pre>
13	getOrganizationRepresentative		<pre> { "id": 15, "created_at": "2025-07-06T23:01:56.986602Z", "updated_at": "2025-07-06T23:01:56.981892Z", "user": { "id": 38, "firstName": "Admin", "lastName": "User", "email": "representer922794@example.com", "role": "orgRep", "password": "pbkdf2_sha256\$6879898\$P76uJ85Njcltj78eAkdax\$P49r81xvz/G1BhZwaF16s8fBg/NaMDU8IIQ2xJDSs=" }, "organization": 7 }, { "id": 15, "created_at": "2025-07-06T23:01:56.986602Z", "updated_at": "2025-07-06T23:01:56.981892Z", "user": { "id": 38, "firstName": "Admin", "lastName": "User", "email": "representer922794@example.com", "role": "orgRep", "password": "pbkdf2_sha256\$6879898\$P76uJ85Njcltj78eAkdax\$P49r81xvz/G1BhZwaF16s8fBg/NaMDU8IIQ2xJDSs=" }, "organization": 7 }, { "id": 15, "created_at": "2025-07-06T23:01:56.986602Z", "updated_at": "2025-07-06T23:01:56.981892Z", "user": { "id": 38, "firstName": "Admin", "lastName": "User", "email": "representer922794@example.com", "role": "orgRep", "password": "pbkdf2_sha256\$6879898\$P76uJ85Njcltj78eAkdax\$P49r81xvz/G1BhZwaF16s8fBg/NaMDU8IIQ2xJDSs=" }, "organization": 7 } </pre> Activate Windows Go to Settings to activate Windows
14	getAllAdmins		<pre> { "firstName": "Hirev", "lastName": "Z", "email": "xronofusion91908@gmail.com", "password": "pbkdf2_sha256\$6879898\$P76uJ85Njcltj78eAkdax\$P49r81xvz/G1BhZwaF16s8fBg/NaMDU8IIQ2xJDSs=", "role": "admin" } </pre>
15	getAllOrganizations		<pre> { "id": 6, "name": "Siemens Digital Industries Software", "phone": "201234567890", "country": "it", "city": "finance", "ceoName": "Ahmed Ibrahim", "eYear": 2017, "industry": "finance,finance1", "organizationSize": "health", "links": [{ "type": "LinkedIn", "url": "https://mail.google.com/mail/u/0/#inbox" }], "updatesToEmail": false, "verificationStatus": true, "photo": "/media/photos/default.png", "created_at": "2025-06-24T17:26:25.719136Z", "role": "orgRep" } </pre>
16	getOrganizationByID	id	<pre> { "id": 2, "name": "Tech Solutions", ... } </pre>
17	getAllSeekers		[Seekers]
18	getUserDetails	id	<pre> { "id": 2, "name": "user1", ... } </pre>

19	getRandomNumbersOfSeekers	Empty	<pre> { "user": { "id": 4, "firstName": "Doaa", "lastName": "Ali", "email": "doal124689@gmail.com", "role": "seeker", "password": "pbkdf2_sha256\$870000\$7qnBCDS0jD0Q6taJf0xrv\$3y91dNNP8i3d95XDj1N0373InQFwsIgw3", "phone": "201019271286", "dob": "2003-08-16", "gender": "female", "title": "it", "nationality": "it", "careerLevel": "it", "employmentStatus": "it", "brief": "Software EngineerSoftware EngineerSoftware EngineerSoftware EngineerSoftware Engineer", "country": "it", "city": "it", "links": [{"type": "Portfolio", "url": "https://oj.com"}], "immediateStart": false, "updatesToEmail": false, "cv": "/media/cvs/mobile_workshops_3_4.pdf", "photo": null, "created_at": "2025-06-08T20:58:57.104137Z", "updated_at": "2025-06-28T11:33:11.420931Z", "skills": [] }, }</pre>
20	getSeekerInfoForProfile	User id	<pre> { "seeker": { "user": { "id": 41, "firstName": "test", "lastName": "test", "email": "esraa70@gmail.com", "role": "seeker", "password": "pbkdf2_sha256\$870000\$6GoxmILHfPzSzcXN34n9R6KJH3o+o08xay08keYvQLQ9YebQc9+70BmlxQpoeA+", "phone": "201111111111", "dob": "2000-01-01", "gender": "male", "title": "Software Engineer", "nationality": "Egyptian", "careerLevel": "Entry level", "employmentStatus": "Unemployed", "brief": "I love coding", "country": "Egypt", "city": "Cairo", "links": [] } } }</pre>
21	deleteAdmin	Admin id	<pre> { "message": "Admin deleted successfully" }</pre>
22	DeleteOrgRepresentative	UserId	<pre> { "message": "Organization representative deleted successfully" }</pre>
23	updateUserDetails		<pre> { "user": { "id": 41, "firstName": "test", "lastName": "test", "email": "esraa70@gmail.com", "role": "seeker", "password": "pbkdf2_sha256\$870000\$6GoxmILHfPzSzcXN34n9R6KJH3o+o08xay08keYvQLQ9YebQc9+70BmlxQpoeA+", "phone": "201111111111", "dob": "2000-01-01", "gender": "male", "title": "Software Engineer", "nationality": "Egyptian", "careerLevel": "Entry level", "employmentStatus": "Unemployed", "brief": "I love coding", "country": "Egypt", "city": "Cairo", "links": [{"type": "LinkedIn", "url": "https://linkedin.com/in/ahmed"}], "immediateStart": true, "updatesToEmail": false, "cv": null, "photo": "https://hireny.pythonanywhere.com/media/photos/default.png", "created_at": "2025-07-03T11:00:00.000000Z", "updated_at": "2025-07-03T11:00:00.000000Z" } }</pre>
24	update Admin	<pre> [{ "key": "firstName", "value": "test", "type": "text" }, { "key": "lastName", "value": "test", "type": "text" }]</pre>	<pre> { "firstName": "test", "lastName": "test", "email": "doal124689@gmail.com", "password": "pbkdf2_sha256\$870000\$7qnBCDS0jD0Q6taJf0xrv\$3y91dNNP8i3d95XDj1N0373InQFwsIgw3/9smdY-", "role": "seeker" }</pre>
25	ContactUs	<pre> { "name": "Yousef Hussein", "subject": "Feedback", "phone": "01012345678", "message": "Great platform, thank you!" }</pre>	<pre> { "error": "All fields are required." }</pre>

		{ "name": "Yousef Hussein", "email": "yousef@example.com", "subject": "Feedback", "phone": "01012345678", "message": "Great platform, thank you!" }	{ "status": "Message sent successfully!" }
26	Add Education	{ "institutionName": "Computer Science and AI at Cairo University", "degree": "Bachelor's Degree", "fromDate": "2020-01-01", "toDate": "2024-01-01", "description": "Studied Computer Science" }	<pre>{ "id": 3, "institutionName": "Computer Science and AI at Cairo University", "degree": "Bachelor's Degree", "fromDate": "2020-01-01", "toDate": "2024-01-01", "description": "Studied Computer Science", "created_at": "2025-07-02T21:22:00.786179Z", "updated_at": "2025-07-02T21:22:00.786201Z", "user": 41 }</pre>
27	Update Education	{ "institutionName": "Updated University Name", "degree": "Master's Degree", "fromDate": "2020-01-01", "toDate": "2024-01-01", "description": "Updated description" }	<pre>{ "id": 3, "institutionName": "Updated University Name", "degree": "Master's Degree", "fromDate": "2020-01-01", "toDate": "2024-01-01", "description": "Updated description", "created_at": "2025-07-02T21:22:00.786179Z", "updated_at": "2025-07-02T21:26:10.585529Z", "user": 41 }</pre>
28	Delete Education	Id = "3"	204 No Content
29	Add Certificate	{ "certificateName": "Machine learning and AI)", "institutionName": "Udemy", "dateIssued": "2023-05-15", "description": "Certified in ML and AI)" }	<pre>{ "id": 1, "certificateName": "Machine learning and AI)", "institutionName": "Udemy", "dateIssued": "2023-05-15", "description": "Certified in ML and AI)", "created_at": "2025-07-02T21:31:46.366636Z", "updated_at": "2025-07-02T21:31:46.366655Z", "user": 41 }</pre>
30	Update Certificate	{ "certificateName": "Updated Certificate Name", "institutionName": "Updated Institution", "dateIssued": "2023-05-15", "description": "Updated description" }	<pre>1 { 2 "id": 1, 3 "certificateName": "Updated Certificate Name", 4 "institutionName": "Updated Institution", 5 "dateIssued": "2023-05-15", 6 "description": "Updated description", 7 "created_at": "2025-07-02T21:31:46.366636Z", 8 "updated_at": "2025-07-02T21:33:00.427620Z", 9 "user": 41 10 }</pre>
31	Delete Certificate	Id = "3"	204 No Content

32	Add Course	{ "courseName": "Flutter clean architecture)", "institutionName": "Online Course Platform", "dateCompleted": "2023-08-20", "description": "Completed Flutter clean architecture) concepts" }	<pre>{ "id": 1, "courseName": "Flutter clean architecture)", "institutionName": "Online Course Platform", "dateCompleted": "2023-08-20", "description": "Completed Flutter clean architecture) concepts", "created_at": "2025-07-02T21:35:22.537544Z", "updated_at": "2025-07-02T21:35:22.537577Z", "user": 41 }</pre>
33	Update Course	{ "courseName": "Updated Course Name", "institutionName": "Updated Institution", "dateCompleted": "2023-08-20", "description": "Updated description" }	<pre>{ "id": 1, "courseName": "Updated Course Name", "institutionName": "Updated Institution", "dateCompleted": "2023-08-20", "description": "Updated description", "created_at": "2025-07-02T21:35:22.537544Z", "updated_at": "2025-07-02T21:36:48.895809Z", "user": 41 }</pre>
34	Delete Course	Id = "3"	204 No Content
35	Add Experience	{ "companyName": "Tech Company Inc.", "jobTitle": "Software Developer", "fromDate": "2022-01-01", "toDate": "2023-12-31", "jobType": "Full-time", "description": "Developed web applications using Django" }	<pre>{ "id": 1, "companyName": "Tech Company Inc.", "jobTitle": "Software Developer", "fromDate": "2022-01-01", "toDate": "2023-12-31", "jobType": "Full-time", "description": "Developed web applications using Django", "created_at": "2025-07-02T21:38:10.487911Z", "updated_at": "2025-07-02T21:38:10.487940Z", "user": 41 }</pre>
36	Update Experience	{ "companyName": "Updated Company Name", "jobTitle": "Senior Developer", "fromDate": "2022-01-01", "toDate": "2023-12-31", "jobType": "Full-time", "description": "Updated job description" }	<pre>{ "id": 1, "companyName": "Updated Company Name", "jobTitle": "Senior Developer", "fromDate": "2022-01-01", "toDate": "2023-12-31", "jobType": "Full-time", "description": "Updated job description", "created_at": "2025-07-02T21:38:10.487911Z", "updated_at": "2025-07-02T21:39:23.709440Z", "user": 41 }</pre>
37	Delete Experience	Id = "1"	204 No Content

45	Get Course Details		<pre>{ "id": 2, "title": "Python for Beginners", "description": "Learn the basics of Python programming including variables, data types, conditionals, loops, and functions.", "category": ["it", "health"], "instructor": "Eng. Sarah Mohamed", "priceType": "free", "priceAmount": "0.00", "currency": "USD", "created_at": "2025-06-24T15:30:14.917965Z", "updated_at": "2025-06-24T15:30:14.917965Z", "lessons": [{ "id": 1, "title": "Lesson 1", "video": "/media/videos/test_08N0zjb.mp4", "pdf": "/media/pdfs/YousefHussienCV_QCNbAap.pdf", "duration": "00:00:05.318000", "created_at": "2025-06-24T15:30:14.917965Z", "updated_at": "2025-06-24T15:40:13.030400Z" }] }</pre>
46	Get Not Registered Courses		<pre>{ { "id": 2, "title": "updated Python Course", "description": "Learn Python from scratch", "category": ["Category 1", "Category 2"], "instructor": "Ahmed Samir", "priceType": "paid", "priceAmount": "10.00", "currency": "USD", "created_at": "2025-06-24T15:30:14.917965Z", "updated_at": "2025-07-02T22:15:54.117651Z", "lessons": [], "total_duration": 0 } }</pre>
47	Get All Courses	Admin	<pre>{ { "id": 2, "title": "Python for Beginners", "description": "Learn the basics of Python programming including variables, data types, conditionals, loops, and functions.", "category": ["it", "health"], "instructor": "Eng. Sarah Mohamed", "priceType": "free", "priceAmount": "0.00", "currency": "USD", "created_at": "2025-06-24T15:30:14.917965Z", "updated_at": "2025-06-24T15:30:14.917965Z", "lessons": [{ "id": 1, "title": "Lesson 1", </pre>
48	Enroll In Course	Id = "2"	<pre>{ "message": "Successfully enrolled in the course." }</pre>
49	Unenroll in Course	Id = "2"	204 No Content
50	Check Course Enrollment	Id = "2"	<pre>{ "is_enrolled": false }</pre>
51	Add Course	<pre>{ "title": "Python Course", "description": "Learn Python from scratch", "category": ["Category 1", "Category 2"], "instructor": "Ahmed Samir", "priceType": "free", "priceAmount": 0, "currency": "USD" }</pre>	<pre>{ { "id": 3, "title": "Python Course", "description": "Learn Python from scratch", "category": ["Category 1", "Category 2"], "instructor": "Ahmed Samir", "priceType": "free", "priceAmount": "0.00", "currency": "USD", "created_at": "2025-07-02T22:11:54.158570Z", "updated_at": "2025-07-02T22:11:54.158609Z", "lessons": [], "total_duration": 0 } }</pre>
52	Add Lesson	<pre>[{"key": "title", "final lesson": "test", "type": "text"}, {"key": "video", "value": "6546.mp4", "type": "file"}, </pre>	<pre>{ { "id": 3, "title": "final lesson", "video": "/media/videos/4_5857142065145580210_XrcmldF.mp4", "pdf": "/media/pdfs/%D9%86%D9%85%D8%A7%D8%B0%D8%AC_English.pdf", "duration": "00:11:14.490000", "created_at": "2025-07-05T23:21:31.811540Z", "updated_at": "2025-07-05T23:21:31.811540Z" } }</pre>

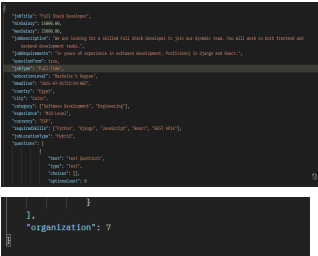
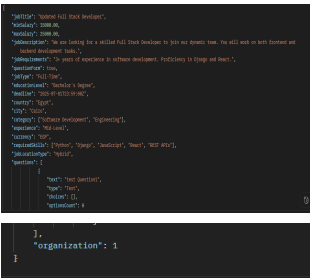
		<pre>{"key": "pdf", "value": "6546.pdf", "type": "file"}</pre>	
53	Update Course		<pre>{ "id": 2, "title": "updated Python Course", "description": "Learn Python from scratch", "category": ["Category 1", "Category 2"], "instructor": "Ahmed Samir", "priceType": "paid", "priceAmount": 10, "currency": "USD", "created_at": "2025-06-24T15:30:14.917905Z", "updated_at": "2025-07-02T22:15:54.117651Z", "lessons": [{ "id": 1, "title": "Lesson 1", "video": "/media/videos/test_OBN0zjb.mp4", "pdf": "/media/pdfs/YousefHussienCV_QCNbAap.pdf", "duration": "00:00:05.310000", }] }</pre>
54	Update Lesson		<pre>{ "id": 1, "title": "updated title", "video": "/media/videos/test_OBN0zjb.mp4", "pdf": "/media/pdfs/YousefHussienCV_QCNbAap.pdf", "duration": "00:00:05.310000", "created_at": "2025-06-24T15:30:34.792840Z", "updated_at": "2025-07-02T22:19:48.401400Z" }</pre>
55	Delete Course	Id = "2"	204 No Content
56	Delete Lesson	Id = "1"	204 No Content
57	getAllConstants		<pre>{ "countries_with_cities": { "Afghanistan": ["Herat", "Kabul", "Kandahar", "Molah", "Rana", "Shar", "Sharif", "Wazir Akbar Khan"], "Albania": ["Elbasan", "Petran", "Pogradec", "Shkoder", "Tirana", "Ura Vajguore"] } }</pre>

58	getAllCountriesWithCities		<pre> { "Afghanistan": ["Herat", "Kabul", "Kandahar", "Molah", "Rana", "Shar", "Sharif", "Wazir Akbar Khan"], "Albania": ["Elbasan", "Petran", "Pogradec", "Shkoder", "Tirana"] } </pre>
59	getAllNationalities		<pre> { "nationalities": ["Afghan", "Albanian", "Algerian", "American", "Andorran", "Angolan", "Anguillan", "Argentine", "Armenian", "Australian", "Austrian", "Azerbaijani", "Bahamian", "Bahraini", "Bangladeshi"] } </pre>
60	getAllLanguages		<pre> { "lang": ["Afrikaans", "Albanian", "Amharic", "Arabic", "Aramaic", "Armenian", "Aymara", "Azerbaijani", "Basque", "Belarusian", "Belizean Creole", "Bengali", "Berber", "Bislama", "Bosnian", "Bulgarian", "Burmese", "Carolinian", "Catalan",] } </pre>

61	getAllSkills		<pre> { "skills": ["supply chain engineering\r\n", "bullet\r\n", "commutations\r\n", "pay equity\r\n", "student retention\r\n", "pulsar", "hevacom\r\n", "travel insurance\r\n", "payback\r\n", "soaps\r\n", "erdf\r\n", "3d simulation\r\n", "中級\r\n", "server side\r\n", "filemaker server\r\n"] } </pre>
62	getAllCountriesForSalaryModel		<pre> { "countries": ["Afghanistan", "Albania", "Algeria", "Angola", "Argentina", "Aruba", "Austria", "Azerbaijan", "Australia", "Bahamas", "Bahrain", "Bangladesh", "Barbados", "Belarus", "Belgium"] } </pre>
63	getAllJobs		<pre> { "jobs": ["Finance Assistant", "Logistics Coordinator", "Food Service Worker", "Logistics Supervisor", "Intelligence Analyst", "Armed Security Guard", "Security Guard", "Accountant", "Finance Manager", "Data Analyst", "Civil Engineer", "Electrical Engineer", "Design Architect", "Mechanical Engineer", "Graphic Designer"] } </pre>

64	getAllExperienceLevels		<pre> { "experience_levels": ["Entry Level", "Early Career", "Mid Career", "Experienced", "Late Career"] } </pre>
65	getAllOrgSizes		<pre> { "org_sizes": ["Self-Employed", "1-10 Employees", "11-50 Employees", "51-100 Employees", "101-200 Employees", "201-500 Employees", "501-1000 Employees", "1001-5000 Employees", "5001-10000 Employees", "10001+ Employees", "Other"] } </pre>
66	getAllJobTypes		<pre> { "job_types": ["Internship", "Full-Time", "Part-Time", "Temporary", "Freelance", "Contract", "Volunteer"] } </pre>
67	getAllJobLocationTypes		<pre> { "job_location_types": ["On-Site", "Remote", "Hybrid"] } </pre>
68	getAllInterviewStatuses		<pre> { "interview_statuses": ["In Review", "Accepted", "Rejected", "Interview Scheduled"] } </pre>
69	getAllIndustries		<pre> { "industries": ["Agriculture", "Automotive", "Aviation & Aerospace", "Banking", "Biotechnology", "Business Supplies & Equipment", "Chemicals", "Computer & Network Security", "Construction", "Consumer Electronics", "Consumer Goods", "Defense & Space", "Design", "Education", "Electrical/Electronic Manufacturing"] } </pre>

70	getAllEmployeeStatuses		<pre> { "employee_statuses": ["Student", "Fresh Graduate", "Intern", "Trainee", "Employee", "Unemployed", "Contractor", "Freelancer", "Part-Time Employee", "Temporary Employee", "Retired", "Volunteer"] } </pre>
71	getAllEducationLevels		<pre> { "education_levels": ["No Formal Education", "Primary School", "Middle School / Junior High School", "High School Incomplete", "High School Diploma / Secondary School", "Vocational Training / Technical Certificate", "Associate's Degree", "Diploma", "Bachelor's Degree", "Postgraduate Diploma / Higher Diploma", "Master's Degree", "Professional Degree (Md, Jd, Pharmd, Etc.)", "Doctorate / Phd", "Post-Doctoral Studies", "Some College - No Degree"] } </pre>
72	getAllCurrencies		<pre> { "currencies": ["Usd", "Aed", "Afn", "All", "Amd", "Ang", "Aoa", "Ars", "Aud", "Awg", "Azn", "Bam", "Bbd", "Bdt", "Bgn"] } </pre>
73	getAllCourseLengths		<pre> { "course_lengths": ["Less Than 1 Hour", "1-5 Hours", "6-10 Hours", "11-20 Hours", "21-30 Hours", "31-50 Hours", "51-75 Hours", "76-100 Hours", "101-150 Hours", "151-200 Hours", "More Than 200 Hours", "Self-Paced", "Other"] } </pre>
74	getAllCategories		<pre> { "categories": ["Administrative & Clerical", "Food Service & Restaurant", "Miscellaneous", "Protective Services", "Accounting & Finance", "Architecture & Engineering", "Art & Design", "General Managers & Executives", "Healthcare Practitioners & Technical", "Human Resources", "Information Technology", "Marketing & Advertising", "Media & Publishing", "Education", "Business Operations"] } </pre>

			<pre> "similarity_score": 35.736698150634766 }, { "id": 4,... "similarity_score": 32.77399826049805}] </pre>
81	Create Job Post	 <pre> { "organization": 7 } </pre>	<pre> { "id": 18, "jobTitle": "Full Stack Developer", "minSalary": "15000.00", "maxSalary": "25000.00", "jobDescription": "We are looking for a skilled Full Stack Developer to join our dynamic team. You will work on both frontend and backend development tasks.", "jobRequirements": "3+ years of experience in software development. Proficiency in Django and React.", "questionForm": true, "jobType": "Full-time", "educationLevel": "Bachelor's Degree", "deadline": "2025-07-01T23:59:00Z", "country": "Egypt", "city": "Cairo", "category": ["Software Development", "Engineering"], "experience": "Mid-level", "currency": "EGP", "requiredSkills": ["Python", "Django", "JavaScript", "React"] } </pre>
82	Update Job Post	Id = "18"  <pre> { "organization": 1 } </pre>	<pre> { "id": 18, "jobTitle": "Updated Full Stack Developer", "minSalary": "15000.00", "maxSalary": "25000.00", "jobDescription": "We are looking for a skilled Full Stack Developer to join our dynamic team. You will work on both frontend and backend development tasks.", "jobRequirements": "3+ years of experience in software development. Proficiency in Django and React.", "questionForm": true, "jobType": "Full-time", "educationLevel": "Bachelor's Degree", "deadline": "2025-07-01T23:59:00Z", "country": "Egypt", "city": "Cairo", "category": ["Software Development", "Engineering"], "experience": "Mid-level", "currency": "EGP", "requiredSkills": ["Python", "Django", "JavaScript", "React"] } </pre>
83	Delete Job Post	Id = "17"	<pre> { "message": "Job post deleted successfully." } </pre>
84	Get All Service Posts for Org	Id = "5"	<pre> [{ "id": 4, "serviceTitle": "gjhghghghghj", "salary": "10.00", "serviceDescription": "202020202\nhnvhngIn", "category": ["it"], "country": "finance", "city": "health", "currency": "USD", "serviceAvailable": "available", "createdAt": "2025-06-25T21:16:11.905700Z", "updatedAt": "2025-06-25T21:16:11.905700Z", "organization": 5, "total_applications": 1, "companyName": "Tech Solutions", "companyLogo": "/media/photos/BACK.jpg" }] </pre>
85	Get Service Post Details	Id = "4"	<pre> { "id": 4, "serviceTitle": "gjhghghghghj", "salary": "10.00", "serviceDescription": "202020202\nhnvhngIn", "category": ["it"], "country": "finance", "city": "health", "currency": "USD", "serviceAvailable": "available", "createdAt": "2025-06-25T21:16:11.905700Z", "updatedAt": "2025-06-25T21:16:11.905700Z", "organization": 5, "total_applications": 1, "companyName": "Tech Solutions", "companyLogo": "/media/photos/BACK.jpg" } </pre>
86	Get Not Applied Service Posts		<pre> [{ </pre>

			<pre> "id": 8,..., "companyLogo": "/media/photos/default.png" },..... { "id": 6,..... "companyLogo": "/media/photos/default.png" }] </pre>
87	Get 3 Not Applied Service		<pre> [{ "id": 8,..... "companyLogo": "/media/photos/default.png" }, { "id": 7,..... "companyLogo": "/media/photos/default.png" }, { "id": 6,.... "companyLogo": "/media/photos/default.png" }] </pre>
88	Get Related Services for a Service	Id =" 4 "	<pre> [{ "id": 8, "serviceTitle": "Digital Marketing Specialist",....., "similarity_score": 0 }, { "id": 7, "serviceTitle": "Digital Marketing Specialist",..... "similarity_score": 0 }, { "id": 6, "serviceTitle": "service",..... "similarity_score": 0 },] </pre>

89	Create Service Post	<pre>{ "serviceTitle": "Digital Marketing Specialist", "salary": 1200.00, "serviceDescription": "We specialize in SEO, social media marketing, and email campaigns to grow your brand online.", "category": ["Marketing", "Advertising"], "country": ["Egypt"], "city": ["Cairo"], "currency": ["EGP"], "serviceAvailable": "available", "organization": 1 }</pre>	<pre>{ "id": 9, "serviceTitle": "Digital Marketing Specialist", "salary": 1200.00, "serviceDescription": "We specialize in SEO, social media marketing, and email campaigns to grow your brand online.", "category": ["Marketing", "Advertising"], "country": "Egypt", "city": "Cairo", "currency": ["EGP"], "serviceAvailable": "available", "createdAt": "2025-07-03T15:14:46.684526Z", "updatedAt": "2025-07-03T15:14:46.684573Z", "organization": 11, "total_applications": 0, "companyPhone": "Tech Solutions", "companyLogo": "/media/photos/default.png" }</pre>
90	Update Service Post	Id = "9" <pre>{ "serviceTitle": "Updated Graphic Design Services", "salary": 5000.00, "serviceDescription": "We offer professional graphic design services including logos, banners, and branding packages.", "category": ["Design", "Creative"], "country": ["Egypt"], "city": ["Alexandria"], "currency": ["EGP"], "serviceAvailable": "available", "organization": 1 }</pre>	<pre>{ "serviceTitle": "Updated Graphic Design Services", "salary": 5000.00, "serviceDescription": "We offer professional graphic design services including logos, banners, and branding packages.", "category": ["Design", "Creative"], "country": ["Egypt"], "city": ["Alexandria"], "currency": ["EGP"], "serviceAvailable": "available", "organization": 1 }</pre>
91	Delete Service Post	Id = "9"	204 No Content
92	Get All Org Reviews	Id = "7"	<pre>[{ "id": 15, "message": "Great support from the organization!", "createdAt": "2025-07-03T15:22:49.494920Z", "updatedAt": "2025-07-03T15:22:49.494947Z", "organization": 7, "reviewer": 38, "reviewerName": "Tech Solutions", "reviewerPhoto": "/media/photos/default.png" }, ]</pre>
93	Create Org Review	Id = "7" <pre>{ "message": "Great support from the organization!" }</pre>	<pre>{ "id": 15, "message": "Great support from the organization!", "createdAt": "2025-07-03T15:22:49.494920Z", "updatedAt": "2025-07-03T15:22:49.494947Z", }</pre>

			<pre> "organization": 7, "reviewer": 38, "reviewerName": "Tech Solutions", "reviewerPhoto": "/media/photos/default.png" } </pre>
94	Delete Org Review	Id ="15"	204 No Content
95	Update Org Review	<pre> Id = "16" { "message": "Updated review message" } </pre>	<pre> { "id": 16, "message": "Updated review message", "createdAt": "2025-07-03T15:27:21.979112Z", "updatedAt": "2025-07-03T15:27:33.825663Z", "organization": 7, "reviewer": 38, "reviewerName": "Tech Solutions", "reviewerPhoto": "/media/photos/default.png" } </pre>
96	Get Course Reviews	Id = "5"	<pre> [{ "id": 6, "message": "This course was very informative and well-structured.", "createdAt": "2025-07-03T15:32:49.983428Z", "updatedAt": "2025-07-03T15:32:49.983474Z", "reviewer": 41, "course": 3, "reviewerName": "test test", "reviewerPhoto": "/media/photos/default.png" },.....] </pre>
97	Create Course Review	<pre> { "message": "This course was very informative and well-structured." } </pre>	<pre> { "id": 5, "message": "This course was very informative and well-structured.", "createdAt": "2025-07-03T15:31:55.491418Z", "updatedAt": "2025-07-03T15:31:55.491458Z", "reviewer": 41, "course": 3, "reviewerName": "test test", "reviewerPhoto": "/media/photos/default.png" } </pre>

98	Update Course Review	Id ="5"	{ "id": 5, "message": "Updated course review message", "createdAt": "2025-07-03T15:31:55.491418Z", "updatedAt": "2025-07-03T15:34:02.289730Z", "reviewer": 41, "course": 3, "reviewerName": "test test", "reviewerPhoto": "/media/photos/default.png" }
99	Delete Course Review	Id ="5"	207 no Content
100	Get seeker Assessments		[{ "id": 10, "score": 0,... "assessment": { "id": 3, "title": "Flutter Developer Exam", "questions": [...],.... "status": "Missing" } }]
101	Get Org Assessments		[{ "id": 6, "assessmentTitle": "Updated", "assessmentDuration": 200, "createdAt": "2025-07-03T15:48:38.395991Z", "updatedAt": "2025-07-03T15:52:47.755651Z",..... "type": "Text", "choices": ["organization": 11, "total_applicants": 0}]]
102	Get Assessment Details	Id = "6"	{ "id": 6, "assessmentTitle": "Updated", "assessmentDuration": 200, "createdAt": "2025-07-03T15:48:38.395991Z", "updatedAt": "2025-07-03T15:52:47.755651Z",..... },]

			<pre>"organization": 11, "total_applicants": 0 }</pre>
103	Get Assessment Applicants	Id="6"	<pre>{ "assessment_title": "Updated", "applicants": [.....] }</pre>
104	Get solution and user assessment for Assessment by assId	Id = "6"	<pre>{ "id": 6, "score": 66.66666666666666, "answers": [...],.... "questions": [...] "status": "Evaluated" }</pre>
105	Assign Assessment to User	<pre>{ "user": 3, "assessment": 3, "answers": null, "score": 0.0, "assessmentStartTime": "2025-05-17T19:27", "assessmentEndTime": "2025-05-19T20:31" }</pre>	<pre>{ "id": 9, "score": 0, "answers": null, "is_corrected": false, "created_at": "2025-07-03T15:40:46.166399Z", "updated_at": "2025-07-03T15:40:46.166449Z",..... "status": "Missing" }</pre>
106	Start assessment for user	Id = "10"	<pre>{ "message": "Assessment started successfully." }</pre>
107	Submit User Assessment	Id = "10"	<pre>{ "id": 10, "score": 16.666666666666664, "answers": [...],.... "user": { "id": 41, ...}. "status": "Evaluated" }</pre>
108	Create Assessment	<pre>{ "assessmentTitle": "Flutter Developer Exam", "assessmentDuration": "5", "questions": [{</pre>	<pre>{ "id": 6,... "type": "Checkbox", "choices": ["initState", "dispose", "setState",</pre>

		<pre> "text": "What does the 'hot reload' feature in Flutter do?", "type": "Text", "choices": ["", "", "", ""], "correctAnswer": "It updates the UI with code changes without restarting the app", "optionsCount": 4, { "text": "Which of the following..",.... "optionsCount": 4 } } </pre>	<pre> "build"], "correctAnswer": [true, true, false, true], "optionsCount": 4 }], "organization": 11, "total_applicants": 0 } </pre>
109	Update Assessment	<pre> Id ="6" { "questions": [{ "text": "12", "type": "Text", "choices": ["", "", "", ""], "correctAnswer": "125328525", "optionsCount": 4 }], "assessmentTitle": "Updated", "assessmentDuration": "200" } </pre>	<pre> { "id": 6, "assessmentTitle": "Updated", "assessmentDuration": 200, "createdAt": "2025-07-03T15:48:38.395991Z", "updatedAt": "2025-07-03T15:52:47.755651Z", "questions": [{ "text": "12", "type": "Text",.... "correctAnswer": "125328525", "optionsCount": 4 }], "organization": 11, "total_applicants": 0 } </pre>
110	Delete Assessment	Id ="3"	204 no Content
111	Get Job Applicants	Job id ="4"	<pre> { "job_applications": [{ "id": 5, "hasApplication": false, "answers": {}, "createdAt": "2025-06-28T01:52:53.557433Z", "updatedAt": "2025-06-29T03:58:44.003763Z", </pre>

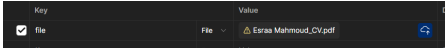
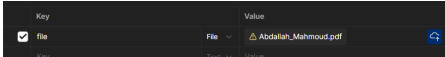
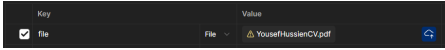
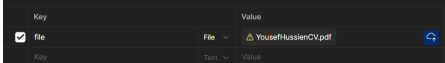
			<pre> "status": "Interview Scheduled", "user": 3, "jobPost": { "id": 4,.... "deadline": "2026-08-30T00:00:00Z" }, "applicant": { "id": 3, "name": "Yusef Yusefghghgh",.... "dob": "2025-06-06" }, "similarity_score": 64.82160186767578 }, { "id": 8, "hasApplication": false, "answers": {},..... }, "applicant": { "id": 4, "name": "Doaa Ali",... }, "similarity_score": 26.78459930419922 }], "post_title": "Software Engineer" } </pre>
112	Apply for Job	Job id = "19"	<pre> { "id": 9, "hasApplication": true, "answers": [{ "answer": "fghghgh" }, { "answer": "0" }, { "answer": [1] }],..... "jobPost": { "id": 19,.... "deadline": "2025-07-01T23:59:00Z" }, "applicant": { "id": 41,.... </pre>

			<pre> "dob": "2000-01-01" } } </pre>
113	Get Job Application Details	Id = "5"	<pre> { "job_application": { "id": 5, "hasApplication": false, "answers": {}, "createdAt": "2025-06-28T01:52:53.557433Z", "updatedAt": "2025-06-29T03:58:44.003763Z", "status": "Interview Scheduled", "user": 3, "jobPost": {... }, "applicant": { "id": 3, "name": "Yusef Yusefghghgh", "email": "yousefhussien5656@gmail.com",.... "dob": "2025-06-06" } }, "job_post_title": "Software Engineer" } </pre>
114	Update Job Application	Job Application Id="10" <pre> { "hasApplication": true, "answers": [{ "answer": "Updated fghghgh" }, { "answer": "0" }, { "answer": [1] }] } </pre>	<pre> { "id": 10, "hasApplication": true, "answers": [{ "answer": "Updated fghghgh" }, { "answer": "0" }, { "answer": [1] }],.... "questions": [{ "text": "test Question1", "type": "Text", "choices": [], "optionsCount": 0 }], } </pre>

			<pre> "companyName": "Tech Solutions", "companyId": 11, "photo": "/media/photos/default.png", "deadline": "2025-07-01T23:59:00Z" }, "applicant": { "id": 41,..... "dob": "2000-01-01" } } </pre>
115	Delete Job Application	Job Application Id="10"	204 No Content
116	Get My Job Applications		<pre> [{ "id": 9, "hasApplication": true, "answers": [{ "answer": "fghghgh" }, { "answer": "0" }, { "answer": [1] }], "createdAt": "2025-07-03T19:49:36.682363Z", "updatedAt": "2025-07-03T19:49:36.682401Z", "status": "Pending", "user": 41, "jobPost": { "id": 19, }, "applicant": { "id": 41,.... } }] </pre>
117	Check Applied To Job Post	Job id ="1"	<pre> { "status": false } </pre>

118	change status of job application for user	Job App Id= "5" { "status": "Accepted" }	{ "message": "Status updated successfully." }
119	Get Service Applicants	Service id ="4"	{ "service_applications": [{ "id": 4, "createdAt": "2025-06-28T02:08:36.750267Z", "updatedAt": "2025-06-28T02:08:36.750267Z", "status": "Pending", "organization": 6, "servicePost": { "id": 4, "title": "gjhghghghghj", "company": "Tech Solutions", "photo": "/media/photos/BACK.jpg", "salary": 10, "currency": "USD" }, "applicant": { "id": 6, "name": "Siemens Digital Industries Software", "email": "yousefhussien919@gmail.com", "photo": "/media/photos/default.png", "phone": "201234567890" }], "post_title": "gjhghghghghj" }
120	Apply for Service	Service id ="4"	{ "id": 5, "createdAt": "2025-07-03T20:23:03.591553Z", "updatedAt": "2025-07-03T20:23:03.591590Z", "status": "Pending", "organization": 11, "servicePost": {...} "applicant": { "id": 11, "name": "Tech Solutions", "email": "esraa12@company.com",

			<pre> "photo": "/media/photos/default.png", "phone": "+201234567890" } } </pre>
121	Delete Service Application	Service id ="4"	204 No Content
122	Get My Organization Service Applications		<pre> [{ "id": 5, "createdAt": "2025-07-03T20:23:03.591553Z", "updatedAt": "2025-07-03T20:23:03.591590Z", "status": "Pending", "organization": 11, "servicePost": { "id": 4, "title": "gjghjghjghj", "company": "Tech Solutions", "photo": "/media/photos/BACK.jpg", "salary": 10, "currency": "USD" }, "applicant": { "id": 11, "name": "Tech Solutions", "email": "esraa12@company.com", "photo": "/media/photos/default.png", "phone": "+201234567890" } }] </pre>
123	Check Applied to Service Post	Service id ="1"	<pre> { "status": false } </pre>
124	change status of service application for user	Service id ="4"	<pre> { "message": "Status updated successfully." } </pre>
125	salaryInsights	<pre> { "Job": "Senior Software Architect", "Country": "Egypt", "Experience": "Experienced" } </pre>	<pre> { "success": true, "predicted_salary_usd_per_month": 436.41 } </pre>

126	Extract Seeker Data from Resume	CV 	{ "firstName": "Esraa", "lastName": "Mahmoud", "email": "esraaamahmoud77@gmail.com", "phone": "+20 1146073609", "dob": null, "gender": null, "title": "Flutter Developer", "nationality": null, "careerLevel": "Entry Level", "employmentStatus": "Employee", "brief": "Passionate.....", "country": "Egypt", "city": "Cairo" }
127	Extract Org Data from Proof	CV 	{ "name": Abdallah, "phone": 0123456789, "email": abdallah@gmail.com, "country": Egypt, "city": cairo, "ceoName": kbvjt, "eYear": 2002, "industry": [SW Engineering], "organizationSize": 100-200, "links": [.....] }
128	Download Resume	CV 	{ "docx_download_url": "resume/test_test_resume.docx", "pdf_download_url": "resume/test_test_resume.pdf" }
129	Download Cover Letter	CV 	{ "docx_download_url": "cover_letters/test_test_cover_letter.docx", "pdf_download_url": "cover_letters/test_test_cover_letter.pdf" }
130	Recommend Titles		{ "success": true, "titles": ["AI Prompt Engineer", "Data Scientist", "Cybersecurity Analyst"] }

			}
131	Create Interview	<pre>{ "name": "Interview with Ali", "date": "2025-07-01T15:00:00", "job_seeker": 41, "organization": 2 }</pre>	<pre>{ "id": 11, "name": "Interview with Ali", "date": "2025-07-01T15:00:00Z", "created_at": "2025-07-03T20:40:16.400304Z", "updated_at": "2025-07-03T20:40:16.400323Z", "job_seeker": 41, "organization": 11, "interviewerName": "Tech Solutions", "jobSeekerName": "test test", "InterviewDetailsForSeeker": "Interview with Ali with Tech Solutions", "InterviewDetailsForOrganization": "Interview with Ali with test test" }</pre>
132	Get Interview	Interview ID="11"	<pre>{ "id": 11, "name": "Interview with Ali", "date": "2025-07-01T15:00:00Z", "created_at": "2025-07-03T20:40:16.400304Z", "updated_at": "2025-07-03T20:40:16.400323Z", "job_seeker": 41, "organization": 11, "interviewerName": "Tech Solutions", "jobSeekerName": "test test", "InterviewDetailsForSeeker": "Interview with Ali with Tech Solutions", "InterviewDetailsForOrganization": "Interview with Ali with test test" }</pre>
133	List Interviews		<pre>[{ "id": 11, "name": "Interview with Ali", "date": "2025-07-01T15:00:00Z", "created_at": "2025-07-03T20:40:16.400304Z", "updated_at": "2025-07-03T20:40:16.400323Z", "job_seeker": 41, "organization": 11, "interviewerName": "Tech Solutions", "jobSeekerName": "test test",</pre>

			<pre> "InterviewDetailsForSeeker": "Interview with Ali with Tech Solutions", "InterviewDetailsForOrganization": "Interview with Ali with test test" }] </pre>
134	Check Interview Permission	Interview ID="11"	<pre> { "id": 11, "name": "Interview with Ali", "date": "2025-07-01T15:00:00Z", "created_at": "2025-07-03T20:40:16.400304Z", "updated_at": "2025-07-03T20:40:16.400323Z", "job_seeker": 41, "organization": 11, "interviewerName": "Tech Solutions", "jobSeekerName": "test test", "InterviewDetailsForSeeker": "Interview with Ali with Tech Solutions", "InterviewDetailsForOrganization": "Interview with Ali with test test" } </pre>
135	Delete Interview	Interview ID="11"	204 No Content
136	Get All Notifications		<pre> { "notifications": [{ "id": 80, "message": "New notification message", "title": "Notification Title3000", "is_read": false, "created_at": "2025-07-03T20:49:58.834072Z", "user": 40 }, { "id": 77, "message": "New notification message", "title": "Notification Title3000", "is_read": false, "created_at": "2025-07-03T20:49:24.627322Z", "user": 40 }], "unread_count": 2 } </pre>

137	Send Notification	{ "notifier_role": "admin", //org,seeker,admin "message": "New notification message", "title": "Notification Title3000" }	{ "message": "Notification sent to all admins." }
138	Mark notification as read	empty	{ "notifications": [{ "id": 104, "message": "Hello Tech Solutions,\n\nThank you for registering as an organization on HireNy. We "title": "Welcome to HireNy!", "is_read": false, "created_at": "2025-07-04T14:46:56.100324Z", "user": 32 }], "unread_count": 1 }
139	Get Seeker Events		[{ "id": 11, "title": "event", "link": null, "type": "other", "description": "event 1 day", "startDate": "2025-06-29T08:00:00Z", "endDate": "2025-06-30T08:00:00Z" }]
140	Get Org Events		[{ "id": 12, "title": "event", "link": null, "type": "other", "description": "event 1 day", "startDate": "2025-06-29T08:00:00Z", "endDate": "2025-06-30T08:00:00Z" }]
141	Create Seeker Events	{ "title": "event", "start_date": "2025-06-29T08:00:00Z", "end_date": "2025-06-30T08:00:00Z", // can be null "description": "event 1 day" // can be null }	{ "id": 11, "title": "event", "link": null, "type": "other", "description": "event 1 day", "startDate": "2025-06-29T08:00:00Z", "endDate": "2025-06-30T08:00:00Z" }
142	Create Org Events	{	{

		<pre> "title": "event", "start_date": "2025-06-29T08:00:00Z", "end_date" : "2025-06-30T08:00:00Z", // can be null "description": "event 1 day" // can be null } </pre>	<pre> "id": 12, "title": "event", "link": null, "type": "other", "description": "event 1 day", "startDate": "2025-06-29T08:00:00Z", "endDate": "2025-06-30T08:00:00Z" } </pre>
143	Delete Seeker Event	Event id ="11"	204 No Content
144	Delete Org Event	Event id ="12"	204 No Content
145	Chatbot	<pre> { "message": "give me roadmap about flutter" // available services, explore courses, about hireny, (or any thing else) // could be uppercase or title case } </pre>	<pre> { "mode": "gemini", "message": "Here's a roadmap for learning Flutter," } </pre>
146	Get Conversations		<pre> { "comparisonId": 19, "conversations": [{ "participantId": 18, "participantName": "Tech Solutions", "participantRole": "org", "participantPhoto": "/media/photos/approve.png", "messages": [{ "id": 209, "content": "Hello from Postman!", "created_at": "2025-07-04T15:53:54.467911Z", "sender": 19, "receiver": 18 }], "unreadCount": 0 }] } </pre>
147	Mark Messages As Read	Id ="19"	<pre> { "message": "Messages marked as read." } </pre>
148	Send Message	Id ="18"	{

		<pre>{ "content": "Hello from Postman!" }</pre>	<pre>"message": "Message sent successfully." }</pre>
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5.2. Cost Evaluation and Infrastructure Analysis

The "HireNy" system comprises three main components:

- **Mobile Frontend:** Flutter
- **Web Frontend:** React (used for dashboards and service management)
- **Backend:** Django (REST API, business logic, authentication, payment integration)

Hosting and Infrastructure

Table 45: Hosting, Infrastructure and AI Costs

Component	Provider	Plan	Monthly Cost	Notes
Web Frontend	Vercel	Pro	\$20	Recommended for production; includes 1 TB bandwidth, team collaboration
Backend	PythonAnywhere	Startup	\$99	Supports 3 web apps, 10,000 CPU seconds/day, 50 GB storage, custom domain
Database	PostgreSQL	Add-on	~\$5–10	Paid option on PythonAnywhere; external providers (e.g., Supabase) also viable
Domain	—	—	~\$1.25	Amortized annually (\$10–15/year from providers like Namecheap or GoDaddy)

Apple Developer	Apple Developer	Annual	~\$8.25	Required for Apple Pay support (annual \$99, amortized monthly)
AI Services	OpenAI	ChatGPT (GPT-4o)	\$20	Used for resume parsing, AI matching, feedback scoring

SSL Certificates: Free via Let’s Encrypt on both Vercel and PythonAnywhere.

Payment Note: Credit card (foreign currency enabled) is currently the only payment method available in Egypt for these services.

Payment Gateway Costs

Table 46: Payment Gateway Costs

Payment Method	Setup Cost	Transaction Fees	Notes
Apple Pay	\$99/year (Apple Dev)	Varies (to be confirmed)	Requires Apple Developer subscription
Google Pay	One-time \$25	~2.9%	—
PayPal	Free	~2.99% + \$0.49 per transaction	Recommended for initial deployment

Estimated Monthly Operating Cost (Startup Phase)

Table 47: Estimated Monthly Cost

Component	Plan	Monthly Estimate
Vercel	Pro	\$20
PythonAnywhere	Startup	\$99
PostgreSQL	Add-on/External	~\$5–10
Domain	Annualized	~\$1.25
Apple Developer	Annualized	~\$8.25
ChatGPT (OpenAI)	GPT-4o	\$20
Total Estimate	—	\$153.50 – \$158.50

These costs are based on the startup phase and can be scaled as the platform grows (e.g., upgrading to Vercel Enterprise or higher-tier PythonAnywhere plans).

Chapter 6: Conclusion and Future Work

6.1. Summary of Achievements

"HireNy" has successfully transitioned from an academic project into a fully functional AI-powered recruitment platform. The system integrates smart matching, real-time notifications, resume parsing, and role-based user experiences. The key components developed include:

- Dynamic role-based user registration
- AI-driven job recommendation and candidate matching
- Resume parsing and skill extraction
- Cover Letter AI-generated
- Job tracking dashboards for employers
- Real-time notifications and email integration
- File uploads including CVs and profile photos

6.2. Conclusion

In conclusion, the "HireNy" project successfully addresses key challenges in the recruitment landscape by offering an innovative, dual-sided platform tailored to both job seekers and organizations. Through features such as AI-driven tools, personalized job support, and streamlined hiring workflows, the platform enhances the overall recruitment experience. By integrating B2B services and promoting transparency, efficiency, and user empowerment, "HireNy" not only bridges gaps in the hiring process but also sets a new standard for recruitment technology that is both inclusive and future-ready.

6.3. Planned Enhancements

The following improvements are proposed to increase the platform's value, scalability, and inclusivity.

Technical and AI Enhancements

- AI-powered interview simulations with emotional and linguistic feedback
- Career path recommendations based on current skills and market trends
- Real-time skill gap analysis and personalized course suggestions
- Integration with online education platforms such as Coursera and Udemy
- Fraud detection through behavioral monitoring and anomaly detection

Ecosystem and Platform Expansion

- Collaboration with government job portals for cross-posting
- Employer verification with trust badges
- Embeddable widgets for universities to support student employment

Accessibility and Inclusion

- Full support for users with disabilities through screen reader compatibility, voice input, and keyboard navigation
- Localization support for Arabic and other regional languages

6.4. Financial Strategy

To maintain platform sustainability and cover third-party service costs (hosting, AI services, payment gateways), selected features will be offered as part of a paid plan for both job seekers and organizations.

Paid Features for Job Seekers

Table 48: Paid Features for Job Seeker

Feature	Access Type
AI-powered resume feedback	Paid
Advanced job match recommendations	Paid
Skill gap analysis	Paid
Access to highlighted job listings	Paid
Full access to detailed employer info	Paid
Higher limit for job applications/month	Paid

Paid Features for Organizations

Table 49: Paid Features for Organization

Feature	Access Type
Multiple job posts per month	Tiered Subscription
Access to full candidate details	Paid
Advanced applicant analytics	Paid
Employer trust badge (verified org)	Paid
Custom dashboard and application tracking	Paid
Chat integration with candidates	Paid
Candidate skill match scoring	Paid